

Neur 310 The Neuroscience of Dreams

Numbered Transcript

NEUR 310 • Dr. Baland Jalal

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Lecture 1: Why We Sleep

At a glance — Jalal starts a course on dreams with sleep itself — its stages, function, and why we sleep — as the necessary foundation.

- 1.1 All right. So we'll be talking about sleep in this lecture. Of course, this is a course about dreams, but we have to start somewhere, and sleep is a good place to start. Right, Arnold?
- 1.2 Right.
- 1.3 In this course, I'll go from things like: Why do we sleep? How do we fall asleep? Why is sleep important? Then we'll go into dreams, and the nature of dreams, why dreams are important, how dreams arise in the brain. Then we go farther out to the edges of dreams to talk about things like sleepwalking, why people sometimes will sleepwalk. Then we'll talk about nightmares.
- 1.4 We'll talk about sleep paralysis. Have you heard about sleep paralysis before? This condition where you lie in your bed, paralyzed, unable to move or speak, and suddenly you feel like there's a demon coming from the corner of the room, approaching you, pressing your chest, strangling you. You see all these monsters. We'll talk about that.
- 1.5 We'll talk about dreams and creativity. How dreams are this magical landscape where the neurophysiology of your brain is completely different from the neurophysiology we know from wakefulness, and how this lends itself to creativity. How creative people throughout history have used dreams to harness their creativity and get new ideas and do scientific experiments all through their dreams. So we'll look at that.
- 1.6 Then we'll talk about dream monsters. Why people have these monsters in their dreams, Heavenly, these monsters that attack them, and you try to run and you're paralyzed, you can't move your feet. They're so heavy. You had that dream before? You want to just move, but you're so damn heavy, right? So let's talk about that. Why these dreams come about, why people all around the world have these dreams.
- 1.7 And then finally, we end on the topic of why and how dreams can be very healing and very nurturing. How it is that dreaming heals our emotions and makes us more calm and can be therapeutic. And if you have PTSD, and trauma, and anxiety, and depression, dreaming can erase all the negative emotions within you. So we'll go into that. Does that sound good? All right.
- 1.8 But let's start somewhere. So sleep. Sleep is an interesting topic. I've been traveling lately, having jet lag, so my sleep is all over the place. But I'm not sleep-deprived in the big sense.
- 1.9 There was this guy, Michael Corke. He had issues with sleep for a few days. He couldn't sleep. He was trying to sleep, but he couldn't. And he kept trying, but he felt aroused by —he felt like his mind was all over the place, wandering. He just couldn't sleep. And

then he tried to sleep on the couch for a few days. He thought his wife snoring [snoring noise] was the culprit, the issue. But it really wasn't because he couldn't sleep after four or five days on the couch. But he kept trying, and trying, and trying to sleep on the couch, but nothing happened. He just couldn't fall asleep.

- 1.10 About day 10, he decided to go to his doctor and he said, "Doctor, doctor, I can't sleep, for God's sake. What's going on?" And so the doctor said, "Well, this is very strange. Let's have a look at you." But they couldn't really find a cause. It was so weird; the guy couldn't sleep. I mean, that's the most basic and fundamental thing of all, yet he couldn't sleep. At this point, things kept going. He just couldn't sleep.
- 1.11 Day 20, he was using a cane to walk. He was that fatigued and that tired. Imagine, have you tried to not sleep for a few days, Arnold? Have you tried that before? Imagine for 20 days, no sleep. That's pretty severe. At this point, his predicament continued.
- 1.12 He went all the way to six months, six months of no sleep. And I'm not kidding you. This is no joke, six months. And at this point, he was hallucinating. He was seeing monsters all the time. He could barely move. He could barely shower. He could barely—I mean, when he was communicating, he was using his body to communicate because he couldn't communicate with language and speech. And it shows us how important sleep is, how it deprives our brains of basic function if we lose sleep. And this guy just couldn't sleep. Two years later, the poor chap died from this disease or illness.
- 1.13 Now, I won't tell you right now what happened with Michael Corke and why he died, what happened inside his brain in the circuitry. But I'll keep that to the very end of this lecture. How's that? So we'll keep it as a mystery, and then by the end of the lecture, we'll dig deeper and see why this chap couldn't sleep. Is that a good deal? In recorded history, Michael Corke is not the only guy who actually died from lack of sleep. There are other people.
- 1.14 There was this soccer superfan who died from watching the European Championship. So he watched like—he binge watched all the matches for like 11 days straight and he died. And we know rats as well. So if you deprive rats of sleep— so you take away their sleep for 15 days, about 15, 18—they will die, too. So we know that.
- 1.15 And then one of the stages of sleep called REM sleep—so sleep is fragmented, there's different stages of sleep, one of these stages is called REM sleep. And if I deprive you of REM sleep, about a month of just REM, just this phase of sleep, you will die as well. So a month of lack of REM. So this all shows us how important REM sleep is. So we'll get into REM and what REM is, all this REM: rapid eye movement sleep, the eyes moving back and forth.
- 1.16 By the way, so let me go into the stages of sleep right away. So you have that foundation. Aserinsky—he was a guy in the University of Chicago, 1950s— with his mentor, Kleitman, they discovered REM, rapid eye movement sleep. REM sleep.

Initially, Aserinsky saw that these babies, their eyes were moving side to side, like this. It was like, what's going on?

- 1.17 And they measured the activity of neurons. You could put—this is called an EEG. It's like gel, you put it on the brain. You put it on here, and then you measure the waves in the brain. So you measure the brain activity. And they did that, and the muscle activity, and they saw that the eyes were moving like this during a certain part of sleep. And the activity of the brain was rapid during one stage of sleep, during this stage; they didn't know what it was at this point. But they couldn't believe their own eyes when they saw this.
- 1.18 I mean, this was a great discovery, the fact that you have sleep, and during one stage of sleep, the brain is hyperactive and the eyes are moving from side to side. At another stage where the brain is quiet, it's all like it's slow, the waves slow down, everything is just much more slow. So they found this.
- 1.19 In fact, Kleitman, he wanted to test this on his own daughter, Esther. So he brought in Esther to test: is this really true? Can this be right that you have brain activity firing all over the place during one stage and another slow and steady stage? Can this be right? And then they confirmed that with his own daughter. Would you do that for science? I would, sometimes.
- 1.20 They confirmed there are two stages of sleep: rapid eye movement sleep and non-REM sleep. So at this point, only two overall stages. Not several stages, not like REM, deep, Stage 1, 2, 3, 4. No, not all that. Just two big stages. So this is how it all started in the 1950s, University of Chicago.
- 1.21 During Stage 1 of sleep, it's where it all started. It all starts there. You become drowsy. Temperature drops, heart rate drops, chemicals in the brain, noradrenaline. You know of adrenaline in the body, right? So you have adrenaline when you're hyper and you're active and you are—maybe a car is rushing towards you and you're on the freeway and have a lot of what? Adrenaline. You have its chemical buddy in the brain called noradrenaline or norepinephrine, excuse me, okay, in the brain that serve its similar function, making you aroused, making you hyper-alert. All right? So you have that.
- 1.22 You can imagine when you're reading a book, do you have a lot of noradrenaline or a little bit of noradrenaline? A little bit. And then when your mom comes into your room and says, "You have to do your homework, okay?" Do you have a lot of noradrenaline? A little bit more, right? It rises. And then when you're giving a lecture, more, right? Playing soccer, more. Being chased by a sabre-tooth tiger, much, right? There's a lot. Okay, noradrenaline. So it has all this.
- 1.23 Now, when you are sleeping, when you're sleeping and you're falling asleep during Stage 1 of sleep, noradrenaline is slowly falling in activity. So you have less at this point because you have to be less aroused. So your brain has to drop its level of noradrenaline. Serotonin neurotransmitter also drops—just adding that, but we'll get

into that. So you have other neurochemicals in the brain. You all know this. And then blood pressure, temperature. Temperature has to drop for you to sleep. This is one key rule. Temperature has to drop for you to sleep. The temperature, core body temperature drops. So you have that as well.

- 1.24** Now, during Stage 1, you have all that. The brain is slowly turning off. You have the cortical region. So the brain has different parts— I mentioned that briefly—you have prefrontal lobes out here. You have the cortex, which is the outer layer of the brain, like a helmet. You have that six layers deep—just adding that—so six layers deep.
- 1.25** You have the visual cortex in the back of your brain. You have all that. When you fall asleep, the cortex quiets down. It slowly quiets down a bit. You have the cortex becoming more and more quiet. The visual cortex, however, doesn't. So the visual parts of the brain in the back right here is active, so it stays on a bit. Or actually, it becomes even, you can say, at some point more active as we enter the dreaming world. The brain is becoming more and more and more quiet.
- 1.26** Then you hit deep sleep. Deep sleep. A deep form of sleep. If I were to—AJ. AJ, right? Okay, good. When I say your name, you have to be quick. AJ, if I wake you up from deep sleep, guess what happens? You'll be like this. Oh, groggy. You can't really focus because it's a very, very deep form of sleep.
- 1.27** And interestingly, this is very fascinating. Neurons in the brain, the activity of the cortex go like this: boom, boom, boom— big ocean wave, synchronized wave, up and down, up and down. Remember Kleitman and Aserinsky, the way of measuring brain waves? You can measure that. It's like a symphony, up and down. It's very interesting. It's quite fascinating. I want to contrast that to the wakefulness wave.
- 1.28** So when you are awake, the brain has this frenetic erratic activity like this, [noise], All over the place. But in this deep stage of sleep, so you have this initially, right? These erratic waves, and then everything quiets down, and then suddenly you have deep sleep, the deep stage of sleep, and it goes like this. Is that clear, Sonia? Okay, good.
- 1.29** And by the way, on these big waves of oceans, ocean waves, as I call them, you have these spindles. You can't really—it's very hard to explain, but you have to see this on an actual monitor. Have you seen them, Arnold? Spindles, called sleep spindles. Now, what's going on with the sleep spindles? Is everything clear so far? We said the initial waves are very quick, big waves, and then these spindles, deep sleep.
- 1.30** These spindles are telling us that the brain is transferring information from the memory parts of the brain, the hippocampus—seahorse-like structure behind your ears, tucked behind your ears, that's called the hippocampus. It takes information from the memory part of the brain, the hippocampus, and places that in the cortical region and the outer layer of the brain, which is more like, to use an analogy, perhaps like the hard drive and the thumb, what do you call the memory stick thing, thumb drive? That's like the

hippocampus, the memory part of the brain that allows you to store memories. Or, yeah, store them initially, have access to them.

- 1.31** But then it has to be removed to the outer layer of the brain, the cortex for long term storage. Is that clear? So you have to have that. And these spindles help you with that. And you have these spindles during deep sleep. Interesting so far? Does it make sense? So it's important. You know why? Because you want to have a part of your life dedicated to taking information from my daily life and storing that in some kind of a safe vault so it's there, so I have access to that in the future. And that happens during deep sleep. This deep stage of sleep. Okay?
- 1.32** You have some spindles during some of the earlier stages as well, like Stages 2 and the stages before deep sleep. But it's mainly during deep. During deep sleep, you have release of growth hormones. You have release of testosterone, although you have that in other stages as well, but it's a key stage for that as well.
- 1.33** This is interesting. I find this utterly fascinating. During deep sleep, glial cells. You have neurons and you have glial cells, supporting cells. These brain cells during deep sleep will shrink by about 20%. Can you believe this? They will shrink, allowing cerebrospinal fluid to bathe the brain. So cerebrospinal fluid is this fluid that will bathe the brain, allowing it to wash all its corridors and all its corners. So you can get rid of toxins and beta-amyloid and all these proteins that can build up and become misfolded and lead to dementia and all that. You want to get rid of that, and that happens during deep sleep.
- 1.34** Imagine this. How important is deep sleep? Hormones, growth hormones, hair, skin, all that. Memory: taking things from your everyday life, storing that. Plus removing all the toxins from your daily life. Pretty important, deep sleep. Okay.
- 1.35** But we haven't gotten—still the most important part of sleep is coming. It's called the rapid eye movement sleep, REM sleep. It happens after deep sleep. So Stage 1, 2, 3, 4, then we have REM. And during REM, in the back of the brain, in the brain stem, again, not too— I don't want to throw too many terms at you with the brain stem. You know why? Because when I did the first course, I went back and looked at the comments and they said, "This guy, he's using all these Latin terms, and you have to be more pedagogical." I'm trying to make it more simple. I understand you haven't spent 30 years or 20 years like me studying the brain, so you have expertise in other areas, perhaps. Maybe Arnold, he knows already, but you know.
- 1.36** Brain stem is the lower part of the brain. It's down here. It's very important. Why? Breathing, body temperature, regulating yourself, all that, regulating your core temperature. Mind you, if you have a tiny stroke to the lower part of the brain in the brain stem, guess what happens? Often you will die from that. A tiny stroke there, tiny trauma—you're dead.

- 1.37** In contrast, the prefrontal lobes, meaning the part of the brain up here that has to do with decision making, and logical thinking, and rational thinking, and higher functions like poetry, pity, infinity, Shakespeare, all that—you can have massive stroke. You can have a big spear running through that and you will survive. In fact, you will function in society. You will go around and just be a bit rude and be a bit a little bit frivolous and disinhibited and all that. But you can function. Imagine that the part of the brain that makes us most human, this part of the brain, with all the higher functions, strokes, all that, you will survive and just have a crazy personality. But here, a tiny stroke in that lower part of the brain, and you are probably dead.
- 1.38** Now, as we enter rapid eye movement sleep, you have neurons in the brain stem in a structure called the pons, a ball-like structure, and the medulla oblongata, a long structure here, part of this brain stem, lower brain circuit, releasing chemicals glycine and GABA. Glycine and GABA. So you have in the brain neurochemicals that can either be inhibitory or excitatory.
- 1.39** Inhibitory or excitatory. The excitatory neurons. Dopamine, right—there is dopamine, right? You have dopamine. Glutamate is the most famous of them all. GABA is the most abundant inhibitory neurotransmitter. GABA. You have GABA and glycine being spit out by the brain stem, dripping down your neck to the alpha motor neurons—fancy name for dripping down your spinal cord.
- 1.40** In your spinal cord, you have connections to your entire peripheral system, your entire body, controlling your movement. And so if you have these neurotransmitters kicking in, they will paralyze you during rapid eye movement sleep. And for that reason, you are paralyzed in REM. You're wearing a physiological straitjacket. You can't move during REM.
- 1.41** And why can't you move? Why is it important to be paralyzed during rapid eye movement sleep? Why is it so important not to be able to move? Because you have your most lifelike, crisp, vivid, and emotional dreams during REM sleep. So your brain is very clever. It says, "Look, Heavenly, I can't have you act out your dreams and hurt your sleeping partner or whatever. You don't want that. So for that reason, I will paralyze you completely from head toe during rapid eye movement sleep."
- 1.42** But then Heavenly here, because she has experience in psychology, worked at major universities, she will say, "Well, Baland, you can move your eyes from side to side. What's going on there? What's happening there? You're not paralyzed, are you? Is that true? Can it be?" And then I'll say, "Well, look, you have a different circuitry for that." There's a different circuitry in the brainstem going up to the eyes, allowing you to move your eyes.
- 1.43** By the way, some people will say, "How come you can move your eyes? What's the purpose of your eyes moving in REM?" It's a good question, right? You want to know, perhaps, right? Some will say, some have said that these eye movements are, in fact,

the person, the dreaming person scanning his dreams. Like, here's a dinosaur. Here's my little brother. Here's the moon. So that's perhaps a function. Others have said that it's merely physiology of REM. It's not really anything important. It's more of a byproduct of REM. So it's not really clear why the eyes are moving during rapid eye movement sleep.

- 1.44** But here's the kicker. We said the brain waves during wakefulness are what, AJ? Smart. Heavenly Sun is smart over here, okay? The brain waves are going like this, right? Very active, right? Okay. [Noise]. Very active. Now, if you eavesdrop on the activity of neurons during REM sleep, guess what? They look indistinguishable from the guy who's awake. So if I was to look at a guy who's awake, and I look at a guy who's in REM, their brains look the same. The brain suddenly speeds up. Everything becomes hyperactive. All the neurons are firing in this erratic pattern. In fact, some parts of the brain are up to 30% more active during rapid eye movement sleep.
- 1.45** There goes the theory that when you sleep, your brain, your neurons are just becoming calm, and everything is just firing down. No, no. What you're seeing here is actually your brain being hyperactive, and about 30% more active in certain parts of the brain. In the next lecture, we'll get into what parts of the brain, because that's important for understanding dreams. That's very important for understanding dreams. What parts of the brain are active during dreams and what part slows down. Now, you have all that. That's the cycle. That's the stages of sleep.
- 1.46** Then, of course, you have all these stages in one cycle. It's about 90 minutes, nine-zero minutes. You have that. But of course, it's not like 90 per se. It can be 100 minutes, 110 minutes. You have these "boom, boom, boom" throughout the night. If you're sleeping eight hours like you should, you would have how many of these? Maybe four or five, right? Of blocks of these.
- 1.47** Interestingly, during the first half of the night, during the first half of the night, you have more of deep sleep. During the second half of the night, you have more of REM sleep. It's almost as if these two stages of sleep are competing like rivaling sports teams. They're battling out. It's like a wrestling match, and they're competing. The two big hitters, the REM and the deep sleep. And that's important, but I'll get to that why it's important in a minute.
- 1.48** I just want to add something here, which is: what's important is, REM, you have release of things like growth hormone and testosterone as well. So there's that part of it, too. You also have— what you can say—an interesting function is that REM is all about taking your experiences from the day and weaving that into a whole, a narrative-like structure, and extracting wisdom. Does that make sense?
- 1.49** So if you think about deep sleep as the boring librarian, just taking information and storing it in different places, REM is the visionary, is the Steve Jobs of sleep. It takes stuff from daily life and creates gist of information and knowledge and wisdom in a way

that's almost ineffable. It's this wisdom that we have. If you deprive people of REM, they will lack creativity and the spark of wisdom and being able to see the bigger whole in things. Is that clear? There's that contrast.

- 1.50** REM is important for cognitive function. Being able to be extremely cognitively acute and fast and having mental agility and all that, that's in REM. Memory, IQ test, all that is a part of the REM sleep. It's a REM phenomenon. Now, given our equation, our equation was these two stages of battling it out each night, right? If I was to deprive you of—let me actually specify. So, more deep sleep, you can have maybe 10 minutes of REM in the 1st cycle of sleep. Ten minutes of REM in the 1st cycle. 2nd, a little bit more. 3rd, a little bit more. 5th, the end, the final cycle of sleep, you have maybe one hour of REM. Does that make sense?
- 1.51** Now, if I was to split your sleep in two, if I split your sleep in two. And I say, "Joe, you have to wake up at 7 AM or 5 AM even, or 3 AM." Let's say you sleep by midnight. I would have caught you, I would have deprived you of all the REM sleep of the early morning. Does that make sense? It's not an equal equation, I'm saying. The point I'm making: it's not an equal equation. If I was to cut 10 inches or 10% or 20% of the final portion of your cycles of sleep, I would have removed a lot of REM sleep. Can you see that? Does that make sense?
- 1.52** And then what happens when you have very little REM? What happens then? Which functions are lost? Memory, cognition, wisdom, all these things, right? Creativity, all these things go away. That shows you that biology, overall, it's— biology is very messy. It's complex. So you can't just go, "Oh, I'm just going to—I'm clever. I'm just going to cheat biology and get a few hours here and there." You can't. If you miss a few hours because you have to catch a plane, you would have lost a lot of REM, is what I'm saying.
- 1.53** Perhaps a rule of thumb, perhaps, and this is oversimplifying things, but if you're an athlete, you might say, "Okay, I can get away with, let's say, having my initial, my initial hours of sleep. If I have my initial hours of sleep, then at least I'm physically ready to run away from a tiger." And that's the idea, actually. So some people argue that the reason why we have all the physiological benefits coming first is that if you have little sleep, let's say you actually wake up 3 AM, 4 AM, something happens, let's say you hear some sounds in the little village you live in. I'm talking about the Stone Age person, right?
- 1.54** Okay, so he's living there, and he's awakened at 4 AM. At least he has the physical ability to run away from tigers and people chasing him and all that. Does that make sense? So arguably, arguably, some will say that having all the other creative parts are not that important. It comes later. If you get it, great. You have to have that on a consistent basis. But once in a while, if you only have the physiological aspects as well, well, then you at least you can survive in nature. Does that argument make sense? I think so. I think it's a good argument.

- 1.55** Importance of sleep. During sleep, during deep sleep—I have to add this because this is my notes. You have lowering of blood pressure during deep sleep—very important aspect. Things like glucose regulation: deep sleep. Things like immune system: deep sleep. In fact, if I deprive you of deep sleep, your sympathetic nervous system, meaning you have your sympathetic nervous system, meaning the stress, the nervous system that's active when your heart is "boom, boom, boom, boom" pounding and your adrenaline adrenaline is all over the place. That part of the nervous system is highly active if I don't get a lot of deep sleep. In fact, heart disease and all that can often emerge from not having enough of deep sleep.›
- 1.56** On the lifespan part of things, sleep will vary a bit. So you've noticed babies, they sleep a lot. You notice that? Babies, they sleep a lot. And so an infant will spend a lot of time sleeping, 17 hours sometimes, sleeping. Like a three months old baby. Inside the mother's womb, they will sleep all the time, pretty much. And interestingly, the brains of babies are often—it's often REM sleep, so they will spend a lot of time in REM, have a lot of REM sleep. So that infant that's sleeping 17 hours, half of that is pure REM. It's just like, REM, REM, REM.
- 1.57** As we age, as we go along to become a bit older, at years, let's say, four or five, you have 13 hours. You become a teenager, then you have 10 hours. A full-grown adult, you have seven to eight hours. So sleep varies. But then you might ask, why do you have all that REM? Why do you have all that REM in the baby brain? And how come REM then decreases by the time you reach adolescence and you become a teenager and you become a full-grown adult? Why do you have that change in REM sleep?
- 1.58** Well, one idea is that during REM, you have creativity and all that, right? In fact, you have a lot of synaptic plasticity, meaning that the synapses and the neurons in the brain are starting to become linked up. There's a lot of neural activity that allows for—this has to do with a chemical called acetylcholine, but you have a lot of plasticity, meaning a lot of synapses being built in the brain. The new connection's being built. That's all where all the creativity and knowledge and wisdom is coming about.
- 1.59** You have that during REM sleep. In a baby brain, does it make sense to have a lot of new synapses and connections and a lot of new information being created? Does that make sense in a baby brain? It does, right? Then you might ask: by adolescence, you have deep sleep emerging and entering the scene and coming in and saying, "Look, I want my part now. I want to have more time." Why is that important? Well, you can think of deep sleep as also—not only the librarian with the books and all that, but also as something that cuts these connections and removes the excess and redundancies.
- 1.60** Initially, you want to have a lot of REM sleep to build the baby brain. And then by the time you reach adolescence, you want to have trimming down, called synaptic pruning. You probably already know this, but the brain of a child is hyperconnected. We know this. It's hyperconnected. And then there's genetic processes. There's a gene that's important for pruning, meaning removing excess connections. So it does that through

adolescence. It removes, it's a "use it or lose it" situation here. So all this stuff that you're not using in your brain is being cut away. And a part of that happens during deep sleep, and deep sleep being introduced in the final portions and then coming in at adolescence. Does that make sense here? Okay, good.

1.61 Then maybe you have an objection. You might ask, "How come when you watch a baby—Baland, you said before that babies spend all this time in REM. But you also said that you are paralyzed from head to toe during REM. Didn't you say that, Baland? You did, right?" Do babies move when they sleep? I don't have a baby. Do babies move when they sleep? Is this a paradox then? What's going on? The answer is the following. The circuitry responsible for the paralysis has not yet evolved in these babies. It's simply premature. That's the reason. It's very simple.

1.62 But they are certainly in REM, and when they are kicking and sometimes smiling even, that is—some have argued, like Allan Hobson, my colleague over at Harvard Medical School—he argues that these are ways to train your sensory and motor systems. Train them so the child's brain is more able to use them later on. Does that make sense?

1.63 From a survival perspective, the fact that you want to—like a baby doesn't want to—look, biology is so costly. It's deeply costly. Look, this lady was pregnant for nine months nurturing this thing, this baby. It's heavy. You don't want to lose it. The first thing you want to have that baby is to train your sensory motor system so as soon as possible, it can move and start to be active, animated, and live in the world and survive, essentially.

1.64 Let's go back to the initial question of the guy who died, Michael Corke. This is an important question. How come Michael died? What happened in his brain? He had a rare syndrome called fatal familial insomnia, FFI. It's a genetic disorder. And what happens, Sonia, is that in the brain, there's a part of the brain called the hypothalamus. In the hypothalamus, you have a structure beneath it called the thalamus. Thalamus. Hypothalamus— it's a part of the emotional centers of the brain.

1.65 The thalamus, on the other hand, thalamus. You know what it is? It's the sensory doorman of the brain. It's like the doorman, the gatekeeper of the brain for sensory filtering and information. Everything that comes in through the world— seeing, hearing, touch—all that has to pass through the thalamus. It's the doorman.

1.66 Now, if the thalamus is not working, what happens? Do you become flooded with information and sensory input? You would, right? If the doorman of an event is not there, what happens? Flooding, right? Of sensory information. That's exactly what happens. The thalamus has to shut down for you to sleep. For your sensory parts of your brain in order to keep vision and touch and hear and all that away from you, you need to have that thalamus, the sensory part of the brain, to be shut down. And so in us, in most of us, in almost all of us, you have that.

- 1.67** But Michael Corke—the neurons in his thalamus was dying off. They were being slowly erased. The neurons were dying. So his thalamus, his sensory doorman was weak. It was weak, it was eroding. And so for that reason, he just couldn't—the sensory filter to the world was not there. He didn't have a filter. And so he was perpetually on. I mean, imagine a car, the engine always being on or whatever. You having headphones on all the time, 24 hours. It's some form of torture, in fact. So this is the thalamus, and this is the mystery of Michael Corke and what happened.
- 1.68** Here's my question.
- 1.69** Go ahead, Larry.
- 1.70** I'm 62. You have an amazing memory with names. It was very good.
- 1.71** You look 22.
- 1.72** So when I rounded 50, my sleep habits changed rapidly. I thought it was maybe only due to me. But then my wife, who's nine years younger than me, just hit her early 50s. She went from being a very deep sleeper— I've always had a very hard time getting to sleep until I approach my early '50s. Now I sleep like a baby, and she has historically slept like a baby. Do people often experience different—is that a predictable physiological reaction to sleep?
- 1.73** Absolutely, it is. You know what? It's so important this portion of how aging affects sleep, and I want to talk about it in the next lecture. I'll dedicate maybe 5, 10 minutes to go into depth. I'll answer that. Is that okay, Larry? It's a very important question. Sleep definitely changes as we age.
- 1.74** You spoke about REM sleep being important for things like creativity and cognitive flexibility and how babies have more REM sleep than general adult population. Are there any times I could think, clinically, where somebody wouldn't be getting the REM sleep and we're prioritizing things like deep sleep for physiological functioning with things like trauma and things like that? Do you ever see cases where that seems to be happening?
- 1.75** The brain sleep is favoring deep sleep in trauma?
- 1.76** Just in any case, clinically, where you would have a situation where somebody would be favoring that deep sleep over the more creative REM sleep.
- 1.77** Have you watched my lectures online? [laughter]. This guy. It's going to be— even that point, I'm going to keep it maybe 10 minutes in the lectures. But there is. There is. There is actually a condition. It's called depression. I'll cover that in-depth. Now, in case I forget, I need you to give me some signal or something in the lecture. I want to talk about this, about deep sleep and depression. It's very important. Thank you, AJ.

Lecture 2: Why We Dream

At a glance — Why we dream: theories of dreaming's purpose and what happens in the dreaming brain.

- 2.1 Took a little nap in the makeup room. They say naps are good for you. Do you believe that? A nap is good? Naps are very good, actually. Do you know that napping, maybe let's say 90 minutes, just like a 90 minutes deep nap, can be as effective as sleeping five or six hours? Did you know that? Naps allow you to bypass some of the, let's call it the redundancies of sleep, the stuff that's on the surface you don't really need. It gets to the crux of the matter. So naps are very good.
- 2.2 And on the basis of this whole idea of redundancies, let's answer one question that Larry had. Larry wanted to know, as you all noticed, as we age, it seems like our sleep is less deep. Almost as if when we were teenagers and just sleeping, it felt so good. We just put our heads on the pillows, and there we were, and we're sleeping 10 hours. But now it's all fragmented, shallow. We hear the slightest sound, and then we wake up and we can't keep asleep almost. Do you agree? What's going on? Let's cover that before we dive into dreams.
- 2.3 Now, it turns out as we age, the quality of sleep lessens. It becomes worse. And to be exact, we have the stages of sleep, and the stage we termed or called deep sleep, the deep stage of sleep, becomes less as we age. It literally goes away. It becomes less—we have less of it. At your late 40s and 50s, deep sleep will be gone by maybe 60 or 70%. Think about that. By only late 40s. Then by the time you hit, let's say, 70, you would have lost 90% of deep sleep.
- 2.4 Think about this. So when my dad tells me once in a while, he tells me, I often bring him for lectures, and when I give a podcast or something, I bring my dad along. I love my dad. He's getting older. And he tells me, actually, what he does is he wakes up in the middle of the night and he can't sleep and he wanders around, and then he wakes me up and it all gets a bit crazy. But he always tells me that his sleep is so shallow; he can't really get a lot of good sleep. And one explanation is that as we age, we have less of deep sleep. Instead, you know what we have? Stage 1 and Stage 2, the more shallow types of sleep where you're just lying there. You're sleeping, but it's all shallow.
- 2.5 Now, there is a population out there, and I'm not one of them, but 1% of the population, you know what? They can get five hours of sleep, and they're completely refreshed and active and invigorated. Only five hours of sleep. These guys are called deep sleepers. It's said that Donald Trump has this genetic mutation that allows him to get a lot of deep sleep. I'm serious.
- 2.6 This is actually a word. When I was out in university, we talked about this. We said there's some people, politicians, that might have this. They sleep five hours and they have genetic mutations in genes called DEC2. That bounce on you? DEC2. These genes

allow you to get a lot of deep sleep, a lot of nourishing, fulfilling deep sleep. So all your cells shrink, as we talked about, to clean your brain, and then you have all the skin and hair and all that being replenished and all your organs and all that. So you have a lot of that deep sleep. So these are lucky guys, very lucky people, deep sleepers. And then I want to add before I go into the dreaming world—because that's going to be really crazy—I want to talk to you about caffeine because we have a few caffeine drinkers in here. So what does caffeine do to you? So you notice that when you drink caffeine, you feel all energetic and you want to move around and be active in the day and you feel fresh. Then if you drink caffeine, let's say at 9 PM and 10 PM or something like that, what happens then? You can't sleep, perhaps. What's going on? Is that really energy? What is happening in the brain? Well, it turns out in the brain, there are certain things that allow us to fall asleep. There are certain parameters, there are certain rules, algorithms in the brain that allow us to sleep and tell us: sleep now, be awake now, be fresh now, be active now, and all that.

2.7 And one of these factors is a chemical called adenosine. Adenosine is a chemical that builds up in your brain. It's like a little clock. It builds up as we are awake, and the more we are awake, it builds up more and more and more in the system, in the brain. Adenosine. Adenosine builds up, builds up, builds up—13 hours, 14 hours. There's a lot of adenosine in the system, and you want to sleep, right? That's adenosine doing that. But then, caffeine.

2.8 What caffeine does to adenosine is the following. In the brain, you have neurons, and on neurons, you have receptors. Receptors are little doorways, little parking spots by which other neurons can communicate with that particular neuron. These parking spots, we call receptors. Now, adenosine receptors in the brain, accumulated, and they are all over the brain. You have that.

2.9 These are the places where, parking spots where adenosine will bind onto, latch onto, and allow adenosine to affect the neurons and the entire brain, and the parts of the brain that allows it to be active and the parts of the brain that calms things down. So adenosine has that function. There's receptors for adenosine. Caffeine, what it does is it comes into the brain, has sharp elbows, goes in there and blocks adenosine receptors. It latches onto the adenosine receptors, so adenosine no longer can go inside the neuron and spread in the brain.

2.10 For that reason, guess what happens? For that reason, adenosine accumulates outside the neurons and in the system. It's still around. It's not like it's going away. It's not going anywhere. It's still there, but it's being temporarily blocked. It's not there for the moment. But then as caffeine wears off, as these adenosine receptors, the caffeine no longer are there, glued to the receptors, guess what happens? You have what? Am I being clear, Arnold, here? I feel like I'm not being completely crisp and clear.

2.11 What I'm saying is the following: you have adenosine building up as hours pass by, more adenosine, more adenosine, more adenosine. Now, you have a lot of adenosine

in the system, right? But because of caffeine latching onto the receptor, blocking it, adenosine can't actually make you tired, right? Okay. But there's a lot of adenosine in the system ready to hit you like a tidal wave, right? Okay.

2.12 Guess what happens then when caffeine wears off and there's tons of adenosine in the bloody system, what happens then? The caffeine crash. That's the end. That's why caffeine, yes, it makes you alert, it makes you fresh. It blocks sleep temporarily, but really what it does is that it's masking sleep. It's really masking it. And adenosine, this chemical parameter, this clock is still doing its work. And whenever there's no more adenosine in the system, or caffeine in the system, excuse me, then adenosine—punch. This is punch, right? Okay.

2.13 All right, so let's dive in to dreams. This might be one of my favorite topics. I love dreams. I worked on dreams for a long time. I was fortunate to work with dream researchers that have done a lot of interesting work throughout the years. But there's nothing like a dream. I mean, dreams are profound. You're lying there. You have rivers, mountains, planets. People, maybe your mother, your grandmother might be your dream. In fact, a lot of dreams are with people we know. They have people we know by name for some reason.

2.14 And the setting is epic. It can be like epic castles. So it's like an epic castle. You're there, but it feels oddly familiar. It feels like home, like your bedroom, but it's a castle at the same time. So you have that. Things are hyperconnected. So you're having tea with the Queen on Mars, wearing a suit maybe, or maybe a space suit or whatever. Everything is strange. Things are hyperassociative. Things are hyperconnected. So you have that in dreams. Populated by a lot of people. There's a lot of people in your dreams. So we're going to that, why there are so many people in your dreams. Emotional, hyper-emotional. Dreams are hyper-emotional. But these are parts of dreams.

2.15 So you have different types of dreams. You have REM dreams, for example. These would fit what I just described. These are complex narrative that have story plots. Your brain is a storyteller and you have story. Story means you have this beginning, middle, and end, typically. There's a sense of a cohesion, like a narrative. You have that. These are REM dreams, complex, crisp, lifelike types of dreams.

2.16 But if you wake up during Stage 1 or 2, the early shallow types of parts of sleep, and I wake you up and I ask you, "What's going on? Are you having any mental activity, AJ?" And you will say, "Baland, I am. Yes, in fact, I was chatting with a buddy on somewhere at my old school or whatever." So you still have what you call dream fragments during the other stages of sleep. Is that clear? Deep sleep as well, you have types of dreams.

2.17 In fact, you have the most terrifying and vivid nightmares during deep sleep. So deep sleep is also important. However, the actual longer nightmares we're familiar with is during REM. It's during REM. But you have these other types of dream activity in the other stages of sleep. Occasionally during sleep, during these early stages of sleep, you

would have these jerks. You feel like you're falling. Have you had that before? You're falling. What's going on? You've had that, Larry? What's going on in the brain?

- 2.18** Why would you feel like you're suddenly falling? Turns out these are called hypnagogic jerks. It's because—think about it this way. As we fall asleep, parts of the brain quiet down, dampen down. This could be, let's say, the outer regions of the brain. It has to do with cognitive and awareness of the world and sensory information, the sensory regions of the brain, quiets down. You also have motor regions quieting down. You're not moving around. So these are quieting down.
- 2.19** But the whole organization of how it works, it's not like—the brain is not this perfect machine where everything just happens in a perfect match, synchronous manner. It's more like it's an organ. It's human. It's messy, it's complex, like a computer that shuts down and then parts shut down. And then occasionally, you might have a page that pops up like that. The brain is like that, too.
- 2.20** So even though you're quieting down the brain, the brain is quieting down as you enter the stages of sleep, you still have parts of the brain perhaps becoming hyperactive at points where it shouldn't. So if the motor cortex of the brain happens to flare up at that point when you're falling asleep, guess what? Hypnagogic jerks might happen, right? So you feel like you're falling. Now, some people have argued that human beings used to live in high places and sleep in trees because of lions and sabre-tooth tigers and all these animals that were dangerous. So it was good to have that reflex as well to be able to quickly move if you needed to, right?
- 2.21** Occasionally, as you fall asleep, you might have a vision, you might see something, or maybe it's not a full-blown hallucination, but you see an image of something on a screen and you see it could be a car. Have you had that before? It's like not a dream. How can I explain? It's not a dream. It's not a hallucination, but it's more like a flashing image. Have you had that before? Hypnagogic visions. Same thing. The visual cortex of the brain, not necessarily all organized and quiet, quieting down in a synchronized and matched fashion. So you might have these as well.
- 2.22** Then we enter REM. Then we enter REM. You know when you've heard about the biblical story of Joseph? He saw in his dream, it's eleven stars, the moon and the sun prostrating to him. Was that it? I think so. Pretty bizarre, like pretty powerful metaphor. It's that—that's what REM dreams are about. It's showing you something like a visual poet. It's like walking about in an art museum and showing you something that you cannot comprehend using language. Indescribable. That's what REM dreams are about.
- 2.23** Let's say the right part of the brain—the brain has two hemispheres—you have the right and the left hemisphere. The right hemisphere is all about poetry. It's about holistic processing things, seeing things in a bigger picture. It's about having a grasp of the world that language cannot express. In fact, the right hemisphere is mute. It cannot speak. It has no way of speaking. If you're to communicate with the right hemisphere

alone, it wouldn't be able to answer you using language because it doesn't have language.

- 2.24** Language is in the left hemisphere. There's a region of the brain called the Wernicke's area, which is a fancy name for a cluster of neurons over here where you have understanding of language. If there's a stroke there, you cannot understand language. There's a part of the brain called the Broca's area—this is in front of it in the left hemisphere as well, where speech and syntactic use of language comes about. Again, this is the left hemisphere.
- 2.25** The right poetry, metaphors, things like that. Even artists, artists who have, let's say, a greater use of the right hemisphere will be better at painting, art, and things like that. It's the more—it's the artistic part of the brain. I'm being overly—I'm overly simplifying it a bit. There is a bit of, obviously, both hemisphere do a bit of both, but there is that distinction. It's almost as if dreams is using that circuitry, is using the right hemisphere. It's showing you something that you cannot use, you cannot—language cannot describe. Am I being clear here? Okay, good.
- 2.26** In dreams, certain parts—when you dream, certain parts of the brain shut down. For example, you have the prefrontal lobes, the outer layer of the brain, this part that has to do with logic, thinking in a serial cohesive manner. That part shuts down. Okay? Does that make sense? It does make sense because dreams are illogical. They lack that logical, cohesive flow. People, places, time, everything is warped. Makes sense when you think about it, given that the logical part of the brain shuts down, right? In fact, that part of the brain allows you to know that you're dreaming as well. So self-agency comes about in that region as well. So given the fact that it shuts down, you don't know you're dreaming when you're dreaming, right? Is that true? Especially the sides here.
- 2.27** So on the side, so the prefrontal lobes, the outer layer of the brain, and the front part, has two major departments. Department one is in the center right here. It's more emotion-based. It's more emotion-based, making emotional decisions. That's in the middle. That region of the brain has a lot of centers going to the emotional core of the brain. If that part of the brain, for example—when you sleep, when you're sleep-deprived, that part of the brain shuts down and cannot control your emotions, for example. A lot of emotional choices and decisions is in the middle part of the prefrontal lobes.
- 2.28** In the outer layers here, on the other hand, you have things like cold cognition, what you might call cold cognition, calculated, cold, the more mathematical almost. Is that clear? That's out here in this outer layer. It's quite important when you think about it because a lot of the choices we make in life are based on this. For example, if I was to make an emotional decision, that would require this part of the brain.
- 2.29** Even though it's not the emotional part of the brain, the prefrontal lobes in the middle part is that bridge between emotion and thought, emotion and higher cognition,

whereas out here, it's more calculated. A psychopath, for example, a pure psychopath that just wants to kill people, guess what? This part of the brain is not working. It's not doing its job.

2.30 But out here, it's pretty calculated. It's pretty, you know what it's doing. It has logic that's out here. In dreams this part of the brain, out here, we call it the DLPFC, dorsolateral prefrontal cortex. A mouthful. You don't have to remember this, okay? Just saying for the people who want to know, it's here. And that part shuts down during dreams. And for that reason, dreams are illogical. You don't know you're dreaming when you're dreaming. Your self-awareness goes away. And yeah, you have that, so that's a part of it.

2.31 Another region of the brain that changes its activity is a region called the hippocampus. So as we hit REM sleep, the hippocampus, the memory part of the brain, becomes more active, so it becomes activated. Does that make sense when you think about it? Do dreams have a narrative component? Story, your life. It blends your life experiences. So that's a part of it, too. You have the hippocampal, the memory part of the brain becomes active.

2.32 Then you have motor regions of the brain. It has to do with motor activity, moving, movement. They become active, too. Here is a paradox of sorts, again, because when you think about it, you're not moving, you're paralyzed. But you are still moving inside your dream. That's the thing—there's actually— we think when we are moving about— obviously, when I move my hands from A to B in space like this, my motor neurons, the regions, the motor neurons in my brain, so we have, these are out here. They are active when I'm moving, correct? They would be sending signals to my body to move. It turns out when I close my eyes and I move my—not even closing my eyes. I just imagine. When I imagine me moving, I'm seeing myself right now moving, kicking, running. Again, it turns out the motor neurons are actually active. Not as heavily, but they're still active. It turns out when I'm dreaming, the motor neurons are also active when I'm imagining myself moving. Is that clear? All right.

2.33 You have motor neurons being active. You have the neurons behind your ears right here. It's called the vestibular sense, your sense of balance. That part of the brain is also hyperactivated. And that's the reason why you can sometimes feel like you're in a spaceship and things are flying through air and all that. That's because of these neurons that has to do with sense of balance becoming hyperactivated. Explains that part.

2.34 But perhaps the most profound part is the fact that our emotional core of the brain, the emotional centers, the limbic centers—I know in this lecture, there's going to be a lot of brain regions. Forgive me. Is it clear, by the way? Does it make sense? All right. So we said the prefrontal, we said the hippocampus memory part of the brain, and now we're saying the limbic. The limbic part of the brain is the emotional part of the brain behind your ears, and that processes emotions. And it turns out that part of the brain is hyperactive. It flares up with activity during REM sleep, during dreams.

- 2.35** That makes sense when you think about it. Dreams are hyper-emotional, hyper-instinctual. You see yourself on the moon, but then there's all these people, and it's all these emotional scenarios going on. What's going on there? Then we have an explanation. The emotional core of the brain is hyperactivated during dreams. Of course, we mentioned before that you have to have that that brain shutdown of your body, the paralysis. You need that. That's important. Otherwise, you would act out your dreams.
- 2.36** But there's another key player in this. And that key player is in the brain stem. We talked about the brain stem before, the lower parts of the brain. Now, that region of the brain will send signals, almost like fireworks, like active signals, to the visual parts of the brain and other parts of the cortex, lighting up your brain during dreams. And it's almost like a compensation for the fact that we talked—the sensory regions of the brain, the thalamus and all that shutting down. We said that before. You need a compensation.
- 2.37** Here you have something called PGO waves. PGO waves. These are waves. These are neural signals from the lower part of the brain, firing up and igniting the brain, making it more active during the dream stage. Does that make sense? I know this is a bit of a heavy few concepts here, but we'll get through it. You remember now, right? You have all the structures.
- 2.38** Prefrontal, shuts down. Things become more loosened up, less logical, less clear. You lose self-awareness. Right? Memory part of the brain comes up. You have memory and narrative woven into the story. Emotional aspect. Emotional part of the brain flares up. What happens? Dreams are emotional. Balanced part of the brain flares up. Dreams are what? You're on the moon on a spaceship. So you have all these components.
- 2.39** More the regions of the brain flares up. What happens? You can move in your dream. But can you always move? Can you always move? Or do you sometimes feel like you're being controlled like these puppets on a string? Have you had that before? What is that about? Why do you feel at times that you can move and other times that you cannot move and you're being controlled? Is there a brain explanation for that too, Larry? You tell me.
- 2.40** Well, there is, perhaps. So even though the motor part of the brain, which is out here that allows you to do voluntary movement, is activated in the dream, in the brain stem, again, that lower part of the brain, the vital part of the brain, has certain activity of neurons that is important for automatic movements, sporadic movements out of your control. Whenever these neurons fire, what happens is that you find yourself in a state when you cannot act out and move in a controlled manner.
- 2.41** There's a constant battle going on in the brain between neurons that has to do with voluntary movement. I'm moving my arm from A to B in space, and the involuntary neurons of the brain stem. A competition that's duelling it out, so to speak. Is that

clear? And that simple brain explanation accounts for the fact that occasionally I feel like I can move; occasionally I feel like I'm being controlled.

- 2.42** You are hyper-associative in dreams. That is really what's going on, first of all. And how do we know? Has that been shown in experiments? It turns out, yes. So if somebody's sleeping in REM, he's sleeping, and you ascertain that he's in REM sleep. You know that by measuring his muscle activity, you can look at the brain waves we said before, the brain waves speed up and become erratic. You have that, and the guy's in REM, sleeping. Now, the guy's in REM sleeping. You wake him up. And as you wake him up, you show him two words: sun and moon, and then sun and chair. And then you measure how fast he is at pairing these words.
- 2.43** Guess what? If you and I were to do this pairing, obviously, it would be much faster to say sun and moon. There's a conceptual semantic link. The poor chap who's in REM—you wake him up—he'll be much faster at sun and chair, sun and stone, sun and so forth. So his brain is geared towards hyper-associative connections, so to speak. That really shows you and clinches the whole idea that we are hyper-associative in dreams. Our neural networks are much more likely to link things in seemingly unrelated ways. And this experiment shows it.
- 2.44** What about the passage of time? Is time slowed down during dreams? Or is it the same as this? Have you seen Inception before? In Inception, the guy, he doesn't want to fall into his dream and be locked there because he would—like a few hours in wakefulness would be thousands of hours in dreams. Is there truth about that? Have you ever slept on the couch and then you're sleeping, and then you look at the clock and you felt like you slept for a few minutes, but it's been like hours. Have you had that before? There's something about it, right? It seems like time has slowed down. Things are slower in dream. What do you think? They're nodding a little bit. Okay.
- 2.45** Well, there's actually experiments on this. They measure the activity of neurons of rats. They look at rat neurons. They're firing away. You measure these. In fact, in rats, by the way, you can measure neurons directly. You put the electrode inside a skull, and then you just measure the neurons. No, ethical barriers, so to speak. Okay. Now, you measure the activity of these rat neurons as they are awake. In the hippocampus, in other regions, [sound] neurons are firing.
- 2.46** You look at the rats, you put them in REM sleep, and lo and behold, in REM sleep, the same neurons are firing as when the rat was awake. Same neurons. When the rat is running around in his maze, when he's in REM, the same neurons are firing [sound]. But here's the catch. Here's the amazing part. You know what? You know what that amazing part is? The neurons in REM sleep were slowed down by up to 50%. There was a slowing down of neurons and their firing rate in REM. Isn't that fascinating? Think about this. Neurons actually slowed down their firing rate during REM sleep. It's profound when you think about this.

- 2.47 I wanted to do an experiment on this very point. A few years ago, I was working with my mentor in California, V. S. Ramachandran. You may know him. Some of you might not. Who said he knew Ramachandran? Anybody? AJ? No. Okay. I worked with a guy, Ramachandran, one of the great neuroscientists. I was very lucky to have him as a mentor. He was a great friend, and he wanted to do an experiment, something like this. We said, "Look, what if you have a guy and he's sleeping? And as he's sleeping, he's dreaming away. And in REM sleep, while in REM sleep, you sprinkle water on him." Okay.
- 2.48 Mischievous, right? You've sprinkled some water on him. And then you wake him up instantaneously and ask this chap, you say, "Look, Joe, look. What happened? You were in your dream. We know that because you were in REM dream. You were dreaming in REM sleep. What happened there? What did you see? So he might say, "Well, I was walking about and enjoying myself. And then suddenly I noticed the clouds coming out, and I thought, well, let me text my mom and say that it's going to be a rainy day. And then took up my umbrella and it slowly started to rain. Scenario A, okay? That's scenario A. Scenario B, you know what that is? The guy said, "Oh, my God, I was hit by a tsunami as I was sleeping!" That's scenario B, right? Which one is it?
- 2.49 In other words, what I'm asking here is: would this guy have enough time to incorporate the information, the water, into his dream narrative or not? I mean, we know it's been done tons of experiments showing that you can actually infiltrate dreams. So if you sleep in a cold room, for example, you'll feel like you're in an ice cold place and it's snowy and all that. We know that. Or if there's smoke outside your window, you feel like you're in a war situation or something like that. We know that. It's true. But what about this chap? Will he have enough time to incorporate that information into his dream? That's the question we're asking. You know what? We still don't know the answer to this very question. Can you believe this?
- 2.50 I took this question to the greatest dream researcher of maybe the last 100 years. His name is Hobson. I said, "Look, Hobson, what about this experiment?" He said, "You have to do it. It's so important." And still to this day, we haven't done the experiment. Can you believe it? It's such an important experiment. There's so many unanswered questions in science like this.
- 2.51 Now, I want to hit you with something more because we said that dreams are—the neurons in the rat's brain are active when they're awake. And then when you measure them when they're sleeping, it's the same neurons that are firing away. Did we say that? We said that these neurons are in the memory parts of the brain, and they're kind of slowed down. This would suggest to us, Heavenly, that perhaps dreams are a mere replay of wakefulness in the sense of the same neurons being active, and it's a mere replay. Is that the case? In your experience, perhaps? Yes, no. Is it a replay or is it something else?

- 2.52** Well, it turns out it's only a tiny percent of dreams that are actual replay of daily activities. Most dreams are something else. It's about the emotional concerns of our daily lives, like what I did and my troubles and my problems, but woven into this magical magistry of craziness. That's what dreams is about. So it's not a mere replay, even though the neurons are acting in this way. It's something else. It's our emotional concerns being put into this crazy story.
- 2.53** Now, I want to touch upon our body image quickly because I'm running out of time a little bit, but I have so much to say about dreams. I love dreams. But this is important. So we all have a sense of an "I", a sense of a self, a sense of a "me". Baland, here, I feel like I'm anchored in this body. I don't feel like I'm anchored in Arnold's body or Brad Pitt's body, unfortunately, okay? That was a joke. You have to laugh, otherwise my joke will become worse, okay?
- 2.54** So I feel anchored in this body. I have a body image. And that part of the brain is created in the parietal lobes. So that is a different part of the brain. Frontal, parietal. Parietal is involved in building a sense of a me, a sense a Baland, a sense of a body image. That's the feeling of, "My body feels here, anchored here, now, in this vessel right here." It's flesh and bone.
- 2.55** That part of the brain shuts down. Regions within that parietal lobe shuts down during REM. And that's why I feel like my sense of self is shifted. I might have a dream where I feel like I'm Baland, walking in this, I'm walking, I'm seeing my body like this, my hands and feet. And other times I feel like I occupy somebody else's body? Have you had that before? Have you occupied somebody else's body in your dream? Maybe you feel like you are inside, I don't know, George Clooney's body, or I don't know, somebody else's body, your mom's body or something. Have you had that before?
- 2.56** In fact, your sense of perspective and the sense of self can shift, so you look at yourself like a Netflix movie occasionally. Have you had that? So there's a shift in sense of personhood and sense of perspective. And that has to do with the fact that that part of the brain, the parietal lobes, shuts down. It dampens down, becomes less activated.
- 2.57** But why do we dream? That's a major question. What are dreams about? Why do we dream? Why do we have dreams? Why do we have these narratives? I mean, when you look at biology overall, it's not wasteful, Heavenly. Biology is not wasteful. It always does things for a purpose. So why have all this time spent in dreams? First of all, you might say, why spend one-third of our lives sleeping in the first place and then spending one-third of that in REM, which is crazy.
- 2.58** Why? You could be out there in the savanna hunting for gazelles and mates and all that. Why spend all this time dreaming? You're vulnerable to your predators, lions, and all that. It makes no sense when you think about it in the grand scheme of things. But it serves a purpose, perhaps. What purpose does it serve? And why do we dream? So first of all, one theory about dreams is that dreams serve as emotional healing. And that's so

important, in fact, I'm going to have an entire lecture on that alone: how dream serves as emotional healing. It's helping us. If we have depression, for example, as AJ was saying, for depression, trauma, and things like that. So it has that emotional healing. So that's one point.

- 2.59** But there's something else. This has to do with threat and threat simulation. So as I walk about in everyday life and I'm facing predators, perhaps not now, as much, but the Stone Age, the person, the Asian guy, right? He was facing all these predators all the time, lions and all that. So for him, having a virtual reality-like testing space free of deadly consequences, would that be helpful? Would it be good to have some kind of a lab where you could test out how to throw a spear or, when you're running from that tiger, how to jump over a stone and all that. Would that be helpful for survival? Or should you go out and test it in real life?
- 2.60** My point is that the circuitry in the brain, circuits in the brain, they work by practice. The more I do certain things, the more I practice a certain skill, the more circuits are being laid down in the brain. The more I have these circuits being solidified and laid down. Now, for me to actually go out in real life and experience these things would be deadly, fatal, perhaps. So having this place where I can play with ideas and perhaps train certain skills and motor skills, would that be important? Quite important.
- 2.61** That's what dreams is about. That's called threat simulation theory. In fact, when they look at people that live in threatening environments, they have more threatening dreams as well. They have more threatened dreams—there's more of these threatening dreams where they can practice certain things. There's something about this theory, about threat simulation.
- 2.62** One classic idea, one traditional idea of dreams is that it's simply this: the fact that dreams activate certain parts of the brain and shut down others. That's it—the brain is trying to make sense of all that—that's it. That's what dreams is about. That's one theory—the fact that certain parts shut down, certain parts are activated. Your dreams, your brain is trying to filter through all this, and he's trying to synthesize this and integrate this, and that's what dreams is about. That's the whole idea. But I don't think that fully can account for dreams. But that's one theory. You have to know it. It's important to know.
- 2.63** Now, I want to touch upon—I'm not a huge big fan of Freud, but I have to mention him. Do you like Freud? Some people like him. There's a lot of debate and controversy around Freud. Some people like him, some people feel like his ideas are outdated, can't be tested. And it's a huge problem, the fact that you can't really test his theories empirically. Some argue that you can test some theories, but mainly it's very hard to test. He's talking about superego, ego, and id and all that. How do you really test something, a memory that's being repressed? It's very hard. This is a problem with Freud's theories.

- 2.64** But basically, Freud argues—and I have to respect Freud for this because in a time where people thought of dreams as merely mental, I want to say, activity that's sort of supernatural. I'm not negating supernatural explanations, by the way. I'm a spiritual person myself. But people that say dreams, you used to say dreams are only magical stuff, superstitious stuff. He actually said, "Well, look, in the brain, there's something going on. We have to look at the brain." So you got to respect him for that.
- 2.65** And specifically, what he said was, he said that dreams are symbols. There's latent and manifest content in dreams, and latent content you have that is disguised symbolism that the brain is using because it cannot confront these ideas in the waking state. They would be too powerful and too evocative and too strong, frankly.
- 2.66** The brain is trying to deal with that in a way of using symbols, again, using things that language cannot express directly, so it uses this metaphorical language. Yeah, it's an interesting idea, right? The fact that that's what dreams is about. It's too much anxiety there, and so the brain has to deal with these symbols using these metaphors. It's an interesting idea, but it's very hard to test directly.
- 2.67** You can think about dreams as a form of exploration, too. Dreams are— it's almost as if in the dream scenario, you are being put in this place where you have all these crazy things going on, and your brain is exploring. It's exploring this new landscape of ideas that it couldn't explore in wakefulness. It would be too overwhelming and harsh to deal with all this, in fact, let alone remember it, but just deal with it. It would be too overwhelming. But as an experimenter, as an explorer, it goes down dreams to see specifically what makes you more agitated and emotionally aroused. There's something about this.
- 2.68** This is something we know that it seems your brain is taking you, the dreamer, as an Indiana Jones explorer and putting you in front of all these scenarios and then watching you react to that. And then whenever you say, "My gosh, this is fascinating." I mean, your brain is saying that. At that point, your brain said, "This is important for the sleeper. This has to do with his emotions. Let me go down that route a little bit more. Let me explore this a little bit more. There's something important here." So the brain does that as an explorer, showing you scenarios, testing different things that wouldn't be possible in wakefulness, and showing it to you. So there's a big component of that.
- 2.69** And then one final bit I want to say—this is an important technical point. You might say when you do these experiments, when you wake people up from sleep, and you ask them all these questions, how is that possible if the guy is awake now? He was sleeping and dreaming, and now he's awake. How come? He should technically think like somebody's awake, right? Does that make sense? What I'm saying is these experiments can be done for a simple explanation is that— actually when you wake up from sleep, there's a few minutes of the neurochemical environment that spills over into wakefulness. So there's a spillover.

- 2.70** And the neurochemical environment is crucial because during REM, you have noradrenaline, that chemical important for arousal and being active and all that, quieting down. It quiets down during REM sleep. Is that clear? It does that. It not only quiets down, it completely shuts down, actually.
- 2.71** So what you have is the following scenario. You have wakefulness. You have noradrenaline during wakefulness. And noradrenaline during wakefulness is what? Aroused at different levels, right? Depending on what you're doing, correct? Context dependent, right? As you enter, as you fall asleep, less noradrenaline. The neurons in your brain, in the brainstem, producing noradrenaline slowly turns off, slowly turns off as you enter sleep. And then by the time you hit REM sleep, they are completely shut off. You completely shut off this crucial chemical called noradrenaline.
- 2.72** And then when you don't have noradrenaline in the brain, things are less focused. I mean, think about it, when you are high on adrenaline and noradrenaline, you are very narrow-focused and you are very clear in your thinking. And when that chemical is gone, when you don't have that, things are more spacey, things are more—the realm is opened up, and that's what dreaming is about, having this neurochemical being less—actually not even being available during REM sleep. And so that is also part of the explanation, the fact that REM dreams are so spacey. Everything is all over the place, and things are crazy like that. Does that make sense?
- 2.73** Okay, I hope this was clear. We had a lot of names for different things in this lecture, but as you will see, I will mention them again as we go along. This was just a basis, and it will become much more clear as we move along. Thank you.
- 2.74** Hi. I was wondering how sleep aids affect dreams.
- 2.75** Sleep aids?
- 2.76** Yeah, like Benadryl, melatonin.
- 2.77** That's a good question. Melatonin, I'm not sure. I know some of these sleep SSRIs and things like that, used for depression. These can actually block REM sleep altogether. So you have no REM. What can happen is REM can come back with a vengeance. You have what's called epic dreams, epic dreams. These are very powerfully intense, vivid dreams that are unlike anything. It's like the Christopher Nolan type movie. It's extremely intense. You can have that if REM is suppressed through SSRIs. But things like melatonin, I don't think that melatonin would—things of the other types of sleep aids. I can't see them affecting REM too much. Yeah, interesting question.
- 2.78** My mom told me this, so it must be true, but she said that you can't remember a dream if you're not woken up during it. But I'm just curious what—you know the answer to that. My mom didn't.
- 2.79** You can't remember a dream if you're woken up from it?
- 2.80** Can you remember a dream if you sleep through the whole thing?

- 2.81** That's a great question.
- 2.82** She told me that in her experience, and it's been my experience, I have to be woken up in order to remember the dream.
- 2.83** There's something—well, again, it's a great question about forgetting dreams. There's something there that I want to cover in the next lecture. Add five minutes to that, because why do we forget dreams? It's such an important—I feel like it's running too fast, this clock, because I have a lot of things to say. Why do we forget dreams? Why did we forget this? It's such an important question. Let's cover that in the next lecture before we—it's very insightful questions you have there.
- 2.84** I wanted to know the difference between a nap and more mindful quieting of the mind and how that affects energy levels.
- 2.85** Basically, you're saying that meditating can be as powerful or even more powerful than naps? Is that your experience? I would say I'm not sure if that's true. If you're sleep-deprived, I think the nap can be more powerful, I would say. But just sleeping or just resting and meditating, that'd be very powerful because you enter some of the same brain— you have the same brain waves, some of the slower, called alpha waves and things like that, when you meditate. And so by resting, you can rest your brain, and it's important. But I wouldn't say that necessarily it's more powerful than a nap.
- 2.86** But of course, what can happen if you take a nap is that naps can be very tiring, too. If you go through all the stages of sleep in a full one hour, hour and a half nap, and you wake up and you're groggy and all that. I can definitely see just resting your cortex a bit. So that's what a meditation would do, for example, just rest your cortex, the outer layer, the more superficial, let's say, parts. Then I could see that meditation could be very powerful.
- 2.87** Yeah, I'll tend to fight the nap in order to just instead be mindful because I do 20 minutes or so. I'll get pretty tired when I wake up. But if I do the mindfulness, it's like—
- 2.88** No, it's powerful. I do it, too. It's very powerful.
- 2.89** You mentioned alpha waves, and on that state, the different—I know there's several, and I'm curious to find out: do different dream states—are they characteristic of specific brain waves taking place in your brain at various times? Do you have more alpha, beta, theta waves in a REM state versus a deep state? If we know anything about that and why.
- 2.90** Yeah, we do know a lot about that. In the deep stage of sleep, we have the slow wave sleep. We have that. Theta waves, to be technical. In dream sleep, we have beta waves. These are the same waves as wakefulness. I mentioned before that wakefulness and dreams show a similar pattern because these are the highly sped up, fast waves called beta waves. You have that. Definitely, in the early parts of sleep, as you're dozing down

and falling asleep, these are the alpha waves you have slowly. Delta waves you have as well. You have in deep sleep a lot of delta.

2.91 It's interesting in REM, and we'll get into this in the final lecture, you have a lot of theta emerging from the emotional and the memory parts of the brain. We'll get into that, but that has a calming effect on the brain. Everything becomes calmer and things become more peaceful and you can weave your memories into a larger picture. So you have theta in that.

2.92 You said in REM sleep, that's when you have the most theta wave functionality through those different sleep states.

2.93 I wouldn't say you have—I can't say most, but I would say you definitely—there's beta waves. That's what it's most known for. Then you have some theta activity that's important for the emotional in REM. Thank you.

Lecture 3: Sleep Gone Wrong

At a glance — Sleep gone wrong: sleepwalking, narcolepsy, and other disorders of sleep.

- 3.1 All right. In this lecture, we'll cover things like sleepwalking, what sleepwalking is all about, something called narcolepsy, really fascinating disorder, and something called REM behavior disorder. Quite fascinating.
- 3.2 But before I go there, I had an interesting question from Larry. See, Larry asked me, he said, "Baland, my mother told me that whenever I'm sleeping and somebody awakens me from sleep, I can remember the dream. But if I sleep through the night, my dreams are forgotten." So that was the question.
- 3.3 How come we forget our dreams? Why are dreams forgotten? What's happening in the brain? And are they indeed forgotten? And it turns out that 95% of dreams are in fact forgotten. So think about this. We have REM sleep for one-third of sleep is REM. In REM, you dream. So you have all these hours of REM sleep, of you dreaming away about all these crazy things, yet you forget a lot of them.
- 3.4 Then AJ might ask, "How come you forget your dreams? What's going on? Why do you forget your dreams? What's going on there?" And the answer is, we don't know for sure, but it seems like having all these Hollywood Inception-like crazy dreams flood your wakefulness would be too overwhelming. So the brain seems to have a mechanism to inhibit that information to accessing your conscious awareness, in order for you not to go crazy, pretty much. That's one idea.
- 3.5 Now, obviously, the last fragments of dreams is what you might remember if I was to wake you up. So you have that. Or in fact, if you wake up yourself, what happens is that you remember a tiny bit of the last portion of the last cycle and the last REM phase.
- 3.6 Now, what's happening in the brain, well, it turns out that the chemicals, the noradrenaline we talked about, that chemical being important for arousal and all that, that chemical, we said, is shut down during REM. So it's not available in dreams, and we said that is why dreams are so crazy and bizarre and spacey and all that. So that's because of noradrenaline.
- 3.7 And for the technical types, noradrenaline is produced in the locus coeruleus, in the brain stem, the lower part of the brain. It's around, I think, 32,000 pitch-black cells there, creating noradrenaline. And that region shuts down, and so you don't have noradrenaline. So you don't have that.
- 3.8 Serotonin, another chemical in the brain, is also not available in dreams. And so when you don't have noradrenaline and you don't have serotonin, you cannot store your memory. There's no memory storage going on, pretty much. So the hippocampus, the memory part of the brain, has trouble storing memory. In other words, the brain is not output— it's not focusing on storing information, but generating new information,

which is what I'm trying to say here. It's generating new information data, you moving around in space, but not really storing it.

- 3.9** So that's the answer. So that's the answer, right? So we forget our dreams, and 95% of them are forgotten, and simply, they're too overwhelming. Some have actually said that—there's a condition called, so we say there's narcolepsy.
- 3.10** So narcolepsy is a very fascinating condition. It's where you are, let's say, you walk about in everyday life, you go to a park, and then suddenly you fall asleep, in the middle of wakefulness, while walking in the park. It's crazy. That happens during this narcolepsy. Or you might be in the middle of eating, and then you fall asleep. Or you might be in the middle of lecturing, and you fall asleep. Like me, right? Half asleep, sometimes. I get a bit tired. But that's narcolepsy, right? So that's narcolepsy.
- 3.11** And in fact, people with narcolepsy, they, it turns out, cannot differentiate between dreams and wakefulness. So the memories that they lay down in wakefulness, and the memories they have from their daily activities, they cannot differentiate it. Did I send that email to my boss? Did I text my friend? They don't know, they can't remember, so they mix it all up. They don't have this barrier that allows them to keep these worlds apart. So they mix everything. And that's a feature of narcolepsy.
- 3.12** And some people have even argued for this reason that the circuitry of dreams are very different from the circuitry of wakefulness. Obviously, we talked about some of these brain centers, but the very circuit, the neurons and their systems, are fundamentally different. And that's why we said they're cosmic, they are overwhelming, they are poetic, right? And that's the dream circuitry that these people have trouble keeping away and having them. It blends into their daily activities. Whereas you and I seem to have an inhibition going on there, so we can keep them apart.
- 3.13** Interestingly, déjà vu. You know déjà vu? So you walk about in somewhere, let's say, in France, and you see this epic castle, you go, "Oh, my God! I've been here before. Wow, it's so fascinating. There's something going on beyond this realm. It seems like in another life or something, or I don't know, but I am experiencing déjà vu."
- 3.14** Now, some scientists will argue that déjà vu is in fact the dream circuitry being activated spontaneously while you are awake. Does that make sense? It's the same thinking, that same logic where things are hyper-cosmic and spiritual and overwhelming. It seems like perhaps déjà vu is that. It's that dream circuitry, those wires, becoming hyperactive in the wakefulness type of, wakefulness situation, pretty much.
- 3.15** Now, during narcolepsy, you have interesting symptoms. One is that excessive daytime sleepiness, which is, as we said, you walk about and suddenly you fall and you can't keep awake. But I want to add, it's not merely falling asleep. It's like a wave that hits you. It's like those, when you've been awake for four nights in a row, kind of thing, and then suddenly sleep hits you. It's more like that than simply being tired.

- 3.16** Imagine actually going about your daily life and you think you maybe had a good night of sleep. And for the sleep, their sleep is pretty fragmented at night. But let's say you, normal routine, and then suddenly you walk about and then boom, you fall asleep. It's quite overwhelming.
- 3.17** And they have another symptom, too, which is cataplexy, which is this paralysis and the crumbling of their legs and their arms and everything. So they fall down and they're paralyzed. It's REM paralysis during wakefulness. So that REM paralysis we have when we are all sleeping and we are paralyzed heavily in our dreams and all that in REM, imagine having that during wakefulness.
- 3.18** And you know what's more bizarre than that? It's triggered by the tiniest emotional surge. So if my kids are coming to the room and playing with me, I might fall down paralyzed and crumple to the ground. Or if I'm making love or something, I may suddenly get paralyzed. Sorry to be immodest here, but I have to say it's the truth, right? This is what happens. You get paralyzed. Imagine that symptom.
- 3.19** In fact, there was a chap who knew Robin Williams, the comedian. And each time he saw him, he would preemptively lie down on the floor because he knew that, whenever he came in, his jokes would just rile him up and he would become paralyzed. So it's quite a cute story. But point is, it's a very debilitating thing. I mean, you can't engage in activities and stuff. It's very hard.
- 3.20** Another key symptom of narcolepsy is sleep paralysis. So they have a lot of sleep paralysis, and we'll go into that in-depth. But essentially, because of their sleep being very messy, and we'll go into what's going on in their brain, but the fragmentation and that REM sleep being all over the place makes it so that they have a lot of sleep paralysis. Every night, they might have several episodes of sleep paralysis.
- 3.21** And sleep paralysis, again, is paralysis upon waking from sleep. You wake up and you're completely paralyzed from head to toe. You can't move, you can't speak. And then occasionally, you might have a ghost or a monster approaching you. Pretty scary stuff. But that's what sleep paralysis is about. And they have this on a continuous basis.
- 3.22** What's going on in the brain of somebody with narcolepsy? In the brainstem, we said you have neurons there that send signals down your spinal cord to paralyze you. Did we say that or no? We said it, right? And these are important because, in REM, you want to have that paralysis. You want to be an in-body prisoner, so you can't move and speak and act out your dreams. We said that, right?
- 3.23** Now, in narcolepsy, those neurons are not compromised, per se, in the brainstem. But just above that, there's a region called the hypothalamus. And in the hypothalamus, you have neurons there that are responsible for sensory awareness. And if these neurons die off, then you have basically the REM—you can think of it a little bit like a switch in the hypothalamus. There's a switch there. It's like you switch sleep on and off.

- 3.24** The hypothalamus is critical for sleep. It's like a marble-like structure inside you. It's deep inside your brain. But it's important for turning sleep on and off. And it has a lot of signals going down to the brainstem, and the paralysis region of the brain. There's a lot of wires going down there. Because of that place, the hypothalamus is not working as it should.
- 3.25** The neurons there, specifically, orexin-producing neurons, to be technical, so we have that on record, orexin-producing neurons, die off. These are called the vampire neurons because these neurons, or the vampire chemical, orexin, because it keeps you awake, it keeps you alert, it keeps you agitated and fresh and all that.
- 3.26** Because these neurons die off and you don't have orexin, you don't have this chemical, this vampire chemical, you have sleep hitting you, hitting you, hitting you, at maybe 200 times per day, up to 200 times, imagine, because, simply because this chemical switch, this orexin, these orexin neurons, that vampire chemical, is off. The neurons are dying.
- 3.27** Also, because of the neurons dying in the hypothalamus, they cannot communicate with the brainstem and the paralysis regions of the brain we just talked about. And so for that reason, the paralysis will hit you at various point. Is this clear? Let me say that again. Orexin die off, so you can—sleep is going like this, like the switch is broken. It gets stuck between on and off all the time. Awake, sleep, awake, sleep, awake, sleep. Why? Because the vampire chemical is dying. The neurons that produce it are dying.
- 3.28** And the communication, the freeway down to the brainstem that allows for the paralysis, is also—it doesn't work. The signal is not working. And so you have paralysis that's hitting you at various times. This is what narcolepsy is about, this debilitating disorder.
- 3.29** But then you have a very similar condition in some respects, in that it entails some of the same regions of the brain. And it's basically called RBD, REM behavior disorder. It's quite a fascinating one, actually, especially for me, because the research on sleep paralysis is one of the things that I do.
- 3.30** And so in sleep paralysis, obviously what happens is that you have REM paralysis, that paralysis of dream—the dreaming realm, spilling over into wakefulness. So you are sleeping, but then suddenly you are awake, but that REM paralysis that's supposed to stay in the sleeping state spills over into your wakefulness state, and so you find yourself paralyzed even though you are awake. So you have paralysis on the one hand, and then you have wakefulness colliding. Collision between wakefulness and paralysis, wakefulness and REM paralysis colliding.
- 3.31** Now, imagine if you had no paralysis at all. What would happen then? Not too much paralysis, but too little paralysis. What would happen? You would have REM behavior disorder. This disorder, when you are dreaming and you're lying in your bed and you're acting out your dreams. You see yourself on that moon wrestling with that alligator, and you see it all, and you act all the stuff out.

- 3.32** That's because you don't have—the neurons that are supposed to paralyze you, that are responsible for this in the brainstem, are dying off. They are dying off, and so you don't have paralysis. That signal going down, dripping down to the brainstem, is not dripping down there and telling your body, "Look, body, you have to be paralyzed now." The signal is not there because the neurons are dying in the brainstem. And so for that reason, Larry, you have what? No paralysis. You're acting out your dreams.
- 3.33** And if you go to YouTube or something and watch some videos—I'm not making a commercial, by the way, OK? But you go online and watch a video and look, see, right? RBD. And you'll see them shouting, and it's often very violent, extremely violent and aggressive.
- 3.34** And so some scientists will say, why is it so bloody aggressive? Why all the violence and all the—people will say that they wake up from sleep and grab rifles and do all kinds of crazy stuff. So it has that, Sonia, it's a violent affair for some reason, and they want to know why, what's going on. We don't know for sure.
- 3.35** Some will say that the reason that this condition, where you're not paralyzed during dreams, the reason it's so—the dreams are so violent and all the patients experience all these violent dreams is that perhaps, let's say you're moving all the time in your dream. So you're acting out your dream and moving.
- 3.36** That movement itself is interpreted by the brain as something aggressive, as something that's physical in nature. We mentioned that you incorporate things into your dreams. Perhaps what your brain is doing here is incorporating that movement into a violent narrative because it makes most sense, given the context of your body. Is that clear? That's one interpretation.
- 3.37** Or others would say, perhaps the reason is that only the most violent of dreams make it through the filter. The rest are forgotten. So that's another explanation for why the violent and the dark nature of these dreams that these people have, these poor people.
- 3.38** By the way, people with RBD, I think it's about 80% of these guys with RBD, will go, will have Parkinson's disease later in life. So it's an early predictor. Can you guess why? Think the brainstem. The brainstem is involved in things like producing dopamine, especially involved in movement. So if the neurons are dying off there, producing the sleep symptom, might that be a sign that you will have more neurons die and eventually affect all the motor regions of the brain and kill off the dopaminergic cells in what's called the basal ganglia, meaning another nearby structure. Does that make sense? It's often a predictor of that.
- 3.39** Yeah, Parkinson's is a terrible condition. My mentor experiences Parkinson's. It can take sort of a, you can have a powerful punch there. I mean, become rigid, you can't move, you feel a prisoner in your body. Quite something. Your mask expression you have. The whole circuit that has to do with automatic and naturalistic movement is dying off.

- 3.40** Basically, the brain has one center for natural movement, when I move my arm from A to B's base. That's one circuit. That's one part of the brain, the motor regions. Then you have another part of the brain that has to do with more natural movements. When I walk down the stairs, I don't go right first, left first. I just move. It feels natural. That is because this is encoded in my brain, and this is other regions that is responsible for that. And those neurons die off in people with Parkinson's.
- 3.41** Now, what is sleepwalking all about? I often get this question. So if people are paralyzed, AJ, during REM sleep, then how come people sleepwalk? Is something going, why do people sleepwalk? Aren't they just acting out their dreams? Isn't that the case? What do you think? Are sleepwalkers acting out their dreams? Yes, no? It turns out the answer is, unfortunately, no.
- 3.42** They're not acting out their dreams, although there are a few cases in the literature where there have been people, reportedly, having acted out their dreams and done horrible stuff. So there was a case that was incredible of this person who tied up somebody in their family during sleepwalking, and it was all a dream and all that. They did it during a dream. But really, it's not.
- 3.43** Sleepwalking, it happens during deep sleep. So during deep sleep, are you paralyzed or not paralyzed? Not paralyzed, right? So, deep sleep happens when you wake up from deep sleep. That's when you sleepwalk.
- 3.44** So let me give you a case of this sleepwalking. It's an interesting case. There was this guy, Kenneth Parks. He was sleeping. But let me go—before he slept—actually, let me give you some context. He had a lot of problems, a lot of financial problems, a lot of gambling, debt, a lot of issues with his family, and he didn't know, he had to deal with these problems. And so he was planning to go to speak to his in-laws about how to deal with some of his problems, and his debt and all that. And so that was the plan, to go talk to them.
- 3.45** But then going back to the fact that he was sleeping on the couch. Big guy, by the way, huge muscles, TV buzzing in the background. Suddenly, in the middle of the night, in the thick of the night, he woke up and started to walk while sleeping. Opened the door, went to his car, still sleeping, drove his car, like, 14 miles, while sleeping. Going, OK? At this point, he broke into the house of his in-laws, went straight to the kitchen, and grabbed a kitchen knife, went up to the bedroom, and killed his mother-in-law.
- 3.46** He's still sleeping. No joke. He's still sleeping. Then he went for the father-in-law, but the chap was able to save himself. Kenneth went down, drove his car, still sleeping, and drove around. And then at some point, he woke up and said, "My god, what's happening? I have blood. I have blood on my tendons and wrist and all that. What's going on? What's happening?" And then he went straight to the police station and he said, "Look, I think I've hurt somebody. What's happening? I have blood all over."

- 3.47** Of course, they didn't believe him in the beginning. Who would, right? They found out that, look, he actually had. He had killed this person, his mother-in-law, and he went through the whole court system. And he claimed that he was sleeping, but nobody would believe him, initially, until they looked at his brain.
- 3.48** They measured his brain, and they saw that, each night—again, the electrodes on the brain with the gels that measure the activity, they saw that, each night, he goes in and out of deep sleep up to 10 times, completely cleansing the fact that this guy was sleeping. He did it all in an agitated frenzy while asleep, and this saved him. But imagine this. This is possible. This is possible. So we have a case like that that's quite profound.
- 3.49** Sleepwalking tends to affect children more. I think it's like 2% of the general population might have had it, but up to 14% of kids. What's going on there? Any guesses? I want you to think about this. Why do kids sleepwalk more? Larry, what do you think?
- 3.50** You mentioned earlier that young children, I can't remember if they were inhibited or they were undeveloped, but I think that those kids would—the sleep paralysis would not be inhibiting them walking because they were younger. Is that why?
- 3.51** You remember the lecture. Very good. Exactly. It turns out, children, children, kids, the regions of the brain that's supposed to act like a brake, that prefrontal, specifically, that part of the brain is not fully developed in these kids. And so they have things like sleepwalking more. They have things like frequent urination because of circuitry not being fully developed. Sorry, I have to say it. They have all these symptoms because of a less mature prefrontal cortex. So it's more rampant. You have more in kids.
- 3.52** But there's another—it can come in flavors. So you can have sleepwalking in the natural sense, where you walk about or you drive your car or you go to the balcony or whatever. Pretty scary stuff. But there's a version of sleepwalking where you go out to them. This is a very specific version, and these guys tend to have this very specific version. They will take a knife. First of all, before the knife, they will go to the kitchen and take a knife and chop vegetables, raw vegetables often, and eat them. And raw foods, they will eat raw foods. They will eat whatever raw foods, often carbohydrates, they tend to eat. But it's called sleep-related eating disorder. So it's a real phenomenon, and it affects a group of these people that have this. It's quite fascinating, isn't it?
- 3.53** Now, you might ask, and this is a reasonable question to ask, are these guys awake? Are they partly awake? Are their memory parts of the brain active? Are they conscious? What's going on? And the answer is, they're like half-robots in a way. They're acting very robotically, like on autopilot. They're walking about, but their brain is in a set algorithm where they don't have—they're not consciously aware. So for example, sleep paralysis, you are partly aware. You know your surroundings a little bit. You feel like you are awake. But during sleepwalking, it's pretty much a robot there. So the whole

region that has to do with conscious awareness is shut down. It's not working as it should. It's still down.

- 3.54 But it turns out, the memory parts of the brain, I guess, are what? Also shut down. So it's not based on memory either. But the fact that the memory part of the brain is shut down explains why, if you wake this guy up and say, "Wake up," he will have forgotten his entire sleepwalking expedition. No memories, no recollection at all.
- 3.55 Turns out, the balance part of the brain, in the brainstem, it's called a cerebellum. It has to do with sense of balance. That region of the brain is working. That's why they can walk without falling. And the motor region, obviously, is also active. That's why they can walk in the first place, but the cerebellum is just making sure that they don't fall. So you have these specific regions being active, and that's why you have this phenomenon of sleepwalking. So it comes in these various forms.
- 3.56 And by the way, for a lot of these disorders, we don't really have treatments. That's a huge problem. How do you treat sleepwalking? How do you treat it? The guy is sleepwalking, he's sleeping. It's very hard to treat. The one way you can do it is often targeting sleep itself, making sure he gets restful, deep sleep overall. His architecture of his sleep is overall of good quality.
- 3.57 How do you do that? Well, stress reduction. Things like going for walks in nature and avoiding too much alcohol. Alcohol can fragment your sleep. That's known to be something that can make your sleep more choppy. Things like that are really often the resort and the treatment options you have.
- 3.58 Sleep talking happens, also, in this deep stage of sleep. Often can be, oh, we have a sleep talker. OK. All right. All right. So your sleep talk can be a few words or full conversations, perhaps. It can be full conversations where you wake up and you just talk gibberish.
- 3.59 And the point of sleepwalking is, and I guess a huge part of this lecture, is that these conditions like sleepwalking, sleep talking, now they call teeth grinding, grind your teeth, or sleep-related eating disorder, are all conditions that have to do with you waking up from deep sleep. And so you have this signature of deep sleep, which is that you are deeply sleeping.
- 3.60 But if, and this is the crucial part, the emotional part of the brain becomes hyperactive while you are sleeping, this is called nocturnal arousal, meaning the emotional part of the brain has to be at a certain level of threshold. It has to be shut down. But if it's titillated a bit, if it's hyperactivated a bit, what can happen is, it can tickle the outer layer of the cortex just enough for you to wake up and start moving around and doing physical activities.
- 3.61 That's what these conditions are about, these parasomnias. They're about your emotional part of your brain being able to interfere with the sleeping state by titillating

the sensory part of the arousal part of the brain, just enough for you to wake up and walk around, but not enough for you to actually be fully awake.

3.62 Now, you can imagine then that, if you have an anxiety disorder, or you have PTSD, or if you have trauma or anything that would cause your emotional brain, the limbic emotional brain, to be hyperactive more naturally, then you can imagine that you are predisposed to having more of this condition. Does that make sense?

3.63 What I'm saying here, to repeat myself, is that, if your emotional core is already hypersensitized, it's hypersensitive because of a trauma which can cause the emotional part of the brain to balloon up and become greater in volume. The cells can literally, the amygdala, the emotional core of the brain, the fear department, can literally become oversize, balloon up in trauma, in PTSD patients. So if it's already hypersensitized, and you are sleeping, and you tend to have a very fragmented and shallow sleep, then you are more likely to wake up partially during deep sleep and have these walking expeditions. Clear so far? It's very interesting.

3.64 These parasomnias are called arousal disorders. They occur on the edges of sleep, on the edges of sleep. Just like sleep paralysis would occur in REM, this other interesting condition occurs during deep sleep.

3.65 Let's finish off with a final condition, which I think is quite interesting. It's called exploding head syndrome. Quite the name. It sounds like the name of a rock band or, I don't know, horror movie or something. But these are seconds of explosions that occur as you fall asleep. Imagine you're falling asleep and suddenly you're feeling like a plane was to crash in your backyard. You're like, "boom", and there's this explosion. It's exploding head syndrome. What's going on? Quite a mystery.

3.66 Now, this is again a parasomnia arousal disorder. It has to do with the fact that, as we mentioned before, when the brain powers down to go to sleep, it works more like an old computer, not even the new one, not even the newer computer, the old ones where it slowly, stage by stage, goes down. It doesn't have that—it's not a synchronized, perfect, matched thing. So same thing here.

3.67 If you imagine if you go to sleep, and you are in, you are powering down, hitting the different stages of sleep, becoming more and more tired, but then suddenly you have an explosion. I mean, you have a hyperactivation of the sensory region of the brain, specifically the auditory cortex. What would happen then? You might hear a large sound. That's simply what it is.

3.68 The fact that the brain shuts down, you have sensory neurons, you have neurons for auditory, vision, hearing, touch. And so, if the auditory neurons are hyperstimulated when you fall asleep, then you might have something like exploding head syndrome. Not surprisingly, it's very common in people with sleep paralysis and other conditions where your sleep is already fragmented and your brain already doesn't know what to do with the whole transition to sleep.

- 3.69** I hope this isn't too silly of a question. I'm curious. I know my dog sleep talks, and I know I sleep talk. Do we know if these functions are primarily limited to mammals? Or what's the extent of the study on dreams and sleep states for—?
- 3.70** In animals? It's an interesting question. So your dog sleep talks?
- 3.71** Oh, yeah. He'll howl, he'll move, he'll try to run.
- 3.72** Oh, so he walks around, is you're saying?
- 3.73** He doesn't get up and move, but in my mind, I'm imagining he's having a full conversation with somebody. It's not just one howl, it's several, and long.
- 3.74** Interesting.
- 3.75** So are mine.
- 3.76** Yeah. Now, I think we know that they have the REM paralysis as well, that you can mess with their REM paralysis and they start acting out their dreams. We know this as well. I can imagine that, yes, since you can mess with their REM paralysis, so they would act out their dreams or not, in this case, they would be able to sleep talking or—we can't call it talking, but yeah, I think that would be my answer. They're very similar to us.
- 3.77** Whether animals dream, we don't know for sure. It seems like they could have fragments of visions and things that might correspond to a dream. But how complex would it be? Their brains are different than ours, so I think it would be different. But definitely, they would have something, since they have REM sleep, and they can have the paralysis and all that. But it's very hard to ask them directly; this is a problem.
- 3.78** For some of those things that you mentioned in the first lecture, like caffeine, and I would imagine alcohol as well, do those additives affect the— do they affect your dreams in the same way that they affect your ability to fall asleep, or do they alter the states?
- 3.79** That's a very good question related to a question we had about sleep aid. I don't think caffeine would affect your dreams. But what's the other one you said? Alcohol? Well, the thing is alcohol fragments. It's also a REM suppressant. So alcohol will suppress REM sleep. And so that can affect your sleep because if REM is interrupted, often you can have rebound of REM later, and then it will affect your dream. And REM rebound dreams can be very intense. I don't think caffeine, though, would affect it. I think caffeine, probably, it can make you hyperaroused, it's hard to sleep, but not directly. I don't see any way that would be possible. Interesting questions, though.
- 3.80** So, in the lecture, you talked about different states that appear to cause these parasomnia effects and how memory is related to a lot of these sleep disorders. I know this is a burgeoning field, but can you talk about some of the big differences between something like a sleep dream and a psychedelic dream?
- 3.81** Psychedelic dream?

- 3.82** Yeah, in regards to psychedelics. Again, I know it's new.
- 3.83** You want to know the difference between a normal dream and a psychedelic dream?
- 3.84** They do the same terminology, and I know that there's probably not that much overlap, but with regard to the relationship between—
- 3.85** Yeah, good question AJ. This is getting into some rabbit hole stuff here, OK? Because psychedelic dreams has to do with a whole—so what happens is, when you take psychedelics, you activate certain chemicals in the brain, OK? So one chemical that you activate is called the serotonin 2A receptor. It's a receptor by which serotonin talks to the cell. Serotonin 2A.
- 3.86** Now, if your brain is flooded with serotonin 2A receptors in a powerful way, you can have very odd and mystical experiences. Why? Well, it seems that, in the brain, the areas of the emotional core, the amygdala and all that, have a lot of receptors for serotonin 2A. I call it the ghost receptor in the brain in my articles, serotonin 2A receptor.
- 3.87** Now, it seems like, during dreams, you have a hyperactivation of these serotonin 2A receptors, and you also have them during psychedelic experiences. There's quite an overlap, and I believe that the states are very similar. The brain can only have so many states. It doesn't have a thousand consciousness states. And one of these states is this mystical state where you feel like you're on a trip, you're tripping. It's like—
- 3.88** Now, the difference is, sometimes, you can have the trip vary. It can sometimes be extremely scary and you get paranoia and all that. And the reason for that is context-dependent. If you're already scared and terrified and fearful, then the whole trip can flip on its head and you get a more terrifying trip. So, similar circuitry, I believe, is involved.
- 3.89** Thank you.
- 3.90** Deep question. OK. All right. OK. Time to sleep now. Go home and sleep.

Lecture 4: Nightmares

At a glance — Nightmares — intensely emotional, threatening dream experiences and what drives them.

- 4.1** In this lecture, we're going to be talking about nightmares. These are the terrifying experiences of—you have some dangerous, intensely emotional experience in your dream, and then you wake up and you're terrified. So this is the topic of this lecture. But let me address some of the questions first.
- 4.2** Reoccurring dreams. Can you have a dream on one particular day? So let's say I dream about, I don't know, myself doing something interesting with my friends and family. And then a week later, I have the same dream. And then a week later, I have the same dream again and again. Is that possible, a reoccurring dream like this?
- 4.3** And the answer is yes. I've had that myself many times. It's very common. In fact, I've had nightmares, terrifying nightmares, reoccur again and again and again on the same night. So I remember as a kid, I was sleeping. And so I lived in Copenhagen. I grew up in Copenhagen. I'm Kurdish originally. Some of you guys know I'm Kurdish from Iraq, originally. I grew up in Copenhagen. In fact, I grew up in the ghetto in Copenhagen, just outside the inner city. It was tough neighborhood. Perhaps that's the reason I had a lot of nightmares.
- 4.4** But I had this reoccurring nightmare, and there was this witch-like lady in my dream that would just come about and scare me. I was four or five, and I was a skinny kid, and I had this witch come into my dreams and just terrify me. I remember her, she was wearing this wig and this scary face and all that. And then I'd have this dream occur again and again and again, sometimes on the same night, and I was so terrified. Until at some point, it just went away.
- 4.5** So it partially answers your question, while contextually embedding it in today's talk about nightmares. So it's a reoccurring nightmare. So that can certainly occur. In fact, in PTSD, people with trauma, people that have experienced extensive traumas, let's say they were driving in a car and then a car hit them, they had a car accident—these guys will have this replay in their minds over and over and over again. They have these flashbacks, and then when they sleep, the trauma resurfaces. They have this. This is very common. This would be a reoccurring dream, wouldn't it? What do you think, Larry? We have that.
- 4.6** One thing is that, and I want to touch on this, too, in the whole dream scenario thing. So dreams are heavily populated by people. That's something I didn't cover yesterday, but it's important. Also important for the whole threatening nightmare situation, Sonia. So dreams have a lot of people in them. A lot of people. Often people we know by name, like Naomi, Amanda, and so forth. We know them by name. They feel close to us, but There are lots of them in our dreams.

- 4.7 And the reason for this is that we think in the brain, there are circuits called theory of mind circuits. Some people refer to these circuits or these particular neurons as mirror neurons because they're involved in mirroring other people's perspective. I, Baland, I want to know what Arnold is thinking. I'm doing a theory of mind around Arnold. I'll say Arnold right now, he's writing notes, he's being very studious, and he's probably thinking, "What is this speaker saying?" And all of that, right?
- 4.8 This is called a theory of mind. I create that. It's actually quite magical the fact that human beings can create a theory of mind, can put themselves in other people's neurons or shoes and see the world from their perspective. Think about this. This is profound, this social cognition that we all have. To see what is—to gauge, what are you thinking right now? What's your agenda right now? All that.
- 4.9 That is a profound thing, and it's quite important for survival. When the caveman, the Stone Ages on the savanna, when he was running around there, it was important to know what is this animal doing right now? What is the agenda? So that's a theory of mind or another human over there with a spear. Is this person trying to kill me and take away my resources and all that?
- 4.10 So having a theory of mind is very important. So we believe there are mirror neurons in the brain. Mirror neurons, these are neurons where if I [slapping] feel pain in my hand, for example, pain neurons in my brain fire away. But it turns out by watching you, can you poke your hand? Okay, a bit harder. I don't want to get sued not that hard. Okay. All right. Okay.
- 4.11 It turns out neurons in my brain will become activated by watching him afflicting pain to his hand. Can you believe this? Neurons in my brain will become activated as if I was afflicting pain to my hand by merely watching AJ afflicting pain to his hand over there. Mirror neurons, mirroring the other person's perspective. We all have these mirror neurons or mirror systems, circuitry in the brain that allows us to gauge the other person's perspective, create a theory of mind.
- 4.12 But it turns out as human beings, we need to practice these circuits. The brain works on the principle that whenever we use certain skills, they become better. The circuits, the neurons, neuron A to B, they become more solidified. They crystallize whenever we use a certain circuit more and more and more. In order to crystallize the circuit, we need to use a particular skill.
- 4.13 And so social skills, for example, how do I use social skills if I'm out all day hunting and I'm not maybe interacting with people that much and I'm not actually seeing the animals that I want to see? Well, would it make sense to dream that you are interacting with people and stuff like that? Would that be quite important? Could it be a way to exploit? I don't know.
- 4.14 We said a third of our lives is spent sleeping, and a third of our lives is spent within the sleeping scenario, dreaming. Would it make sense to utilize that time on something

useful, like thinking about other people, interacting with them, having chats with people, and really refining our social skills? Would that be quite important? Would it be a good use of biology, like our brains?

- 4.15** And it turns out that's the case. So a lot of dreams are social, a lot of them. And it turns out most of them are threatening in nature. So although it feels good to have a chat with Jasmine on a Hawaii beach, that would be quite nice, most dreams are threatening. So it's usually some monster or some negative interaction or something unpleasant, some person being very threatening to me and stuff like that. So that's more conducive to survival. So social dreams, social interactions.
- 4.16** And one other thing here is that it turns out dreams have a lot to do with the self and the ego, and this is quite important to emphasize. Even the fact all the nightmares we have and all the sleep paralysis, monsters we see and hallucinations, it's usually about a demon or a ghost or an entity or a being or a creature wanting to devour our souls. So what is that about?
- 4.17** Why is the creature not interested in my Rolex? Even though I don't have a Rolex, but let's say I had one, why wouldn't it be interested in my Rolex or in my bank account? That's what a burglar would do, right? Especially in Ohio and New York, and these places. They would go for the money. They wouldn't go for my soul or devouring my soul and contaminating me, right, Larry? They wouldn't. They would go for what? They would go for the money, but they go for the soul.
- 4.18** So it tells you that dreams, the dream state, the nightmaric state, the sleep paralysis hallucinations is all about the self, the ego being updated, the self, the consciousness, I, the me within me. All that seems to be getting a massage in dreams.
- 4.19** There are different types of nightmares. There are these idiopathic nightmares, which are nightmares with no apparent reason. I have a nightmare. I see a ghost. I see witches. I see all this. I'm a little boy in the ghetto, and I'm seeing all these monsters coming at me. What is that about?
- 4.20** Is that the monsters in the streets? Or these are the bullies that are trying to take away my lunch money or my whatever. Is that it? Or is it something more? And what about the dreams of reoccurring, mugging, and traumatic memories. What is that about? So let's tackle each.
- 4.21** For idiopathic nightmares, nightmares with no apparent reason, no apparent trauma, no apparent cause, right? These nightmares seem—are very common, first of all. They tend to occur in the rapid eye movement stage of sleep. So in REM sleep, the stage where we have lifelike, vivid and crisp and emotional dreams. These are the idiopathic unknown caused nightmares.
- 4.22** They tend to be less frightening. Oh, they are, they are more frightening than just a frightening dream. So you have a dream that have negative emotions. So you have a

dream with negative emotions, Heavenly, sadness, anger, guilt, maybe embarrassment. You have that.

- 4.23 But then you have a nightmare. The nightmare is what? Is one that is much more terrifying and intense. So that's level two. And then level three, you have the intensely powerfully vivid, I wake up in cold sweat and my heart is beating [sound]. That is the post-traumatic stress nightmare.
- 4.24 Now, the first, dreams occur in REM. Idiopathic nightmares occur in REM. PTSD nightmares occur in deep sleep. This is a deep sleep phenomenon. "Baland, why is it a deep sleep phenomenon? You told us that dreams occur in REM. Didn't you say that?" Right. Okay. Did I mention yesterday that noradrenaline, this chemical cousin to adrenaline in the brain—and then you have noradrenaline in the body as well—but in the brain, that makes you hyper-focused and hyper-attentive and hyper-intense.
- 4.25 Did we say that noradrenaline shuts off during REM? Did we say that yesterday? We said the locus coeruleus in the brainstem that produces noradrenaline shuts off. We said that, right? In the REM stage of sleep, you have no noradrenaline, and that explains why dreams are so spacey and everything is all over the place. There's no logic and cohesion and all that. Okay.
- 4.26 PTSD. Is PTSD characterized by an aroused state where there might be a lot of adrenaline and noradrenaline in the body? Are these guys hyper-manic and running around? Not hyper-manic in the clinical sense, but are they aroused and always terrified and your heart is beating and all that? Do they have a lot of noradrenaline, perhaps in their system? What do you think?
- 4.27 Might they have so much noradrenaline in their system that they cannot enter REM sleep? Now, imagine if you had that much noradrenaline in your body, Heavenly, so much that you cannot go into REM. Your brain refuses to go into the REM state because you have so much noradrenaline circulating in your system. Imagine that.
- 4.28 Now, I mentioned briefly yesterday that in REM sleep, we deal with emotions. When you have negative emotions, so let's say you run about in my day, I have interactions with people, some good, some positive. I may experience something negative. All that in my dreams, especially in REM, you deal with the emotional content, and you take away the negative edge of emotional experiences. How and why is that the case?
- 4.29 Well, you experience a memory without the stress chemical noradrenaline. So your brain says, "Oh, even though it was stressful in real life, I don't feel any stress now." So for that reason, perhaps it's not that dangerous anyway. I can deal with it. I can put it in a greater context of my life and say, "Oh, It doesn't really matter. It's not that dangerous. I've had experiences like that before. I didn't die" and so forth. Is that clear? What I'm basically saying is that in REM sleep, because you have no adrenaline and no stress, neurochemicals, you can experience the world or re-experience the world in a more benign and benevolent and softer way.

- 4.30 Looping back to PTSD. They cannot enter REM sleep. And how tragic. Who needs REM sleep more than these guys, right? I mean, motorcycle accidents, Cameron, all of these terrifying things, the emotion just being so powerful of that traumatic event. And now you can't even go to REM and remove the emotional sting. How tragic. So their brains get into this infinite loop where they try to relive the experience over and over and over and over and over again. To no avail, REM won't allow them to go into that state.
- 4.31 So it happens during deep sleep. So it reemerges and resurfaces in deep sleep. And so that's the nightmares of people with trauma and PTSD. Deep sleep phenomena. Can you believe this? Their brain, every single night, each night, is trying to relive the nightmare and deal with the emotions and remove the emotional sting, but is unable to, but it keeps trying and trying and trying. Yet it fails over and over and over again each night. So you have this infinite nightmaric loop in your brain. Can you believe it?
- 4.32 And then, Larry, guess what? The doctor, a nice doctor, comes out and says, "Here, buddy, here's a pill." It blocks noradrenaline. It blocks that little mischievous chemical that won't allow you to get REM sleep. And the patient will take it, take the pill. Guess what happens? The nightmares disappear because now REM is there again. There's REM, they have normal REM sleep, and for that reason, the nightmares seem to disappear. Isn't that interesting? Isn't that profound?
- 4.33 Look at the prediction. You have noradrenaline, you know what's going on in the brain. Based on that prediction, in neuroscience, you can make specific treatments based on that and potentially help people all around the world who suffer from something like PTSD and nightmares. What do you think, Naomi? It's amazing, isn't it? I think so, too. I think so, too.
- 4.34 How do you treat nightmares? Can you treat them? Can you deal with them? What do you think, Sonia? Is there a way? Beyond the pill, though. The pill is one thing, but other than that, generally, can you deal with nightmares? Is there a therapy for that? Sonia, what if we reduce your stress? Would that be helpful?
- 4.35 With stress, for example—and we know, by the way, people that are stressed, that have a lot of neurotic traits, that are stressed, that are anxious have a more sensitive amygdala, the emotional part of the brain. Again, we mentioned that yesterday. If it's more sensitized, they will have more night time awakening and things like nightmares as well. We know that. Reducing stress, might that be an option? What do you think? It turns out, yes. Going for walks, resting, reducing stress. What is it? Drinking tea?
- 4.36 Mindfulness.
- 4.37 Mindfulness, right? All that. We have that in our treatments. We use that. I've written on that. But then, AJ, we can do something else. We do this sometimes. We address the catastrophic cognitions by telling the patients, "Look, buddy, you dreamt about monsters from your culture. Don't worry, these are not real monsters. These are not

real dreams. You're not being haunted and possessed. Nothing terrifying is going to happen to you."

- 4.38 Or if somebody thinks they will actually die from the events, it's pretty powerful. Look, your heart is beating and you're waking up and all that, often. That can be terrifying. So it makes sense to remove some of the catastrophic thoughts about it. So we do that. That's one step. That's one way to deal with it. But the coolest way, I think, to deal with it is the following. This is very cool. This is very interesting.
- 4.39 Now, we said the brain works in a way where I have events occurring in the day. Things happen to me in the day as I walk about. I go around and meet people and experience all these things. The brain in my dreams, in deep sleep, we have the librarian, right? Putting things in order and consolidating and reconsolidating my memories, meaning my memories are put in places at various points, and then enters REM sleep, and you have wisdom and greater meaning emerging from the experience. This is all part of the brain plasticity process, the brain rewiring itself in response to the world.
- 4.40 Now, especially when experiences are novel, when they are new, when they have just occurred, there's a window of plasticity where you can attack the event and change it. You can change the way your brain deals with it. When you think about it, yes, trauma is a horrible thing, but it's a subjective thing.
- 4.41 It turns out trauma is more common now because we are more sensitive. We are not used to all the traumatic things that are happening to us, Larry. So people in the past that were fighting wars and the gladiator type people and all that, living in the Roman, ancient Romans and all that—they had people like chopping up heads, and things were crazy at that time. There were wars and swords and everything, right? They were less sensitive to these things.
- 4.42 Versus us, we have become very sensitized to the world. So the way we interpret the world is in a very fragile way. In fact, you can even get trauma these days from hearing about a trauma from a second person. Think about this. So Brance's over here telling me something terrible that happened to him, and here is Baland getting trauma. Think about a mirror situation like that. That's pretty profound.
- 4.43 Now, what I'm trying to say here is that the brain has a subjective way of dealing with the world and putting down circuitry and wiring in that into the system. Now, there's a window, especially in the beginning, where experiences are particularly plastic. The brain is particularly able to reshape its form and structure.
- 4.44 In fact, if you play Tetris—this is actually a treatment for PTSD. When the trauma occurs and within 4 hours of the trauma, you sit down with a psychologist and play the video game Tetris, you know this, with the thing coming down, right? You play this—you can actually affect the memory. So you hijacked your attentional system so you don't think about the trauma, and that can actually help. Isn't that profound? That's actually Emily Holmes and others at Oxford have done work like this. Quite interesting.

- 4.45 I'm going to get to the punchline now. I know I keep dragging it out. So if Baland has undergone some trauma, I was mugged. I had this terrible experience, or better yet, I was in a car accident. My car crashed into this other car. It was horrible. It was terrible. I sit down with the psychologist now, and the psychologist will say, "Look, Baland, tell me about your experience. I'll say, "Okay, I was driving. Nancy, our friend over here, she ordered me a robot to go to my home. Self-driving car. And on the way, the self-driving car got crazy and ballistic and drove into, let's say, another car." Traumatic experience, extremely traumatic.
- 4.46 Now, then the therapist will say, "Well, let's reinterpret that. We create an alternative, fictitious ending where the accident actually didn't take place, but the robot was nice and humanlike, and instead of driving into this other car, drove you to your home." You have this alternative ending. And I think about that, and I bring it to mind all the time, and I think about it, and so forth.
- 4.47 Several weeks of me doing this will actually affect my nightmares. Can you believe this? Several weeks of me doing this can affect my nightmares by 66%. Six-six. And this is huge. Nightmares is one of the things you can't really treat, per se. It's very hard. You usually deal with the—you cope with the symptoms. It's not a monster. It's not dangerous. It's not going to kill you. But here we have an actual treatment that removes the nightmares by 66%. Isn't that good? That's pretty much.
- 4.48 But here's the kicker, though. Here's the punchline of it all. Whenever you listen or read your own script of your alternative ending, and you hear a piano chord in the background [sound], every 10 seconds. You do that. And then you listen to the same piano chord as you're sleeping, your nightmares will go away by 90%. Can you believe this? Nine-zero, nine-zero. Simply by rehearsing the alternative ending and then adding this additional layer of you listening to the sound. Is it interesting?
- 4.49 It's Pavlovian conditioning at its best. Here you are, sleeping, and you have this alternative ending to your nightmare, and you just merely listen to piano chord, and your nightmares go away by 90%. Think about that. That's pretty impressive. And the reason, of course, is that in the dream itself, the nightmare, you can now deal with the emotion. The alternative ending is being processed in the dream.
- 4.50 Arnold had a good question, right, about the cause of nightmares. He partially answered it. But one cause is called the—it's a stress acceleration hypothesis. Fancy terminology for the fact that—it was a simple theory, actually, saying, in the first three years of your life, if you have a traumatic-like event happen to you or something negative, something negative happened to you in the first three years.
- 4.51 In the first three years of a life, does our memory—is it working? How does our memory store events in the first three years as an infant? Do you remember all your things you did at two years old? No, right? You don't. You have this infantile amnesia going on. So according to one theory, nightmares is all about you having negative

experiences during this phase, and you can't really cope with it. And so for that reason, for that reason, you create nightmares. You store it in your brain and it resurfaces in nightmares.

- 4.52** So that's one explanation. What do you think? Do you buy this? It's interesting, right? It's an interesting idea. We don't really know if it's true, but there's some data where—and this could be simple things. For example, if I am the first child and I have a sibling come to the world, for me and you, it doesn't matter. But for that child, that could be pretty traumatic. Traumatic in the sense that it's a big deal. My parents and I have to compete for the attention. There's attentional competition going on. So it's a big thing for me.
- 4.53** The argument is that—how to deal with it is that you have nightmares. You have frightening dreams where you deal and you cope with the experience. Then it turns out, in fact, that people that are first born have more nightmares. That's one argument for that.
- 4.54** This is from personal experience. I still remember it, actually. February of 2008, I was 10 years old, and I was having major nightmares constantly over and over and over for no reason. I don't know why. And I was told to pray about it. So I started to pray every night before bed, "Please don't have a nightmare." And it stopped. It just all of a sudden stopped. Any idea?
- 4.55** So our buddy Arnold over here, he was praying. And his prayers were answered and his nightmares vanished. Whatever that means. Does it mean anything? Well, I believe it to. I believe it does have a significant impact. Look, what is prayer about? What does prayer do? Look, I'm a spiritual person. I believe in God and all that. I believe in prayer. So I'm not one of those scientists. But what I'm saying is, what is prayer about?
- 4.56** Think about this. Attentional focus. You are tensely focused on an immensely benevolent being. In your brain, you have fear centers. You have terror departments that can become hyperactive. So you have that. And so attention capacities in your brain are highly scarce. Now, if you focus on a very positive being like a God, that takes away from all the negativity. It calms down the amygdala and the emotional culprits, the cranial gangs that important for negativity. It's dampened down by you thinking about a God. That's the first thing. That's the first thing.
- 4.57** Second, and I'm speaking as a scientist here, is the placebo. Placebo is the fact that I give Arnold a super—I say, "This is a super drink, a pill, the newest pill. Here, take it, pop it." You take it and you go to the gym, and suddenly you come out and you have all these muscles like, what is it called? Popeye or whatever? You know this guy? Okay, you come out like him.
- 4.58** I've experienced that.

- 4.59 Right? Placebo. The placebo affects our physiologies in ways that are profound. You can't even believe it. In fact, the placebo itself can be a placebo. So if I tell you, "This is a placebo"—no, I say, "The placebo is powerful," and I give you a pill and say it's the placebo, it has an effect on you. It's mind-boggling, the placebo. That's one explanation.
- 4.60 Third explanation is that God exists and he's taking away your nightmares. I don't want to take away that possibility because I believe in that myself, too, to be frank. I believe that's possible. But I'm saying scientifically, these are the alternative explanations for your situation. Does that make sense? Great question. Profound question. In fact, in one of—I'll get to sleep paralysis in my next lecture.
- 4.61 I have in my therapy four steps to deal with sleep paralysis, and one of these is attentional focus, which for many people involve prayer, because I found that prayer can make sleep paralysis go away, too. There's something about that emotional state of immense focus on a benevolent creature, a being that's all-powerful and all-good. That can be powerful. And as scientists, we don't talk about whether God is real or not. I'm just saying practically. I'm just talking practical.
- 4.62 That took me off my footing. What were we talking about? Nightmares, right? It's a nightmare. Oh, yeah, night terrors. There's something called night terrors, Arnold. Have you had that before? You seem to have a lot of experiences. [Laughter]. Have you had night terrors before?
- 4.63 Yeah, I've had them, yeah—if you define it.
- 4.64 Night terrors, I'll define it for you. Okay, you get a definition now, a precise one. Okay. So night terrors are these experiences when we're sleeping in bed, and then suddenly you wake up in a state of panic. Your heart is beating, you're sweating. Sometimes you may have a cold sweat even, and you're just terrified and out of your mind. This is called a night terror.
- 4.65 Now, I ask the patient or the person, often kids, by the way, kids often have this, and we talked about why that might be the case, why kids often have these disorders. Lack of maturation, remember? Now, you ask this kid, "What happened? What did you see? Were there ghosts there?" And you know what they'll say? "I don't remember anything. I'm just frightened. I'm frightened, doctor, help me. What's going on?" These are called night terrors.
- 4.66 You can see how they're different from nightmares and sleep paralysis. And although in night terrors, occasionally, you might hallucinate a ghost when you wake up and see something because in that limbo state between wakefulness and dream, it's possible for you to see something. Yes, that is possible. So you might have hallucinations. As we said, the brain has this consciousness state on the edges of dreams.
- 4.67 But usually it's just that sense of terror and waking up and being terrified and frightened. So night terrors. And night terrors belongs to the class of disorders and

conditions we called parasomnias, Heavenly, yesterday. These are things like night walking, teeth grinding, sleep walking, sleep-related eating disorder, and so forth. That's that category. These are, again, where the emotional part of the brain hyperactivate the sensory part of the brain. So that's an important thing.

- 4.68 Now, we talked about Freud yesterday a little bit, and we said that Freud has this idea of the manifest—manifest and latent content of dreams. So the latent content is the stuff that's disguised with symbols, the objects, the things that I see, the experiences that I have that are intangible. And so I use certain metaphors to capture that ineffable experience, and this feeling that I have of dread and anxiety.
- 4.69 By the way, Freud also talked about the royal road to the unconscious, meaning that dreams are a type of wish fulfillment. You know his superego, id, and ego, and all that. And so superego, the higher level of the prefrontal cortex, let's say, as an analogy—that part cannot deal with the frightening stuff. And so we use these metaphors and these things in dreams.
- 4.70 But then in the dream itself, everything is—you can go directly to your unconscious, basically, the royal road. It doesn't have the filters. Am I contradicting myself here? Hold on. I'm saying it's the royal—so your unconscious can use—it's the royal road to the unconscious, but it still uses this metaphor. I just want to be clear so it doesn't get muddled up here. And so his ideas for nightmares actually fit in here perfectly. So the nightmares are simply the stuff from our lives that we cannot deal with, and so we have nightmares. So that's something very important to add.
- 4.71 Yesterday, we mentioned the threat simulation theory. What was threat simulation? Remember? I think, Amanda, what was that about? It's the idea that in dreams, we have negative experiences, and we try to deal with this in a—the brain is trying to deal with it in the dream. So tackle it first head-on in order to simply—what's the word? When you experience it in your dreams first, you can better deal with it in real life. That was the idea.
- 4.72 So much content here. What do we talk about? So many things. Let's talk about the brain. Now, in the brain, what might cause the nightmares in the brain? What circuitry? What circuitry might be responsible for the nightmares? We talked about the amygdala, the almond-like structure behind the ears. That part is partially responsible for nightmares, meaning that people with nightmares have a hyperactive amygdala. That's one part of the circuit.
- 4.73 Second part, the ACC, anterior cingulate cortex. Anterior cingulate cortex. This is an important structure for you to know in the circuit of nightmares. The anterior cingulate cortex, we usually just use ACC, we just use ACC. It's active during empathy. It's the empathy part of the brain. In fact, it's a social empathy part of the brain. Meaning, if you afflict pain to yourself, the ACC in my brain will become hyperactive from merely

watching you. It's part of the social cognition mirror neuron type system as well. You have the ACC. Amygdala ACC, the emotional.

- 4.74 So amygdala, let's call it the fear headquarters of the brain, although you have positive experiences emanating from the amygdala as well, but often negative. Amygdala, ACC. ACC, empathy part of the brain, but often in a negative sense. So it's not positive empathy. So if you have a pleasant experience—AJ, let's drink something nice over there, juice, fruit juice or something—and I watch you and I feel your sensations, I feel, "Wow, that's so nice to watch," that's not what I'm talking about. Although there is some ACC involved, probably, but it's mainly a negative tone that's part of the ACC.
- 4.75 If I were to zap the ACC, you might feel like there's a vague, ominous figure in the room. It's lurking and watching over you. That's what the ACC is doing. It's part of the prefrontal cortex, meaning the outer layer of the brain and the front, the ACC. Does it make sense that it's involved in nightmares now, that it has a role? It also isn't so social, other people, negative feelings, empathy, a little bit of empathy. So that's part of the circuits.
- 4.76 Another key part is the prefrontal lobes, the medial, meaning the outer layer, the CEO of the brain, which is out here. In the middle part, remember yesterday we said in the sides you have what? Cold cognition, calculated thoughts. If, example, I'll give you an example to my gladiator analogy before. If I'm in a fight in a war zone, and I'm in the past, and things are with swords. There's a lot of medial prefrontal cortex in my decision making. I'm fighting with swords—ding, ding—it's intense. I'm seeing people bleed, and blood, and limbs being cut off and all that. A lot of medial prefrontal emotional decision making.
- 4.77 There's a guy in front of me, he's about to die. Should I end his life? Emotional circuits kicking in. Oh, no, just let him survive. I can't do it. It's too much. It's another human being, right? That's the medial prefrontal. In contrast, if I'm sitting in my office somewhere, a nice office, you know, big screen like this, and I'm seeing these guys over there. They're sitting, the enemy, and I have these drones, and I'm going to press a button, make these people die. I'm just going to press it. Is it the medial prefrontal or is it the LPFC? Which one is it now?
- 4.78 I'm thousands of kilometers and miles away from the people that I'm going to kill, and I'm saying, "These guys are probably evil. Let's just press a button." In this case, it's what? It's the outer side. It's a calculated non-emotional decision because it doesn't have that intensity of that encounter. So outer side of the brain, calculated decision-making. Inner, here, is the emotional decision-making. In that emotional decision-making medial part of the prefrontal cortex, you also have the break system of the brain. The break system.
- 4.79 Now, this break system allows you to put a break on your emotions. And so if that part of the brain has thinned out, it's thinner. It literally could shrink. Think about this. We

mentioned yesterday that the amygdala, the emotional core of the brain, can balloon up and become larger in size. As it turns out, the breaks of the brain, the medial prefrontal CEO up here, the seat of consciousness, the head office, can become thinner if you are hyper stressed, if you have PTSD, if you have an anxiety disorder, if you have depression. The breaks can literally become thinner and thinner and thinner over time.

4.80 When you have a thin break, guess what happens? When you have a big emotional center, headquarter, then what? You commit crimes, you become hyper-emotional, hyper-terrified, and you have a lot of, perhaps, nightmares. If you don't have the emotional breaks, you have a lot of nightmares. The nightmare circuit in the brain involves these structures: the CEO, the breaks of the brain thinning out, the emotional core, the amygdala becoming larger, the ACC of brain becoming hyperactivated. This is the algorithm for a lot of nightmares.

Lecture 5: Sleep Paralysis

At a glance — Sleep paralysis, the condition of waking while the body remains immobilized, and its cross-cultural interpretations.

- 5.1** So in this lecture, I'm going to be talking about sleep paralysis. Now, sleep paralysis is this condition where you lie in bed—you're sleeping seemingly peacefully, lying there. Then suddenly, you feel like you're awake. You scan the surroundings of your room. You feel awake. And then you try to move, but you're paralyzed: unable to move or speak. You're trying to shout, "Mom! Dad!" but no words are coming out. Your throat is stuck. Everything is just stuck in there. You can't speak. You can't move a single muscle. Nothing is moving.
- 5.2** In about 40% of cases, a shadow from the corner of the room will approach the helpless sleeper. Usually has a humanlike shape approaching you, coming closer and closer until it's on your chest, strangling you, pressing on your chest. Wild experience, wild experience.
- 5.3** I've had this before myself. Quite similar to this classic scenario. I was in Copenhagen. Again, this time I was a teenager I was sleeping in my bed. And I had this experience. I woke up paralyzed, unable to move and try to shout, "Mom! Dad!" but just no words would come out. I was just completely horrified. And then a monster, a ghost, or a vicious evil is being in the room.
- 5.4** Now, this vicious evil being was what you would call a vague, nebulous, unseen creature. I couldn't really—it was amorphous—I couldn't really pin down where it was, but it was there and it was evil. And the evil of this creature was what you would say, evil of epic proportions. It was like, take all the evil in the universe and concentrate that into one bubble and then have it in your little bedroom. That's what I felt like. It was so evil. And it knew all my intentions. It knew me, Baland. What I was up to, what my life was all about. It wanted to devour my soul. Really, it wanted me. It wanted to kill me. I've never had that much agony and distress in my life. I was terrified.
- 5.5** Now, as the being eventually got closer and started to press on my chest, it strangled me. You know what? That was the first time in my life I was certain that I was going to die. I was sure this was it. This was the moment. It was all over. No more, no more games, no more life, no more fun. Nothing, nothing. It was over. I was going to die. And I thought—I tried to pray, and I did all these things. I was just trying to get the monster to disappear. And eventually, I wrestled myself out of the state. It vanished, quite surprisingly.
- 5.6** As I woke up the next day, I thought, "What do I do? Do I go tell my parents?" At the time, I wasn't a very studious kid. I was, in fact, performing really poorly in school. In high school, I was the worst student. Can you believe it, Cameron? Didn't do any homework. I was the guy always trying to be cool and trying to impress. I don't want to

say who, but I was trying to have fun. I was a slacker. I was a slacker, and that was my life. I was a slacker. I didn't like school. I hated school. I had a 25% absence rate. Can you believe it? 25%. I pretty much didn't appear. Each Monday was like an extended weekend, long weekend. This was my life.

5.7 And when I had this experience of sleep paralysis, it was so profound, I started to Google it, seeing a ghost in my room, seeing a monster attacking me, choking me, what do I do? Google gave me no answer. I thought I was psychotic. What in the world would you do in a situation like that? I couldn't go to my parents, obviously. I couldn't go to my siblings.

5.8 I started to read. I said, "Look, let me go to the books. Let me go to the library." There was a little library in the ghetto, and it didn't have many books, but it had a few of them. I went there and I started to look up various books about certain phenomena, and I was hooked on the brain—like sleep, brain, what can happen when you sleep, what's going on in the brain. And I was hooked and I was changed. And at that point, I got interested in knowledge for the sake of knowledge.

5.9 And my life changed at that point. In many ways, it was a transformation that was intellectual, but it was also spiritual. I started to question things in my life and what's the meaning of things. And that's a long story. But point being that here in my little bedroom, I had the most profound experience of them all, making me question the here and now of conscious experience. Who am I? Who is Baland in this little speck of dust we call the Earth, right? What is me? What is Baland doing? And that experience alone opened up this doorway. It's a quite profound, emotionally life-changing experience.

5.10 Sleep paralysis. Now, this experience took me on a long journey around the world to explore what sleep paralysis is. I had to know what happened in my bedroom, what was going on, what is the monster all about? And I started to ask my friends, and some of them had similar experiences. They would say, "I was sleeping and a ghost came, and I was paralyzed." So they had similar experiences that they would share with me. And commonly people would hide it and not really tell others because you don't want to tell other people that you're going crazy, right? You're bonkers, you're going insane. You don't want to tell. You don't want to share it, but I had other friends that had similar experiences.

5.11 Now, my grades were poor in school. I moved at this point—we moved—I moved. I told my parents, "I have very poor grades. I can't get into any university here in Copenhagen." I had studied some English literature a little bit, but I couldn't really get into psychology or study the brain. My grades were so poor, for God's sake. So I went to Egypt. So I went to Egypt to a private university over there. We got some scholarships from me studying night and day, trying to make up for all my years of not studying. And I was able to get in and get some knowledge. And I became absorbed

with knowledge at that point. I was reading day and night. Can you believe it? I was just devouring books. Sleep paralysis.

- 5.12** Then in Egypt, I was asking people—I was the guy that would always hang out with the common folks. So I would hang out with the porters and the people. There was a lot of—it's a very classed society. There's a lot of class. I would hang out with the common folks, and I would ask them, "Do you have sleep paralysis? Have you experienced this before, this condition of being paralyzed and unable to move and speak?" And they would say, "Yeah, I've had that before. And it's probably, we call it a genie—it's the evil genie." Have you seen Aladdin before? That cartoon? It's the genies that attacks you, that chokes you, that can strangle you and potentially even kill you. Sometimes it sexually violates you. My God, my God. That's what they would say to me. I was like, "Wow, that's so interesting."
- 5.13** And I recall, although my friends in Denmark had sleep paralysis, the explanations weren't that exotic, Larry. I mean, there were explanations, but they weren't that exotic— with like genies, for God's sake. Like the evil genies, the blue thing—it wasn't like that at all. It was just merely people having paralyzed experiences saying, "Oh, it's probably stress. Oh, it's probably the brain. Oh, it's probably physiology. Oh, it's probably this and that."
- 5.14** I got more and more into research. Not in the sense that I want to be a researcher, I want to be an intellectual. No, I was passionate about knowledge, and I had to decipher this experience. And so I did my first research experiment. I never thought that this would actually get published or actually become into a research study. I didn't think that. I was just like, "Okay, I need to compare sleep paralysis in Denmark and Egypt and see what's going on."
- 5.15** First step, I contacted my brother, called him. He was in Copenhagen, a medical student at the time. He was much smarter than me—got into medical school. I couldn't get into medical school. And so I said, "Bamo,"—that's the name of my brother—"Bamo, look, can you collect some data? Ask a lot of people over there what sleep paralysis is, how people understand it, what they think it is." And he did that. And on vacation, on occasion, I would join them as well. I would go to Denmark and fly there for a vacation, see my family, stuff like that. And then we would collect all this data. And we found that indeed, most people would say that sleep paralysis was merely the brain. It was merely physiology. It was merely stress and things.
- 5.16** So my initial hunch, intuition, was on track. People actually thought it was just your brain playing tricks on you. That was the Danish explanation for sleep paralysis. In Egypt, we did the studies. We interviewed tons of people, and we found that indeed, people think that it's Aladdin, it's the Aladdin monsters, the genies, evil genies. They can kill you, attack you, choke you, potentially even kill you.

- 5.17** And it was profound. Like 95% of people in Egypt would say that it was a spiritual experience, whereas you would have almost none in Denmark. Like 5% maybe would say that it was supernatural. So clearly there was a cultural difference here. Clearly. About 50% of the Egyptians would specifically say that it was the genies that caused their sleep paralysis, that the creature was extremely evil and vicious.
- 5.18** At this point, I asked myself: how does cultural ideas and notions affect experience itself? Does it contaminate the experience? Does it change it? Because for me, it did. I said this was the most profound, overwhelmingly supernatural thing that occurred to me. So surely if you are an Egyptian and have this epic encounter with a demonic Aladdin-like figure, surely that would affect your life.
- 5.19** So I thought, "Let's compare. Let's bring in—let's get some participants, interview them, see how it affects them." It turned out that for Egyptians, for an Egyptian who had sleep paralysis—so you have a Danish person with sleep paralysis, and you have an Egyptian with sleep paralysis. For the Egyptian, sleep paralysis occurs three times more frequently in that person. Not only that, he feels the experience is prolonged. It's the paralysis. The whole thing lasts much longer.
- 5.20** Step two. So it lasts much longer, it's much more frequent, and it's much more, as you would expect, fearful. So like 50% of Egyptians thought they could die from it. Now, you might think that's trivial, but imagine you having a trauma. It would be a big deal if you thought you could die from it. Thinking you can die from an actual experience is a huge thing. You have 50% of these guys saying that they could potentially die from sleep paralysis.
- 5.21** Sleep paralysis is a common experience. It affects about 20% of the general population. It's not like a quirk that affects 1% or 2%. It's rampant in society. It's very abundantly prevalent. So the fact that you have within 20%—you have 50% believing it's the evil genies. And in fact, half of that thinking they could die from it, that's a lot of people potentially being traumatized.
- 5.22** I thought, "Wow, this is interesting science. Let me replicate it, though." Because in science, as you know, you always have to replicate your experiments and your studies because you need to be sure that it actually is true. And so for that reason, actually, I was at a conference in Italy. Excuse me. The conference was in South Africa, but there was Italians there.
- 5.23** And there was specifically an Italian audience, an Italian guy who came to me after one of my lectures. He said, "Look, Baland, we have a similar cultural phenomenon in Italy. It's called the pandafeche. It's a pandafeche monster. It comes at night, it attacks you, chokes you." This is what he said, okay? And yes, he was using a lot of hand gestures just like me, right? He says, "Yeah, so we have this. We should do a study." I said, "Okay, brother, let's do this study."

- 5.24 And so we did a study in Italy, and it turned out that Italians, one-third of them, thought that sleep paralysis is caused by this pandafecche monster— giant witch half cat-like creature thing. It's a witch, ghost, and a cat-like creature all wrapped in one. Horrifying. Can you believe this, Sam? Horrifying. One-third, and that's a lot. Think about this. Here you have a secular European country—I know the Pope— it's a religious country and all that. But still, for a secular country, for one-third to believe that it's a witch for God's sake, that's a lot of people, wouldn't you say, Larry? That's quite the number.
- 5.25 Now, similar to Egypt, we found that the Italians thought that they could die from sleep paralysis. They had frequent sleep paralysis. The paralysis was prolonged. So again, we found a similar pattern. Our hypothesis seemed to be supported by our data. Very interesting here. So here you have two countries—cultural ideas of sleep paralysis, and here you have the experience becoming amplified. A salience landscape—the experience has become hyper-salient.
- 5.26 We had to dig further. We had to go deeper in and understand what does this do to you? Because for me—I was highly affected by it. And it wasn't merely fear. I almost—hadn't I been more resilient—and I had great loving parents—my mother and father, even though I came to Denmark as refugees—I was a refugee child, and my parents experienced trauma and all that to some degree, especially my mom. My parents were very caring. They were very nurturing. So I think that made it so that whenever I experienced adverse life experiences, I didn't develop mental disorders. So it was like a buffer.
- 5.27 And this is a side point. In neuroscience and neuropsychiatry, if you have a good environment, like good loving parents, that can prevent you from becoming—developing trauma and anxiety and depression, potentially, because it acts like a buffer.
- 5.28 Now, back to what I was saying. I was wondering— sleep paralysis itself affected me a great deal. I was hugely impacted, almost started to worry about it and become—I didn't become depressed, but it was profound. Then I thought, "Okay, what about the Egyptians and the Italians? The ones who see the monster, do they develop depression and anxiety and PTSD-like symptoms?" And it turns out the answer was, "Yes, potentially. Potentially, it could."
- 5.29 That's a big thing. You know why? I'll tell you. Amanda, it's big. You know why? Because here, lying in your bed, the safest place of all, it should be safe, restful. Yet you have an intergalactic monster or space alien—there's a different cultural interpretation, by the way. One American version is a space alien abduction. The spaceship, the ghost, the being is coming down and abducting you and doing all kinds of things to your body and all that. So that's one version. So you have this intergalactic being in your room and doing all these horrible things to you.

- 5.30 And you think it's real. You're 100% certain it's real. I mean, that should have an impact on you, right? And it seems to be doing that, just that, as we found with the depression, anxiety, and PTSD-like symptoms, especially people who saw the ghost. People who report seeing monsters have more of these psychopathological symptoms, more of this depression and anxiety.
- 5.31 And this fits well with one of my colleagues, Richard McNally. He discovered the following. By the way, this was before me. This was like in 2005, when I was a kid. He was doing his research back then on sleep paralysis. Later, we became friends and mentor—he became my mentor and all that. Great how life can work out in that way. But it was amazing.
- 5.32 So he did a study. He looked at people with PTSD, post-traumatic stress disorder—they've been through a major trauma. And have them listen to the audio scripts that they had of their encounter. So I went through this, and this happened, and the car hit me, or I was in war, and I was shot, and I almost died, and so forth. They listen to the audio scripts of people with trauma. They listen to that on tape—their own encounters, their own encounters. They listen to this and they measure their physiology, their heart rate, their blood pressure, and all that. They listen to all of that.
- 5.33 And then they listen to somebody with sleep paralysis who said, I was abducted by alien, and the alien came down on the ship and did all this stuff to me. And they again, listen to the audio script of the encounter and the narrative of what happened. And it turns out their physiological reactions are comparable—the same.
- 5.34 So here you have a guy who actually went to war. He went to the war zone in Afghanistan or whatever country, and he was in battle. He was physically in a battle zone, and the other guy is presumably sleeping, presumably no aliens in his bedroom, yet their physiology is equivalent. Their post-traumatic-like physiological reactions are the same. That's quite profound. Again, shows you the depth of this phenomenon.
- 5.35 We said it affects about 20% of the population. It lasts usually for a few seconds, sometimes several minutes. Some say—we have found data of like one hour, for example. We've done research on this myself in about seven countries. So we've confirmed a lot of our findings in Denmark, in Egypt—as we said—Italy, Poland. Where else? I've done research in America, Turkey—we found some interesting ideas of entities and ghosts.
- 5.36 So all around the world, there are different cultural interpretations and ideas for what it is, what the monster is all about. So it could be—in Japan, it could be a certain ghost that attacks you, chokes you. In Mexico, another version, and so forth. So there's a cultural variance. But within each culture, there seems to be an overarching idea of a monster that has a humanlike shape—shadow-like being often. Often it wears a hat. And then it inspired my own fashion choice. And then it approaches you and attacks you and all that. It's called the Hat Man and all these various versions of it.

- 5.37** You can have hallucinations in various sensory modalities. So not only is it vision, but it can also be tactile, it can be hearing things, it can be other forms of bodily hallucinations. You might have an out-of-body experience, seeing yourself out in space, a duplicate of yourself out there hovering over you. I've had that before, another Baland, a copy. Usually, the conversation is, "Who's the real me? The Baland here or the one up here?" It's hijacking of identity of sorts. That can commonly occur.
- 5.38** Out-of-body experiences where you see, let's say, another figure. So not you out there, but you see maybe—I've had Gaddafi hovering over me when I was in Egypt, so I had that experience. It was quite intense. It was during the Libyan revolution. I watched a lot of TV and CNN and all that. So it probably leaked into my experience. So you can have all these various hallucinations.
- 5.39** So how does it come about in the brain is the question. What's going on? We want to know this, right? And by the way, a lot of this research came about later. So my initial research was focused a lot on the cultural ideas and how the phenomenon is understood around the world. But later I went into the brain science, and that was with my mentor in California—he's a brain scientist, so I shifted gears a bit.
- 5.40** During the stages of sleep, you cycle through how many stages? Naomi? Four? Was it four? Was it four, right? In which stage do we dream? We dream in REM, right? Absolutely. We dream in REM. Now, if you have vivid and lifelike crisp dreams, we said, your body says what? The neurons in the back's brainstem paralyzes your entire body during a REM, so you don't act out your dreams. We said that, right? And then we said, in certain conditions, you have no paralysis, so you actually do act out your dreams and hurt yourself.
- 5.41** In sleep paralysis, what occurs is that you start to wake up consciously. So the sensory regions of the brain—so your brain, the outer layer has a sensory region, mainly here in the back, the visual cortex, but somatosensory up here as well, and perceptual reasons—they become activated prematurely, even though you're supposed to be sleeping. So you have a decoupling of the centers that are responsible for wakefulness and the centers responsible for REM physiology and sleep, meaning that these two systems go awry. They work separately from each other.
- 5.42** Why would that happen in some people and not other people? What's going on there, you can ask. And the answer to that is we don't really know. So we don't really know why the decoupling happens, but it turns out—again, going back to all the phenomenon that we've covered in this course so far—the brain is not perfect. So in these transition phases, it can have these blunders, glitches in the systems. And so if you came up with too much noradrenaline here, too much serotonin here, too much activation in this circuit, and you end up with this decoupling.
- 5.43** And a key player, obviously, again, is the emotional center of the brain. And that's a core theme of this lecture. If the emotional centers are too much, too active, that can

arouse the higher level sensory regions. Unsurprisingly, college students have some of the highest rates of sleep arousal. Can you believe this? They have some of the highest rates, Brance, especially around exam times. So around exam times, are you stressed and anxious? Does stress and anxiety sensitize your emotional core? It does. It does.

- 5.44** So college students have some of the highest rates. Anxious and depressed people—not so much depression, but more anxiety, PTSD, trauma. All the conditions where the emotional core is hyperactive can lead you to have this glitch of nocturnal arousal and sleep paralysis.
- 5.45** When you hallucinate, right? You hallucinate, and the next lecture is about the hallucinations, about the ghost, what's going on in the brain, how the brain comes, creates these ghosts. But what's really fascinating is that when you're hallucinating, you can actually incorporate the surroundings into your narrative.
- 5.46** So if you are sleeping in a room and there is—and I had this, I had a patient who told me, "I was sleeping," and for some reason, the bed was angled so they could see a mirror of themselves. They could see themselves in a mirror. And this poor chap had sleep paralysis five, six times a day, woke up, opened their eyes, saw themselves in the mirror. Guess what they saw? A crippled version of themselves or like a horror movie. Or you might have a picture of your mom morph into a monster. I mean, this is real. People have these experiences—shows you how tricky it can be.
- 5.47** Now, given how terrifying and potentially distressful it can be, how do you treat it? How do you deal with sleep paralysis? This was my own thought because I really had to do something about sleep paralysis. A lot of patients I was chatting with, and they had a lot of horror coming from this. Some people have it five times a week, three or four times a week.
- 5.48** So basically, we say that if you have sleep paralysis now and then, in the general population, we said it's about 20%, if you have it now and then, it's okay. We can tell you, "Oh, it's not that terrifying. Nothing's going to happen. You won't die" and all that. But if you have it three or four times a week, then we got to do something about it. This is where it becomes—we call it, fearful isolated sleep paralysis. It's when it becomes repetitive and reoccurring.
- 5.49** I thought about this, are there any therapies for sleep paralysis? I couldn't find any. I mean, everything was about CBT, meaning cognitive behavioral therapy. You have to deal with the monster. It's not really going to kill you. It's not really the Aladdin genie. It's not really the space aliens. It's just your brain playing tricks on you. And that can be helpful to some degree. I use it whenever—we use it when we can in our work.
- 5.50** But is there a more direct way to deal with it? Mind you, I have a lot of sleep paralysis. I had a lot of experiences back then after that first one. By a lot, I don't mean four times a week, but once every two months, I might have it. Once every six months, I might have one experience. And I noticed one thing. When you lie there, you're actually

consciously aware of your surroundings. You are cognitively aware. And that means the prefrontal cortex—the outer layer, your consciousness, sensory systems—are intact. And we know this from the description.

- 5.51** But in real life, in vivo, it shows you you are consciously aware. And why don't we actually exploit that? You could do therapy on somebody who's in that state, technically. He can't move, he can't speak. But if you could whisper information to him, technically, you could actually have him do certain things. I thought, "Well, let's do something about this."
- 5.52** And so I developed this therapy and it includes four steps. That's what I thought. This might be the easiest for me. I thought, "let's do four steps." And I thought, the first steps— based on my own experience, based on my own monsters—the first step should be: whenever the monster attacks me, whenever I find myself paralyzed, guess what? I'm going to do something very sophisticated. I'm going to close my eyes, okay? I'm going to close my eyes. Makes sense, right? With all the sensory input infiltrating my visual sight. Close your eyes.
- 5.53** And then you tell yourself, "Look, sleep paralysis is common all around the world." You remind yourself of that. Sleep paralysis is common all around the world. It's benign. It won't kill you. It's not actually dangerous. "I won't die from this." And this sounds very banal, like trivial, right? But when you are in a very terrifying, dangerous situation, Arnold, sometimes just calming yourself down, is that helpful? It is. It grounds you a little bit, right? And just closing your eyes and grounding yourself. So that's the first step.
- 5.54** Second step, it's called—so the first step in technical jargon is called cognitive re-appraisal, meaning you just change the interpretation of it. That's the first step. Second is emotional distancing in technical jargon. Meaning you say, "Okay, now that I know that this is common, I won't die, that people have this all around the world, that it afflicts 20% of the population, what if I emotionally distance myself from the experience?" I say, "Look, I won't die from this, so there's no reason for me to worry about it. Let me just push it away. Let me just say this is just the—let me just be emotionally detached right now." That's the second step.
- 5.55** Third step—okay, Heavenly—the third step here is the key step. It's the key step. It's when you— going back to what I said before that, your attentional capacities are highly limited in the brain. So the frontoparietal region—meaning the front and the parietal region up here, this part of the brain and this part of the brain—you have attentional circuits. Attention is always competing. And there's fear attention—me focusing on potential monsters.
- 5.56** And there's me focusing on something extremely positive, like a prayer, or my mom's face, or a positive experience of some sort. So I said, "Okay, look, I'm going to intensely focus on something positive right now. Right now, I'm going to focus on that. And then

to hell with the monsters"—excuse me—"I'm going to focus on that something very positive right now." So that's step three. And that's the key step.

- 5.57** Fourth step—we have a little fourth step which is just—I'm just going to relax and non-judgmentally deal with this and avoid trying to move my body. Why? We'll get into that in the next lecture. But basically, when you try to move your body in this paralyzed state, you can cause pain and spasms in your body and tighten things up. So you don't want to do that.
- 5.58** Four simple steps. We tried this with a few people. It dramatically reduced sleep paralysis, dramatically. Changed their trauma symptoms— their worry symptoms. And quite promising, we did a study in the University of Bologna with Italians, and we found that in a pilot study, this therapy reduced sleep paralysis about 50%. 50%. This little recipe that I came up with in my bedroom. Okay. A 50% reduction. The first trial ever for sleep paralysis was when we found it. What do you think, Larry? It's good, right? Have you had it before?
- 5.59** I have experienced it. I didn't have a spiritual negative connotation, so I must be in the 60% of people. But it was very scary. I couldn't move. But without the spiritual connotation, it wasn't—it didn't mark me.
- 5.60** You didn't have that. Yeah, that makes sense. It shows you the spiritual connotation, how powerful it can be, how it can affect you. It really is an extreme experience. To this day, when I'm traveling—especially these gothic hotels they book me—I get sleep paralysis once in a while there. It's actually jet lag. Jet lag and other conditions can fragment your sleep and cause you to have sleep paralysis. And so when I have it, I said, "My God, I've developed a therapy and we found promising data. We're working on it now with an app. We're at Harvard and all that." And I say, "Look, this monster feels so real. I can't get rid of it." And what do I do when I try my therapy—and occasionally, I will even fail at it. Think about that. That shows you how intense it is. Yeah, it must be one of the most fascinating, if not the most profound phenomenon in medicine.
- 5.61** And occasionally, you can jolt out of your body and become lucid. So you have a lucid experience during sleep paralysis. So that's profound. Again, it has to do with the fact that if sleep paralysis is your sensory systems turning on, and your REM physiology turning on, and your REM physiology lingering on, and leaping over into wakefulness. Lucid dreams are a similar phenomenon. Your conscious parts of the brain, the sensory parts of the brain, are active inside the dream. That's a lucid dream phenomenon. That's next lecture. My point here is that there's a lot of overlap in these states.
- 5.62** If you have sleep paralysis and you are consciously aware, then you can become so aware that the—I want to say, soul— your neurologically anchored body image can emerge out of your physical body in a dreaming sense. It's all dream imagery. You're hallucinating your body emerging out of yourself. But it can feel quite real.

- 5.63** So I had this once, and I stepped out of my physical body, and I walked around in my apartment. I was walking around there, and I thought, "What do I do right now?" You had that before? I think about it. I looked down on my physical body. Have you seen Patrick Swayze and the movie *Ghost* when he's about to die? It's like that. I saw myself down there and thought, "Okay, now I'm going to do an experiment. I have to do something about this."
- 5.64** So I took a piece of paper, like scrap paper on the floor, and put it in my pajamas. Okay? And I went back to my physical body, and I jumped down there, and I thought, "If the piece of paper is there when I wake up, I would have made some ripple in this scientific world." Was it there or not? Unfortunately, it wasn't there. Otherwise, I would be giving this talk at the Nobel conference, not at Peterson Academy. No, Peterson Academy, of course, is great, but I'm saying. We have to add that notation, right? Profound, right?
- 5.65** Here is an interplay of self, the "I", the physical me, what's seemingly a spiritual dimension, what people call astral-projection, intergalactic meetings with space aliens, ghosts, monsters, all lumped in one experience. Self— amorphous, evil-like, vicious being that have theory of mind. You're using your theory of mind circuits. It shows you all that—how we said this self is anchored in the parietal lobes—how that can loosen up and actually show up in your room as a doppelgänger and all that. That's what's so amazing about this.
- 5.66** Now, before we wrap up, I did want to answer some really good questions we had from Cameron. And the first question I do remember still, and it ties into all of this, he said, "Why is it that nightmares rear their ugly head in deep sleep and not REM." And the answer, I think Brance partially answered it before, is that in REM sleep, you don't have noradrenaline and serotonin. You don't have that. And so when you don't have these stress chemicals, and the brain cannot enter that state full of stress chemicals, the brain has to find an alternative route.
- 5.67** Now, here's the answer to your question then is that—why deep sleep? And the answer is: we don't know for sure, but it seems to be that the place where it latches on and you get the nightmare there instead. One reason is that you can have noradrenaline in deep sleep. You have that, so it doesn't shut off completely. And so for that reason, it's a good place to—if you are really desperate for some way to deal with the traumatic memory, then why not bring it up in deep sleep? And you do indeed have dreams in deep sleep. But these dreams in deep sleep, as I mentioned, are much less intense in the sense of less complex, less narrative in structure, and so forth. That's the answer.
- 5.68** For people who have NPD, narcissistic personality disorder, borderline personality disorder, attention deficit hyperactivity disorder—if they have heightened or decreased levels of activity in their prefrontal—what's it— the anterior singulate cortex or their prefrontal medial cortex—do people who have those senses, or decreased levels of

that activity—are they more predisposed or less predisposed to specific hypersensitive dreams or serious nightmares?

- 5.69** It's a great question. If you have hyperactivity, is it hyper or hypo?
- 5.70** Either or. I can imagine one way or the other wouldn't—
- 5.71** Things like narcissism or psychopaths and things like that—they tend to have the emotional core of the brain being not very active. It's not very active. In fact, the medial prefrontal cortex—remember when the guy was lying and I was about to see if I should kill the enemy with my sword and the medial part of the prefrontal kicked in and said, "Look, he's a human like you, right?" Guess what? In psychopaths, that part of the brain is what? It's just completely dead. It's not responding. You're using the outside. You're saying, "Look, this guy—I'm going to take a sword on his clothing and his shield and all that. Let me just kill him. That's a psychopath. That's the outer layers of the prefrontal doing its job. That's doing very well in psychopaths.
- 5.72** Now, the amygdala is a lot worse. It's shrinking. It's not really activating. So you don't have fear. You don't have emotional responsiveness. You're dead cold. And the question is—and the ACC, the empathy part of the brain, do they have empathy, these psychopaths? No, not really. So that's also typically reduced connectivity.
- 5.73** The answer is: do they have more nightmares? And I'll have to be frank and say, I don't know. Does it affect their nightmares? I don't know. Do psychopaths have nightmares? What I can say that their arousal system with noradrenaline and all that is cool as you can even imagine. Psychopaths are the guys that can go in and rob a bank and be completely cool. No emotions, no heartbeats. Like me, if there's a cop behind me I go, "What's going on?" That's me, right? But the psychopaths, nothing. He's cool, smooth, no emotion, emotionless.
- 5.74** So for him to experience a nightmare, my guess would be he doesn't have nightmares because he wouldn't have these frightening emotions. Has it been studied carefully? I don't know. So maybe you can look it up and let me know. I'd actually be curious to know, but I don't think they have nightmares in that regard. You don't have the emotional triggers in that sense.
- 5.75** It's interesting also—I wanted to add on a final point on sleep paralysis because it adds to the whole experience. Given everything that can happen during sleep paralysis, guess what can happen more? Like sexual dreams or sexual attacks are very common. So we see that as well. And you might ask, "Why is that? What's going on in the brain? Why do people actually claim of being raped?" So people will say, "I was raped by the monster." And it's quite humiliating for somebody, right? They're sleeping, and then the monster is not only trying to kill them, but it's bloody raping them as well.
- 5.76** So this was also something we had to look into and see what's going on in the brain. And it turns out that during REM sleep, the hypothalamus—I mentioned that structure

a few times, the marble-like structures, fairly deep inside the brain, in the middle. They have centers for aggression. They have centers for sexuality. They have centers for sleep. They have centers for body regulation and sweating and various functions.

- 5.77** Interesting that sexuality and aggression is hooked up in the same circuit, isn't it? It shows you potentially how in some people, how sexuality and aggression can be something that turns certain people on, especially in males. You have a circuit here that shows you clearly the same part of the brain mediates aggression and sexuality. And the monster that attacks you during sleep paralysis is pretty aggressive, too.
- 5.78** During REM sleep, this circuit— this monster, that's called the monster circuit, the sexual circuit, actually, the sexual circuit—the hypothalamus is hyperactive during REM. That not only explains why we have sexual dreams, but also explains why the monster can attack you and do sexual things to you. Within that circuit, you also have the emotional part of the brain and the dopamine centers of the brain being hyperactive in dreams.
- 5.79** In REM sleep, your dopamine centers—the centers that have to do with pleasure, with joy, with thrill-seeking, are hyperactive in dreams. No wonder, right? With everything that's going on in dreams. In fact, if that dopamine circuit in my brain—there's one going from the limbic emotional core to my prefrontal—if there's a stroke there, guess what happens? I lose the ability to dream altogether, showing you how important dopamine is for dreams and for REM sleep.
- 5.80** Looping back to sleep paralysis, okay? During sleep paralysis, you have hypothalamus—aggression and sexuality, part of the brain—being hyperactive, given that it's a REM phenomenon. You have dopamine centers—and dopamine, of course, is highly active during sexual encounters—again, being hyperactive in REM, can explain how this horrendous experience can turn into a rape scenario.
- 5.81** I just want to add that as a final little note on sleep paralysis. Does that make sense? Okay, good. Well, that's all I had for now. Thank you very much.

Lecture 6: The Creative Brain

At a glance — Dreams and creativity: how the sleeping brain fuels insight and invention.

- 6.1** In this lecture, in this portion of the course, we'll be talking about dreams and creativity. What comes to mind is this chap, this Indian chap, Ramanujan. Have you heard of Ramanujan before? He was an interesting guy. He lived in India, it was about 100 or more years ago, and he was extremely poor. Extremely poor, had no access to things like books, to libraries, barely any money; poorly educated, couldn't really afford to go to school.
- 6.2** But he had one passion: math. This was one passion that he had, math. But without access to libraries and books and journals and Google and all that, what he would do was he would scribble down his math on the back of envelopes all day. So that's what he would do, scribble them down. At some point he thought, "Well, let me contact mathematicians out there in the world."
- 6.3** So he would contact some of the leading mathematicians of the time, like G. H. Hardy of Cambridge; he was a leading mathematician with John Littlewood. So he'd contact these guys through letters and send them. God knows how long it took back then for the letters to arrive, but he would send the letters, and most people thought he was a crackpot.
- 6.4** I mean, what is this? These weird scribbles look like something from another planet. I mean, they didn't look like the math of that day. They looked different. They looked startling, like from another planet, almost. Hardy had a look at them, initially dismissed them. Then he went for a game of tennis with Littlewood, so they were playing tennis. And as they were playing, he suddenly had the epiphany, the opening. He said to himself, "If these scribbles are not true, none would have had the imagination to make them up." Okay?
- 6.5** So there was something, there was some truth there that he had to investigate. So he went back to his room in Cambridge, in Trinity College, Cambridge, and he looked at the scribbles, and it turned out they were actually on point. They were true. This was actually true math here. He looked at more of his stuff, and eventually, he had to invite this Ramanujan, this poor mathematician, with no access to education—no books, nothing, just his own mind—and invite him to Cambridge.
- 6.6** And it turned out this Ramanujan revolutionized math, and all his math was on point. It was like, to this day, things like black holes and things like the deepest math. I'm not a math person, okay? But he has revolutionized math and things like astronomy and things of that nature. And it was all in his scribbles back then. A true genius, a true genius working on his own. The idea of a lone genius, well, here back then, you actually had one, Ramanujan. It's quite something.

- 6.7** Now, here's the kicker, and the point of all this. Ramanujan would claim that he would get his scribbles, his math, from his dreams. From his dreams. Can you believe that? Isn't that crazy? He had the village goddess, or something like that, whisper to him the math and tell him what it was. And back then in Cambridge, nobody understood. He was like, they were like, "Tell us what your scribbles are. How do you get your math? How do you come to these conclusions?" And he would say, "I don't know. I have no way to prove it. I have these dreams, and it turns out it's right. What do I do about this?" And they couldn't believe it.
- 6.8** He was so profound in math that this guy that was the leading mathematician said, "On a scale from 1 to 100—innate mathematical abilities—I, the greatest mathematician in the world, am about 20. Ramanujan, this guy working alone, is 100. And yet he has his dreams inspiring him.
- 6.9** So Arnold, can dreams inspire us in that way? What is that about, right? What is that about? What happens in dreams that would lead to this conclusion? And by the way, Ramanujan is not alone. You had people like—things like, what is it? Edison, Thomas Edison, right? You know Edison? What he would do—he would sit down, and as he would fall asleep, he would have a steel ball in his hand. Sometimes he would hold a spoon, and on the floor, there would be a plate. He would hold it like this, and as he was drifting off, the spoon would fall down, and he would wake up and write down all his dreams. And that led to a lot of his discoveries as well. So that's what he would do.
- 6.10** And Salvador Dali, the painter, would do some stuff like that as well. There are other people as well. The guy who invented the periodic table, Mendeleev. He claimed that he was awake for three days and three nights. And suddenly in his dreams, the answer just fell into place, and he was able to develop the periodic table. Crazy. Mary Shelley's Frankenstein came— was through dreams. The person had, like Mary Shelley, who wrote this novel, had a dream about the Frankenstein monster and how it would unfold. So in literature and science and in chemistry, there's all these examples.
- 6.11** So what's going on in the brain? Well, we know the prefrontal cortex, the CEO that allows for logical and cohesive and linear thinking, shuts down during REM sleep and during dreams. That's an important key point because it allows us to think in a more divergent, in a more open, in a more fluid manner. Not being imposed by the rules of logic and algorithms of correctness and rigidness, just being open, seeing things from novel perspectives and novel vantage points.
- 6.12** The other point, the other neurochemical explanation—I've mentioned this several times—which neurochemical is not available during REM, and hence dreams? Starts with an "N". Noradrenaline, right? The stress chemical, that one that makes our heart pound [sound], and all that. So when you don't have that, you are less focused and less narrow in your mindset. And so for that reason, you can look at things from a more flexible vantage point and a more less focused manner, less rigid manner as well.

- 6.13** So these are the two key components: the prefrontal lobe, the CEO of the brain, shutting down, as well as the neurochemical explanation being the noradrenaline producing neurons, not producing noradrenaline, and therefore, you are more flexible and open in your way of thinking.
- 6.14** And this explains, too, I mentioned before that REM, the whole thing about REM is seeing things in a bigger picture and bigger perspective; seeing the gist of things, seeing the core essence, extracting the wisdom. And also just seeing the—like being able to have creative insights. Maybe I can help elucidate using an example.
- 6.15** So one study, they had people sleep, and another group was staying awake. They just had about a 90 minutes nap—which is often in experiments that way, 90 minutes, so you can go through a complete cycle of sleep. Then they show them words like "osoeg," okay? That's an anagram for "goose". They show them this word, and they have them solve it; they have to solve it as quickly as possible, that's the whole idea. They find consistently that people that have REM sleep are much faster at solving this. I'm not talking about sleep-deprived people that need sleep because they're fatigued. We're talking about just being able to solve these solutions much faster.
- 6.16** Or they will have people do virtual reality-like mazes on a computer screen where they have to get out of the maze and find the exit. And again, they have them stay awake and then have them sleep. Upon taking a nap and having REM sleep, they will be significantly better, significantly better at solving it— much, much better at solving it. And you know what, Sonia? If they dream about the maze and elements of the maze— so if they see themselves in their dream trying to solve it—they will be up to 10 times better at solving it. Just from REM. So this shows us that REM sleep is important for creativity, having insight, having that spark, what you call seeing the bigger picture in things.
- 6.17** And this is a side note. I often wonder: How do geniuses dream? Imagine you're already creative, you're already a genius, and we use this word "genius" loosely. What is a genius? What is really a genius? I think I met a couple of, maybe one genius in my life. And a genius is not somebody who is fast at thinking and has a lot of book knowledge, but it's somebody who truly sees the gist, has a creative spark that you can't really pin down with words.
- 6.18** It's not merely being able to link things in an unrelated manner, in a quick manner and being just out-of-the-box thinker. It's more about some kind of poetic spark. Like when Tagore, the poet says, "a teardrop on the cheeks of time," an evocative metaphor that just makes you think— something like that. Or Shakespeare—Shakespeare's some of his sonnets, there's true creativity there. It's not merely mixing up words, and it's really that which REM produces that ability. That's what it's about.
- 6.19** But what about a true genius, somebody who has this ability to be creative in this way? How do they dream? Do they have hyper-creative dreams? Do they see themselves

solving things in a manner that us, mortals, would not be able to? Or what do you think? I don't have the answer to it, but what do you guys think? Perhaps, right? Who knows? Larry?

6.20 Well, I don't want to argue, but I have an opinion of what I think genius is. I think a genius is a statistical anomaly in a series of disciplines. I think some people are physical, athletic geniuses. Some people are musical geniuses, some people are mathematical geniuses. I think when you get up and look in the mirror in the morning, you're technically looking at a genius, and most of us don't get to do that. But I think that—you asked a question, so I defined what I think a genius is, and I'll probably be corrected here. But you asked a question about genius, and I had an opinion, but I forgot the question.

6.21 What is a genius? That's what I think.

6.22 I think it's a statistical anomaly of a certain—one in 50,000 or one in 100,000. I think in my rational mind, it could be one in a million. But I think it's technically an IQ of 142 is what I think a genius is. But I think there's more than intellectual genius. I think there's a series of levels of genius.

6.23 You know what I think, Larry? I think for me, I was fortunate to be at Cambridge—I was at Trinity College, Cambridge—the same Cambridge College that these guys went to. I was also—I saw some really smart guy, and every morning I would go and I would see this statue of Ramanujan, Isaac Newton, next to me. I was very inspired by this. And I thought, what is genius? And Newton had an answer. He said for him, he could solve things by merely thinking about it. So by thinking about stuff over and over, he almost became a genius. I don't buy it.

6.24 I think genius is some indescribable ability to look at things from this different perspective. So as I said, at Cambridge, I had mentors and colleagues that were at the top of the field of science and medicine. Top, absolutely. Nobel Laureate levels of knowledge. But I didn't see genius there. I saw a lot of experience. I saw a lot of hustle trying to be at the top, but they lacked some creative spark that you can't explain and can't put words on. Ramanujan had that. He was something else. He came from nothing and just literally went and was 100 times better than the top mathematicians. That's genius for me.

6.25 I have my mentor in the Indian guy I mentioned before, Ramachandran. Some of you guys might know him. He's probably the greatest neuroscientists of the last 100 years. I was lucky to have him as a mentor and a friend. Very lucky. I mean, not that it reflects anything good in me; I thought he saw a good companion, a friend. I saw genius in him because he was different. He was childlike.

6.26 He would forget where he'd park his car every single day. "Where's my car parked, Jalal?" He would call me Jalal. He wouldn't call me Baland for some reason. It's Jalal. "Where's my car, Jalal?" He'd walk about, and he was just different. He was like a child.

He was childlike in his creativity. Please go back and watch his lectures on YouTube or something. This guy, he's a pure genius. He made discoveries. He would sit down and just come out with scientific breakthroughs from just sitting. He was actually from the same town as the Ramanujan. So maybe there's something in that gene pool.

- 6.27** But yeah, I agree. It's hard to put down what it is and what genius really is. But yeah, somebody like Messi, the soccer player, is genius for me. I love him, for example. Do you like Messi? Absolutely. There's something different when he's playing. Anyway, we're talking about dreams here, so we have to go back to our topic.
- 6.28** So that's a huge part of the whole creativity and dreams. And I think the dopamine point is also important. We talked about dopamine and dreams, and if you cut the circuitry of dopamine, then you can't dream. You lose the ability to dream. So dopamine is very important. And dopamine is also crucial in what you would call the flow state. State of flow. Do you know about the flow state? It was described by Csíkszentmihályi. I should know this. I worked with him. Csíkszentmihályi, he's one of the cofounders of Positive Psychology. He developed the concept of flow.
- 6.29** Flow is when you are intensely engaged in an activity like soccer or tennis, and you forget about space and time. A lot of dopamine there. It's almost like the prefrontal seems to be shutting down. You are not focusing on the world and all your distractions. You just engage in that one activity. There's something about that that's very similar to the dream state and creativity. The ability to say, "Look, to hell with everything" and just be focused and intensely, emotionally immersed with your dopamine circuits firing on all cylinders on this one thing and just forget everything else. I think that's a part of the creativity process. I think flow—the flow state of losing yourself is key. I think it's key to understanding what creativity and dreams are like.
- 6.30** Although REM sleep, Brance, is this stage of sleep where we associate dreams and creativity and say that dream is the creative state and that all happens in REM sleep. That's what we're talking about this whole time. But it turns out there's another stage of sleep where you are also hyper-creative. And this comes maybe as a surprise to some of you guys, but it's interesting.
- 6.31** I mentioned before that Edison and others would use the steel ball and the spoon and all that and wake themselves up. But obviously, if you're waking yourself up, initially at the first stage, like when you are falling asleep, that's not REM, right? REM is much later. REM happens later. So what stage are you tapping into here? Stage 1. Stage 1, Larry.
- 6.32** So it turns out Stage 1 of sleep, the very first stage of sleep, is important for creativity as well. Some call it the sweet spot of creativity, and that's a big term and big words when you're competing with the REM. So that's big. But the thing is that during Stage 1 of sleep, your cortex, your CEO, conscious agent, your sense of self and awareness is shutting down and you're entering sleep, but it's not completely shut down.

- 6.33** So you have this mixed state of being awake and asleep at the same time. In some ways, like consciously, you're not fully asleep and you're not fully awake. And so for that reason, people will talk about it as a sweet spot. And if you wake people up during this state, they will have more creative insights. They will have more solutions to problems that require creativity and creative of spark. So that's Stage 1 of sleep.
- 6.34** All right, so I'm going to go into a deep one, but before I do, I want to tackle a couple of questions. I had a good couple of questions. One was—this is from the previous lecture for context—why do we sometimes have reoccurring dreams? Why do dreams reoccur? And this could be me seeing myself, that my teeth are falling out. That's one, having that. Or it could be me running late for an exam. Have you had that before? Running late for a meeting? It turns out it's one of the most common dreams that people have all around the world. Being late for an exam, trying to run and being late. Teeth, have you had the teeth? You've had that? Those are common ones. Obviously, the legs being very heavy, and we'll go into that. That's a whole lecture. But so there are common dreams that are reoccurring.
- 6.35** And scientists will tell you that the reason for this is that your circuitry in your brain, when you initially, let's say you're a child and you're three or four when your teeth are falling out, something like that. So your teeth are falling out. That's a pretty emotionally, I want to say, traumatic, but it's definitely an overwhelming experience for that child. You look at it as a small thing, but I've seen my niece, when we chase her with like, "Come here, we have to look at your teeth and try to pull it out." So when it's like, we run around, she's hiding, we can't catch her. But it's a big thing. Do you want your tooth to be pulled out? No, right? So it's wired into the circuitry there.
- 6.36** It's in your fear circuits, and then you forget about it. But later on, it keeps haunting you and you don't know what's going on. But lo and behold, it's that memory, potentially, that's resurfacing when you're an adult, when you're older, and you keep dreaming about having the same dream. And that goes again for the anxiety and being late for an exam, well, exams are pretty evocative emotionally. So it makes sense that you would have that. You potentially could have that being burned into the circuitry and for that reason have these reoccurring dreams.
- 6.37** So it seems the circuitry and the emotional salience of experiences can account for a lot of this. And a huge point of this is that, and I want you guys to know this and go away with this, is that emotional content, things that are emotionally salient and fearful and arousing and capture your attention, that is much more likely to activate the circuitry in the brain that wires it together and makes it more plastic.
- 6.38** So plasticity, the idea that the brain can change and fold itself after experiences. Well, not everything in the world will change us. If I go about my day and just like a robot and on autopilot pilot, I won't wire that into my circuitry necessarily. It's more when I have an experience and that experience captures my attention and typically might be

emotionally arousing, right? I need to wire that into the circuitry. Does that make sense?

- 6.39** So if I walk about here in Scottsdale, and I just go about the mall and I see things around me, the brain doesn't really care to wire that down. Why is that important? But if I walk around and I see a lion, does it make sense to wire that down? It does. You have full attention on that lion. It pretty much captures attention, and then you have certain neurochemicals like acetylcholine being ramped up, which makes your brain more active. Dopamine will also potentially make your brain more plastic. So you have that.
- 6.40** So you can see now, if you have a major event like your teeth falling out or exams or other things—that is common for most of us—that would be wired into the circuitry and resurface in dreams as these reoccurring nightmare-like dreams. And I wouldn't call them necessarily nightmares. Nightmare category, that category depends on an intensity of fear. But so that would explain it.
- 6.41** What would happen if you were to wake from a dream? Let's say you're dreaming and you're sleeping, and you suddenly become conscious inside your dream. Could that happen? You can meet all the people you want to meet, you're engaged in all the fantasies you want to engage in, right? And one of the most common themes in these dreams are often things like sexual scenarios, flying. Call these lucid dreams. In fact, about 20% of the population, so they will have a lucid dream at least once in their life, so they have these lucid dreams.
- 6.42** And they're often triggered by something strange and weird happening inside the dream. So if I, to use the lion example, I walk about and I see a lion, that captures attention. I need to shift my whole way of thinking. In a dream, to some extent, a similar thing can happen. So if I walk, if I am in my dream and I experience suddenly speaking to my ancestor that's dead, or William Shakespeare, even though dreams are already bizarre, so the bar is pretty—it's quite difficult to experience something unusual in a dream already.
- 6.43** But sometimes that can trigger consciousness circuits in the brain. Again, the outer layer of the brain up here, they become hyper-activated, and they're supposed to be hyper-activated when you are awake and not sleeping. If they become hyper-activated while you are in REM, you start becoming consciously aware inside the dream and have this lucid dream. During lucid dreams, and by the way, you can trigger lucid dreams by using things like goggles and things like that.
- 6.44** So one way to trigger lucid dreams, and I've had a lot, I've had a few lucid dreams in my life. I have a lot of them. Very interesting. But the way you can trigger lucid dreams—when I was a kid, I would sleep in my room and I would often tell myself, I said, "Look —" And I didn't know anything about lucid dreams. I didn't know what lucid dreams were or anything like that. I was five and I was like, I want to become aware inside my

dream that I'm dreaming so I can fly. That was my thing. I wanted to fly inside my dream like Peter Pan. That was my dream.

- 6.45** And so each night I would say, "Okay, Baland, tonight is the night. Okay, prepare yourself. When you're dreaming, become aware that you're dreaming." And I did this over and over. And eventually it hit. It hit, and I flew in my dream. It was a magical experience. I still remember it. It was quite profound. I had a lucid dream as a kid, and it just stuck with me.
- 6.46** And that's actually one of the tricks: tell yourself before you're going to bed that this is a dream, that you are trying to wake yourself up during the dream, become conscious during the dream. If you meet people during the dream, that that should be a sign that you are dreaming. If you meet a weird stranger or something like that, that's impossible in real life. Make that a cue for you so you can jolt yourself out of the zombie sleep state and become more consciously aware.
- 6.47** Other times, as I said, the goggles. So you have a system, where you lie down and you see, you train yourself. I watch like, "bling, bling, bling, bling, bling," like three or four times of light flashes. Then take on the goggles, enter REM, and you see the flashes in your dreams, and you go, "Oh, my God, this is a sign that I'm in REM sleep dreaming. Become conscious now, it's lucid. Then you can generate a lucid dream. Isn't that fascinating? Have you had lucid dreams before? No? You have, Arnold. Lucid dreams are interesting.
- 6.48** Do you know that it's literally that part of the brain up here, the DLPFC, the dorsolateral prefrontal—meaning here, the prefrontal—the sides here, the calculated part of the brain, that part. If that part of the brain becomes a little bit too active, a little bit too activated during the dream state and REM, you literally float into the lucid dream. It's also important to know that the more active it becomes, Brance, the more lucid you become. The more active these structures, the more lucid.
- 6.49** In other words, it's not merely an on-and-off phenomenon. It's literally—it has degrees. I've had lucid dreams where I knew that I was lucid and I thought, "My God, I want to visit this person, and I definitely need to visit this person over here. I want to do some mischievous things here and all that." So I had all these plans. Right, Heavenly? I had all these plans. When I couldn't really do it, I was lucid and aware of the fact that I was dreaming, but I wasn't fully in control.
- 6.50** So we differentiate, we say, okay, lucid dreams, you might have awareness that you're awake, or you might have full control of the scenario. So there's these two things. And it's rare that you have the full thing where you can just do whatever you want and just create paintings and and see like, glorious images and flowers and fly and do whatever you want. I'm over romanticizing it. People are pretty lustful, but I'm just saying. Instincts, what can I do? So that is the situation that's in lucid dreams. And the more

the DLPFC, the more this region becomes active, the more you have that control, the more you have a sense of volition in the lucid state.

- 6.51** For a long time, though, interestingly, they thought that lucid dreams can't be real science. Lucid dreams? What is that about? Is that a real phenomenon? But now it has been shown that actually the guy that's sleeping and dreaming away, you can you communicate with him, Sonia. Can you imagine? He is dreaming in REM and he's saying, "I'm lucid." And whenever he feels like he's lucid, he will move his eyes like this: two times mean "I'm aware now." Two times mean, this way, "I'm moving a stone and I'm jumping over here." So you can do that now. You have that. Can you imagine?
- 6.52** So communication with the sleeper and the dreaming person has now been shown. And through brain scans, we confirmed that the person is actually in REM, having lucid dreams, and then you measure the muscle activity of REM paralysis in the eyes, and you communicate with them using these Morse codes. It's quite something.
- 6.53** Now, is there ways for you to get more lucid dreams without these methods? Can you think of any? Well, the thing is about lucid dreams, the way that I think about lucid dreams and lucidity is that the brain—the more you train it to have that consciousness switch early on as you're falling into sleep, the more you have that, the more you can potentially become lucid. And that goes not only for things like lucid dreams, but sleep paralysis as well, when people jolt themselves out of sleep paralysis.
- 6.54** The more you make yourself able to control and harness dreams and activate that circuitry, the more likely you are to have lucid dreams and have lucid states of dreams. That's one big part of it. But there's no way to actually become a better lucid dreamer. There's no actual thing you can do other than keep trying, using that circuitry over and over again. That's yeah.
- 6.55** I want to talk a little bit about the monster in our dreams and what that means. Now, I won't go into the full details now, but I want us to get into it. In your dreams, you can feel like a monster is chasing you. You can have that reoccurring dream occur over and over. So what is that about? Why do people see a monster chasing them? Why is the monster coming at them, attacking them, and they feel like they want to run and they want to move? So what's going on in the brain?
- 6.56** It turns out that in the brain, you have a place for volitional movement, the motor cortex, the motor regions. And so the emotional core of your brain, the amygdala, represents the monster chasing you, and you want to get away, and you can't. You feel like you can't get away. You want to move, but you can't. And the question is, why can't you get away?
- 6.57** Well, the first layer is that the emotional part of the brain, the limbic, the instinctual part of the brain is simply dominating and outcompeting the motor regions of the brain. There's too much flair there. The 30% active amygdala, 30% more active than wakefulness, it simply outrivals the motor regions of the brain. Even though you're

volitionally trying to get away, you can't. You feel stuck. You're all stuck there, right? First thing.

- 6.58** Second component is the following. The second component is that in the brainstem, in the lower part of the brain, we talked about that yesterday, you have a region for automatic movements. Automatic movements, meaning movements that you cannot control. These are sporadic signals in your brain firing whenever, sporadically, in the REM state. And they are also competing and dueling it out with volitional movement.
- 6.59** So on the one hand, you have the emotional core, which represents the monster of your dreams. Of course, that's being colored by the memory parts of the brain over here, telling you what the monster looks like and all that. It says it looks like this and that. It has all these features. And then that is spilling into the emotional core. And then you have this monster. You can't get away. And at the same time, your automatic movements make it feel like you're stuck and you can't move because you're being controlled. So far, so good.
- 6.60** Here's the interesting part, though. What you can is, this guy in this state, having this monster-like nightmare that's so common around the world, that's reoccurring in all kinds of scenarios—you've had this dream, probably, Cameron. Well, many of us have had this. If you go and you activate the dorsolateral prefrontal cortex using a TMS machine, you zap it like [sound], you zap it up here, you make the chap lose it in his nightmare, in his whatever situation he's in with this monster.
- 6.61** So he now feels like he actually can move and do whatever he wants. He becomes a lucid nightmare person. And at that point, he can turn around and approach the monster. And he can approach the monster now and do whatever he wants with the monster. He can kill it; he can go away. You see that? So by using technology like stimulation of the sides of the brain, making you more lucid inside this nightmarish situation, you can tackle monster.
- 6.62** Now, we know when you as a person engages in what we call approach behavior—approach behavior is that whenever I feel confident and I feel strong and not anxious and not stressed is when I am engaged in a lot of approach behavior. I deal with the world in an approach manner. I am proactive. I do things. I don't have that withdrawal, what you call withdrawal behavior, which is like being timid and shy and being like drawn inside of you.
- 6.63** And one of the ways that you can actually engage to hold approach behavior system—you know what that is? How you can engage that system? By simply walking and facing your fears. So whenever you walk, what happens is you look from side to side—by looking, merely walking and looking, and the visual scenario being like this, you see the world and your eyes are—you have a scenario that's moving like this. This alone will dampen down the activity of the amygdala, that alone.

- 6.64 So now when you engage, when you approach that monster in a lucid-like manner, and you go directly towards it, you dampen the amygdala. And we know from behavioral therapy and therapeutic methods, whenever you face your fears, Arnold, when you have a fear that's deeply burned into the circuitry, whenever you face it and you engage with that fear and don't run away from it, you remove the circuitry, you eliminate the fear, and reinstate a sense of inner harmony and peace. I wanted to add that as a final note before we move to the questions and answers.
- 6.65 You have an extremely cosmopolitan background. I've got to travel to over 50 countries, but I don't have any command of language. But is English unique in using the word "dream" for an awake mental ascent? Because I think it's so important to have a successful life to manipulate your dreams and to embrace dreams. I've noticed that many Americans or many people don't dream. But is there programmable options from—well, I guess it's a two-part question. Is dreams usually split in other languages from your experience, night dreams from intentional dreams? And is there a correlation between the two?
- 6.66 I want to make sure I understand it. So you're saying, are there a differences between night dreams and day dreams?
- 6.67 Yeah, like dreaming as "I'm called to be a great professor, so I have a dream of going to Cambridge." That's a dream that I think that you've experienced and that you walked out. I think that It has huge impact on people. But the English language, like in Greek, I think there's five words for love. We have one. Do other languages share "dream" as an awake state as well as a sleep state? Is that unique to English, or is that common across human experience? Then secondly, do you think that there's a positive learning application from the way that you choose to think when you're awake that can tag into some of the creativity that can come when you sleep?
- 6.68 It's basically learning from the dream, right? It's interesting. So culturally, I had a mentor once. He said—I was talking a lot about my dreams. I was saying— this is back in Copenhagen, my very early days when I started to read books and all that—I went to him and he advised me on reading certain books. And he said, and I told him, "Look, I have these dreams. In my dreams, I see these flashes of pages." I have an okay memory. It's not that good, but it's okay. I saw flashes of pages of text in my dream like this, like flashes of—I would read it. It was quite bizarre because it was quite intense.
- 6.69 I went to him and said, "Look, I have this." And he said, "One thing, always remember, don't blend the barrier between wakefulness and dreams. If you do that, you're in a very strange place. This is the land of what you would call becoming psychotic." So psychotics, for example, the psychotic state of being psychotic, being unable to distinguish reality from dreams, that's a very terrifying state. And it mimics a lot of the dream state.

- 6.70** I mean, think about it. When you're dreaming, you are hallucinating, you have amnesia, you have delusions, you think things are real that are not real. You have all these things that mimic the psychotic state. People that have psychosis, an active psychosis, seem to be living in a dream-like state.
- 6.71** Now, I mentioned yesterday that dreams are so powerfully bizarre and strange and intense that the brain has a system for shutting the door of accessibility to dreams because otherwise, you'd be all overwhelmed. So what you remember is the tiny fragment of the final portion of the dream. Point of all this is that you don't want to be living in a world that mixes dreams and reality too much. That's my answer.
- 6.72** Now, can you harness the creativity from dreams? Can you do that? Can you take things out and extract points? I think you can. I think, as I mentioned, some of these guys have done—like Mary Shelley and Frankenstein and others—have really done that, and have done that myself. Some of my theories, actually, in my research, came from me in dreams. I saw dreams of certain things, and I thought, "My God, this I could use."
- 6.73** In fact, my therapy for sleep paralysis came in a sleep paralysis dream-like scenario. I had a ghost. I had a friend who passed away, and he came as a ghost in my room, and I was quite terrified. And so I thought, "Okay, let me apply some of these steps." I didn't have a recipe at that time, but I applied some of these steps. And lo and behold, the whole experience changed and the monster, my friend turned into a monster, disappeared. So I think you can, yeah, you can harness dreams in that way, and a lot of people do. Great questions.
- 6.74** So you said that REM sleep is helpful for big-picture thinkers. I noticed, I guess, when it comes to Richard Dawkins, intellectuals like that, they're more parts—they think about reality and its parts more than big picture thinking. I'm just trying to think, maybe, is it possible that people like him aren't getting that REM sleep?
- 6.75** I see where you're going. I want to try to interject you. I see where you're going, right? What it is is that, look, the left hemisphere is engaged in event registration, small details, parts. That's the left hemisphere. If you get a stroke to the right hemisphere and the left hemisphere becomes large, you become like a savant, a calculating savant, for example. Right hemisphere: big picture, artistic, poetic, spiritual even, one could argue, right?
- 6.76** Now, that mimics the dreaming state. And it turns out that normally, the left hemisphere dampens the activity of the right. It dampens it so you don't have too much poeticism in your daily life. You don't want to walk around like a crazy poet and just being poetic when you're supposed to fill out a DMV thing. You don't want to do that, right? So you don't want to be overly poetic. You have to be pragmatic. And so the left hemisphere is dominant in that way. Mind you, this distinction is slightly oversimplified, but you get the gist of it.

- 6.77 Now, it turns out in dreams, for example, that inhibition of the left hemisphere and the parts that are looking at events and details seems to inhibit the right so it doesn't get over excited. And so in dreams, that flow of left-to-right of information is shut down a little bit. So the right and left are balancing it out, and the right can flourish and express all its content. So there is that.
- 6.78 Now, somebody like Richard Dawkins, for example, yeah, I do think that people in the extreme scientific worldview can sometimes be so hyper-focused on the details that they miss the bigger picture. They focus so much on details of the theories and the parts that they miss the overall grand scheme of things. Running around and saying, "Well, life is just these molecules and atoms," and all that, but there's no maybe greater purpose or whatever. And they miss the bigger picture somehow. And I think there is this left and right asymmetry in the brain in terms of activation, where, arguably, the Dawkins types are more in the left hemisphere.
- 6.79 Yeah, that's a good question. All right, thank you.

Lecture 7: Monsters in the Mind

At a glance — 'Monsters in the mind' — the frightening figures of dreams and sleep paralysis.

- 7.1** So today we'll be talking about monsters. Dream monsters. You know what happened last night? I was sleeping in my bed. Then suddenly in the middle of the night, can you believe this? I woke up. Complete darkness around me. And I thought, what now? I went to bed early, at 9 PM. No parties outside my room or anything today. So I was good.
- 7.2** And then I woke up and I thought, "What is this?" I looked around in my room, and I saw something in the corner, but there was no monster there. So I just woke up and I started to walk. Went to the restroom. I thought, "Maybe I just need to use the restroom." I had a lot of water with my dinner. So I thought, "That's it. I'm just going to go to the restroom, come back to my bed and just sleep." Get back to good, nourishing, restful sleep.
- 7.3** Now, at this point, it occurred to me. I said, "Look, I didn't tell them yesterday that if you drink a big glass of water before you go to bed, then you can remember your dreams much better." That's actually a technique. So certain people, certain scientists will recommend, if you want to remember your dreams much better, to drink a big glass of water just before going to bed.
- 7.4** And the reason for this is that when we drink that water, we have basically more interruption in our sleep. Our sleep becomes more fragmented. And we said yesterday that dreams are forgotten, and we remember the final portion of REM at the very end. But imagine now, Larry, if your dream is interrupted all the time, your sleep is interrupted, so you go in and out of REM, well, then you are more likely to remember your dreams. Does that make sense?
- 7.5** So I thought about that, and that was actually not a good thing to think about because then I got even more aroused. And it occurred to me that in order to fall asleep, you have to have serotonin levels and noradrenaline levels fall. Why is that the case? Well, noradrenaline is the chemical that keeps you aroused, that keeps you fresh and alert and energetic and vibrant. If you have a lot of that in the system, you can't shut down and go into the stages of sleep, like Stage 1 and 2 and 3 and so forth. In order to fall asleep, you have to dampen the activity of these neurotransmitters.
- 7.6** But here I was thinking about all these things I forgot to tell you, so these chemicals were going up. And then I thought, "Well, I forgot to tell them, and it just came out of the blue. I forgot to tell them about plasticity in dreams. And this was annoying because now I couldn't go back to sleep at all. It was 2 AM or something, the depth of the night."

- 7.7 Now, plasticity and dreams is the idea that—look, we mentioned yesterday, the tiger and all that. You saw the tiger, and you encode that information in your brain, right? That's unusual. And anything that's unusual, you code. You have attentional focus on that thing, and you encode that in your brain, and that become plastically wired into your brain. So your brain is hyper-plastic. And the more you use your brain in certain ways, the more you encode new information.
- 7.8 So the more I, for example, let's say, I like to play soccer. I like Messi. I like soccer. So the more I play soccer, there's a motor strip right here on the cortex of my brain, a motor region right here. And on it is a vertical strip of my entire body on the surface of my brain. And it's actually on the other side. So my right side is on the left and vice versa.
- 7.9 The more I use a certain skill, for example, play soccer, the more I do that, the more my feet area and my—if you use my thigh, for example, or if I use my head and all that—those areas in the map will expand and take over adjacent areas. So we know that. It's a use it or lose it brain. The more you use a certain skill, Naomi, the more those neurons will become ingrained and expand, literally like an invading army, and take away nearby cortical regions, nearby brain regions.
- 7.10 Now, this is relevant for dreams. Why is that relevant? Well, it turns out in the Stone Age, the caveman was living in a bloody cave. And in this cave, was there a lot of electricity and iPhones and games and Pokémon Go and all that? Or was it completely dark? What do you think? Completely dark, right Sonia?
- 7.11 Now, which sensory modality is the most important of them all for survival. Vision is the most important. If I go around on the savanna and I want to spot a lion or a gazelle, or I want to chase a mate behind the bush, having vision is critical. It's of key importance. Without vision, I'm pretty much a lion's next meal. I mean, that's what it is, right?
- 7.12 So, vision. In fact, we have 30 areas in the back of the brain dedicated to vision alone. One-third of our cortex is dedicated for vision, right? So the cortex is the outer layer, we said, of the brain, the six-layered pie-like structure out here. Now, so if you don't have electricity in the cave, you don't have iPads and iPhones and i-this and i-that, then what? Your vision, your visual cortex, what happens to it if it's a use it or lose it brain? Do you use your visual cortex in the dark? You don't, right?
- 7.13 So the risk here is that it might slowly shrink. It might slowly become smaller and smaller. I mean, think about it, it's pretty dark. I'm from Copenhagen, Denmark. It's pretty much dark. The whole winter, it's dark at 4 PM. And then the sun rises at, say, 10 AM next day. So you have darkness upon darkness upon darkness. Your visual cortex would shrink as hell, right? It would become very small. One recent theory suggests that in order for our visual neurons to survive, we have to sleep, and more specifically,

we have to dream, and we have to do that for that many hours each night. That's one idea of why we dream.

- 7.14** Do you know that creative people are more likely to remember their dreams? That's well known, that's been studied. I wouldn't say well known, but there's research for indicating that. If you are more creative, you are more likely to remember your dream in the first place. The thought is that, well, we don't know why that might be the case, but there's something here about having a more creative brain. It seems like creative people tend to be more of lucid dreamers as well. There's a link there as well.
- 7.15** The question is: did lucid dreaming, for example, come about because it makes a certain group of us, like we said, 20% of the population more creative? They have this virtual reality testing space within the lucid dream where they can do all these activities without any fatal consequences and being lucid and having volition at the same time. It's a perfect testing place, wouldn't you say?
- 7.16** There is actually this idea that perhaps having certain people—they are more creative and then can use these lucid dreams for their own exploration. And then, of course, they seem to be able to remember their dreams more. And we don't know. It could also be that their dreams are just more creative, and therefore, they remember them much better. That's a possibility, too. So we don't know that for sure.
- 7.17** Now, as you can imagine, my sleep at this point was completely out of the way. I was lying there and lecturing to myself. I was going there and I was just seeing myself lecturing and telling Larry this and telling Sonia this and telling Arnold this. It wasn't good now. And I thought, "Well, now I'm in trouble. What do I do?" And so I thought, "Okay, the fancy restaurant from the hotel yesterday, I drank a lot of water, but I also had this tiny fish." I asked for fish and they gave me this little piece. I get hungry. I need some protein. And I thought, "Maybe that's it." When you are in a state of ketosis and don't have enough glucose in your system, what can happen is that your sleep can become a bit interrupted.
- 7.18** Why? Well, there's a overall—when you are in a fasted state or semi-fasted state, you have a rise in these stress chemicals, noradrenaline and so forth, just go up, cortisol and so forth. And so I thought, "That's it." Let me get up and get some walnuts. Nancy got me some very nice walnuts, and so I thought, "Well, let me eat some of those. They might help." So I just woke up and grabbed some walnuts and ate them. I had a big mouthful of them, gobbled them down. I'm very good. I went back to bed, and I thought, "What do I do now?" So still, I couldn't sleep at this point. But what occurred to me is that I hadn't told you about dream diaries either, did I?
- 7.19** So we said serotonin and noradrenaline in the brain are absent during REM. The memory part of the brain, the sea horse-like structure behind your ears, the hippocampus, this structure is important for accessing and placing new memories. When you encode new memories in the system, you need the hippocampus. Because

of serotonin and because of noradrenaline not being available during REM sleep, the hippocampus cannot encode new memories. All right? All right. So far, so good.

7.20 By the way, there's this idea that we actually do store our memories in dreams. We just don't—the dreams, we actually do store them. All of our dreams are stored, but we just don't have access to them. And that's why perhaps sometimes you might be walking in the forest or you might be taking a shower, and then suddenly you go, "Oh, my God, I remember that dream from yesterday or a week before." So people talk about that, but it's not been studied carefully, and we don't know for sure. We don't know for sure.

7.21 Anyhow, I was saying, so you have serotonin and noradrenaline not being available in the REM state, so the hippocampus cannot encode new memories, right? But it turns out when you wake up in the morning, you have a surge of serotonin and noradrenaline. In fact, that is part of the wakefulness state. What is wakefulness? We know what sleep is now, but what is wakefulness?

7.22 Wakefulness is basically noradrenaline and serotonin flooding your sensory regions. You have structures in the base of the brain, in the brainstem, dorsal raphe nucleus, fancy name. I'm just going to throw it in there to show I'm smart. And the locus coeruleus here. These will produce serotonin and noradrenaline going up to the sensory regions, we said the cortex, and flood it at wakefulness. That flooding of noradrenaline, you can imagine, it would wake you up. You have that the moment you wake up.

7.23 That's why if you want to keep a dream diary, Larry, that's the point you want to—right the moment when you wake up, that's when you want to write down all your dreams. You want to say, "I did this and that. I saw this and that, and I experienced this and that." In fact, ideally, you want to maybe use your phone to talk because it makes it more—you can explain all the details without—if you write, you might miss out some details. It's hard to really write down, "I did this, and this happened, and then this happened." But when you talk, you can do it much better. Then some people might even use this for information, for creative inspiration, understand their psyche much more. But that's psychoanalysis. I don't know much about that. You can analyze your dreams and so forth.

7.24 When I was in California, living there, I was studying in the laboratory of my mentor, Ramachandran. One night as I was sleeping, I saw this shadow on my wall. I saw the shadow and thought, "What is this shadow?" And I looked carefully. I looked twice, and I thought, "The shadow is actually the shadow of me." It looked like half Peter Pan and half me. I couldn't tell, really. But I sensed it was me. And I thought, "What is my shadow doing?" And it was a half dream, half wakefulness.

7.25 It was in between, like an in-between state. I thought, my God, this is interesting. This is important. It must mean something, the fact that I'm seeing my own humanlike shadow in this sleep-wake type of state. I had thought a lot about sleep paralysis at this

point, Heavenly. A lot of thought. What is sleep paralysis? Why do we see monsters in our dreams? What do these monsters mean? Do they have a significance? Why am I seeing a shadow-like figure, literally a human man-like shadow in my bed?

- 7.26** Then I thought, well, why do people all around the world claim to see shadow-like beings in their room when they sleep? And why are these shadows? Why shadows? And why are they often faceless? Why not just have a face there? So why do you see a faceless dark shadow approaching you from the corner of the room, climbing you, pressing on your chest, choking you, and so forth? What's going on there?
- 7.27** At that point, it occurred to me that perhaps given this shadow looked like me, I thought, "Maybe that shadow is me. That shadow is me." That's what I thought. It's a crazy thought. But then I drove to the laboratory and I had a chat with my mentor, my colleagues at this Center for Brain and Cognition, our center there. And we thought about this, and I thought about this, and I said, "Look, perhaps we all know that human beings have an internal body map, an internal body image, a neurologically rooted sense of self. How do we know this?"
- 7.28** Well, it turns out if I get a stroke to the right side of the brain here, a blow, a stroke here, it could be trauma even to that part of the brain, guess what happens? Often you have paralysis on the left side, especially if it hits the motor centers or the somatosensory centers, the regions important for sensing. It might feel like my hand is dead, or literally, I might have paralysis in the arm and the leg. So this is well known.
- 7.29** But in some cases, guess what happens? In certain cases, when people have a stroke, particularly in regions in the back of the parietal lobe, the place that has to do with body image—if you have the stroke there, specifically in a region called the SPL, superior parietal lobule, SPL, if you have it there, it hits there, you might say, "Doctor, doctor, this arm doesn't belong to me, but it belongs to you." So you develop these delusions where you no longer can identify with your sense of body.
- 7.30** Think about this. The person is fine in all other respects. You can sit down and play chess with him, and he plays computer and iPad, he goes on iPad, but he has one quirk, one delusion: the fact that his arm doesn't belong to him. Why? Because in that center, in that part of the brain, your sense of self, your body image, is mapped. It's mapped onto that space. It's mapped onto that space.
- 7.31** And I also knew that in a structure nearby, just the neighboring structure, also in the parietal lobe, it's called a TPJ—I'm just adding the names for the technical people. It's called the temporoparietal junction. This is the next by, in the parietal lobes. If I was to stimulate that region of the brain using an electrical current, then you might feel like your sense of self is hovering over you. It's above you out there somewhere. It's looking at you. You might even feel like it's mimicking your postures.
- 7.32** It feels like if I move, it's moving too. If you do like this, it's also doing the same thing. You feel like it's me, but yet it's not me. It's this mix of a shadow-like being that's you,

but not you at the same time. This shows you, again, we have a sense of a body image encoded in that part of the brain. This was interesting clues for me to think, "Okay, well, perhaps during sleep paralysis, something similar is going on. There might be something there that's equivalent, that's bringing out this body image."

- 7.33** Now, I thought about this further. I thought, "Okay, what do we know about the normal construction of a body image, a sense of a self?" Well, we know that me, if I, Baland, I want to build my body image, I move my hand from A to B in space like this, what happens? I close my eyes. Signals are sent from the motor regions, first from the premotor regions, signaling intentions, going to the nearby motor regions, saying, "Move, Baland, move." Signal's going down my spinal cord and I move.
- 7.34** As I move, as I move, signals from my joint, skin, muscles travel back, all the way back to my spinal cord, up to my brain, up to the parietal regions, telling my brain, "Look, this is your body image, Baland. You should feel anchored in this body here and now. Your body is out here. Your sense of self, the limit of the self goes to here." It's telling me how to update a body image. Is this clear now? This is how the normal body image is constructed in a normal wakeful person.
- 7.35** But then I thought, "Well, during sleep paralysis, you have this interesting state." You're lying there paralyzed. You can't move. So you're lying there, you can't move. And then you're sending all these signals—move, move, move, get out of here, which makes sense because you fear that—you think there's a monster, right? It's scary. The whole thing is terrifying. But during that state, because you are paralyzed—obviously, it's all involuntary—but you are paralyzed. And because of this paralysis, the feedback that would normally tell your brain how to build a sense of a body image is out of the picture.
- 7.36** Is this clear? I know this is a bit of a—you have to get your mind around it because even for me, you have to. But the thing is, normal body image, you send signals, motor cortex—move—and then you move. How does your brain know? How does your brain know where the limits of your body is? It does that by sending information from the moving arm and legs or whatever, back to the body image centers of the brain telling you, "Oh, this is your body. Oh, here it is. Oh, Baland, you're moving it now in this way."
- 7.37** This function is lacking during sleep paralysis, and so this decoupling leads potentially to the fact that you see—your own sense of shadow is projected out there. This is my theory for sleep paralysis that I proposed 10 years ago. But then you might say, Larry, says, "I object. Why the faceless monster? Why the shadow-like monster? I mean, your brain is smarter than that. It can hallucinate a completely complex monster like Freddy Kruger with all the details and scary elements of a burned face and all that. Why just a faceless shadow?"
- 7.38** In fact, during dreams itself, so during dreams, the monster you see in your dreams, they tend to have those details. They actually tend to be like Freddy Kruger and what

else you see in your dreams, like different monsters. They tend to have very specific visual details like that. The reason we think that is the case, well, that is because you want to be able to remember the monsters in your dream. That's an important critical function of monsters in your dream.

- 7.39** You want to see a monster and go, "Oh, I saw this monster. It was scary," and then go tell other people in the village. So it has this symbolic meaning, and you tell other people. So it has this—it's memorable. It has to be memorable in your brain. So that's one key feature. But the monster in sleep paralysis is faceless. And we think the reason that it's faceless is that— well, I think I've proposed that as you are sleeping, you are in a terrified panic-like state. And computation, vision, visual computation is very costly. When I see the world out there, I'm not actually seeing the world as it is. I'm hallucinating the world. Let me explain.
- 7.40** When I look at a table or a chair or this black screen over there, I've seen this a hundred times in my life. I've seen it. I see a screen like this all the time. So does it make sense for my brain to process that from scratch each time? So what it has is that in the memory parts of the brain, there are representations of a screen like that. So it's already there in the representation in the higher order memory part of the brain.
- 7.41** So when I see something like a screen, information creates an initial outline, goes to the back of my brain, the primary visual areas, the lower level sensory visual areas, the most basic visual areas, and says, "Oh, here's a screen. Here's a screen." And then information travels up to higher areas like things like depth, color. If it's moving, then movement. Higher meaning and memory. This is how it normally travels. And for a long time, we thought that's how it works. You just see it and it travels all the way back and then to memory and so forth.
- 7.42** But now we know there are as many fibers going from the higher area, like memory and meaning and representation of higher understanding, back to the primary visual cortex in the primary visual areas. Meaning when I see that, I merely create the initial outline. I know, "Okay, this is a screen," and then I say, "Oh, I've seen that before." So information travels quickly from the memory part of the region and says, "Oh, here, this is what you're seeing. This is it. You don't have to hallucinate. You don't have to see the whole thing. We just tell you what it is, and then you see it."
- 7.43** That's why people with a condition called Charles Bonnet Syndrome, when they have damage to the visual parts of the brain or even the eye and there's holes in their visual cortex— when they see the world out there, their brain will make things up because they don't have the sensory information coming from the world. It's lacking, it's messy, it's muddled up. And so their brain will just say, "Oh, this is probably what you're seeing. Maybe you are seeing this." So you go about in the daily life and they might see an elephant suddenly, and they see these exotic things or cartoon figures or things like that.

- 7.44 Their brain is making up from—the sensory gap that's there in their vision and in the visual cortex, the brain makes up for that. When I look at a table, I've seen a table 100 million times. My brain just extrapolates and says, "Oh, here it is for you. Here it is. I have a little bit of sensory data and I have tons of past information, and here's the table for you. Here's this and here's thatf."
- 7.45 Now, going back to our monsters. Computation is very, very costly. And so if I'm lying there in a panic-like state and I want to get out of there and I want to survive, this is my chance. I don't want to die. This could be fatal. The smartest thing right now could be what? Would be to think about where are the windows? Where are the windows in this house? Where's the door? Where are the escape routes?
- 7.46 I only need to know that there's a monster in the corner of the room. There's something evil there. That's enough information. So information travels to the initial visual areas where I just created an outline and I say, "To hell with the rest. I don't need all the details right now." My survival is important because your emotional core, your fear centers are flaring up with activity. You want to get out, you want to survive.
- 7.47 I argue pretty much that this is the reason, this is the explanation, that if you look at the whole visual system and the visual hierarchy and how the brain constructs your view of the world, then this makes the most sense. The fact that people see faceless shadows, that this would be the explanation. What do you think? Does this theory make sense? A little bit, right? That's what I think. I think it makes a bit of sense.
- 7.48 And then I thought, "Well, this is a problem because in some sense, it's a compromise." In some sense, knowing that having that information quickly where the monster is and all that and being able to escape quickly is good. But the cost is when you don't see the face, you become even more terrified, don't you? Are you more terrified of, let's say somebody was to break into your house and they were wearing a hoodie and you could see there was just a shadow? Would that be more terrifying or if this just like goofy guy came in? When the brain doesn't have sensory information, it often can run amok and just make up things and become even more terrified and more scared.
- 7.49 Then, of course, we tend to project agency and intentions into this being as well. As I mentioned in my previous lecture, it's almost like the monster knows so much about you, doesn't it? It just knows who you are, your intentions, your life, and it just wants to devour you. It's like it has a special lens into your brain. It's terrifying, scary.
- 7.50 And what we think is going on in our theory is that we think that the theories involved in creating other minds, theory of mind, is hyper-activated here. We talked about the mirror neuron system and sense of self, and the fact that these systems are highly active in REM. But more than that, these systems tend to be highly activated in this state in sleep paralysis. The reason is the following.
- 7.51 Normally, when I go about and I have conversations with people, I have a chat with Arnold, "How are you doing? Doing well?" "Okay, I'm fine." Then what I do is my mirror

neurons and my mirror neuron system, let's call it the theory of mind modules, they create a representation of your mental state. What are you doing? What are you up to?

- 7.52** Now, the reason why we're not completely blending and merging is because my prefrontal cortex, my breaks on the brain—remember the breaks in the middle? They are saying, no, this is a separate person. This is another person. It's an external agent. Then I have information from my own body as well, the sensory information in my moving limbs and all that. We call that proprioception, that movement and that information that travels back. It's telling me this is a separate entity.
- 7.53** But in sleep paralysis and in dreams, when you see a monster, or in sleep paralysis, when you see a monster, that is lacking, you are paralyzed. So that information from your body is not there. And your prefrontal cortex, is it active or dampened down in sleep paralysis and dreams? It's dampened. So without the cortical breaks and the sensory information from your body, your social cognition neurons become hyper-activated. And so I can no longer differentiate between you and me. It becomes a blur.
- 7.54** When I merely imagine a creature, an external agent, it becomes real. It manifests as a real being with an agency and intentions to understand my very soul and so on. So we believe that is the case. And then the whole thing is embedded in a cosmic terrifying milieu, in a world that's scary and highly, powerfully intense. I want to say spiritual. It's just otherworldly. And what's going on here?
- 7.55** Well, we think that both in—first of all, in dreams, here's the thing. So in dreams, in dreams, for example, you have lack of serotonin. So serotonin is lacking in dreams, in REM. But it seems like—and we know this from when people take antidepressants and so forth, medicines that can activate serotonin, serotonin have certain receptors. It has certain versions, variants.
- 7.56** Now, when I say that serotonin-producing neurons go down, well, the activity of certain receptors of other types of serotonin, like serotonin 2A, certain types of serotonin, can actually become hyper-activated. From our chat, remember we said serotonin 2A receptors are what? The type of psychedelic world. It's almost like when you go into REM sleep, sleep paralysis, you enter the psychedelic bizarre world, what I call the ghost receptors, because of the neurochemical environment, becomes all flared up, all active, all hyper-titillated.
- 7.57** And that is because of the neurochemical shift. So now the whole environment is cosmic, too. So it's much easier to say, "Oh, this is, let's say, a monster from another world. It's a space alien that's so terrifying and otherworldly, I can't explain it," versus just saying, "Oh, it's a burglar or something."
- 7.58** I had a reporter. I spoke with a reporter about 10 years ago. She told me that she lived in New York, and she said that her sleep paralysis was a burglar, somebody who broke into the house, tried to get in. That was her explanation. You see? So the mundane explanations are there, too, for things like sleep paralysis. It's just a burglar, or it's my

friend that passed away. That's there, too. It's part of the spectrum and the whole realm. But often, most of the time, it has this epic, otherworldly, tinge to the whole thing. And that is because we think, I think, because of these neurochemical shifts that's going on.

7.59 And in fact, I didn't mention this, but you can have positive sleep paralysis, for example. Did you know that? You can see big white horses and angelic figures coming down and talking to you and having prophetic visions and things like that. That's crazy. Again, it's always embedded in this spiritual feeling. It's always spiritual. It's always like, it's not of this world, but it's fine. You can have that. 10% have that. But most of the time, though, it is this negative dark picture.

7.60 Why might that be the case, by the way? Some people are embracing it. Yeah, I think that's right. I think if you have a more positive mindset, both towards monsters in your dreams—obviously, when you're dreaming, you don't have the conscious awareness, but certainly if you're lucid or half lucid. But sleep paralysis, if you have a more positive emotion, then yes, I think so. I think that goes back to what we talked about yesterday with prayer and having positive feelings and how that can change your circuits and the activation of circuits.

7.61 Let's think about this for a minute. The fact that we have sleep is such a vulnerable state. You're lying there, you're just ready to be a meal for the next thing that comes by. That's the whole thing. You are very vulnerable. It does make sense that you want to be in a hyper-threat situation, hyper-vigilant situation. Then if sleep paralysis is also triggered by your emotional centers being hyperactive, then it makes sense that you are already a bit agitated.

7.62 We know that the emotional parts of the brain, the emotional centers of the brain, can actually color vision. They can, not color vision, they can taint vision. They can contaminate vision in what you see. We know that. If I come into sleep paralysis with a fierce state already, I'm already terrified, well, then it makes sense that you could potentially— the ghost, even if it was a shadow that's benign and potentially benevolent, then you would color the whole thing and make it scary and terrifying and create a whole horror scenario out of it. That makes sense.

7.63 It turns out if you have anxiety, you have PTSD, you have things like trauma, just like with nightmares, you are more likely to have things like sleep paralysis as well. Just goes with the territory. We found that if you have these symptoms, if you have sleep that's shallow, if you have a lot of fear and anxiety already making you hypersensitive to the wind, like the branches hitting your window and things like that, then you're more likely to have things like sleep paralysis.

7.64 Think about sleep paralysis, what it really is, and these conditions overall when you wake up, and whether deep sleep, sleep talking, sleep walking—it's all about the person being in a shallow state and being on the edge of what it means to be awake.

Really, what can wake you up and put you in that shallow state is emotions and hyper-emotionality. If you have anxiety, depression, depression to a certain degree, mostly trauma, then you are more likely to have sleep paralysis as well.

- 7.65** Let me illustrate this whole thing, actually. I often use this example of Little Lisa. I think this makes a lot of sense. This is a story about a girl called Little Lisa, and it encapsulates all of this in a very, I think, important way. Little Lisa was living in a fictitious island. On this island, they have a certain cultural idea for what sleep paralysis is. They say sleep paralysis looks like this and that. It's this monster. It comes at night. It has these features. It has scary teeth and fangs. It looks like this and that. So there's all this elaborate idea for it.
- 7.66** Now, Little Lisa is having dinner with her grandmother, and the grandmother is telling Little Lisa, "Be careful. At night, this monster could come. It can attack you, choke you. It can potentially sexually violate you. It looks like this and that. It has all these features." And mind you, little Lisa has never had sleep paralysis in her life. She has no idea what the grandmother is talking about, but now she goes to bed. Now she goes to bed.
- 7.67** As she is sleeping, as she is sleeping, she's imagining, she's dreaming potentially about monsters, about creatures and things like that. Obviously, she's a bit scared now. We know, by the way, that you can incubate. If you think about a certain thing before sleeping, then you're more likely to think about that as well. Just wanted to add that here. We have that. Now, back here.
- 7.68** Now, if little Lisa is hypervigilant now to any potential threat in the room, and her body —she's monitoring her physical sensations. So she goes, "Is something holding me down right now? Is there something in the room right now?" So she's monitoring her body. And then suddenly, at some point, she goes, "Oh, my God, there's something here." And she wakes up, and she becomes alert. And the idea here is that it became a self-confirmatory behavior.
- 7.69** She woke herself up during REM where she's paralyzed because of her fear and anxiety centers being all ramped up. She woke herself up where she is paralyzed, thinking something would hold her down. So that's the first step. Then she sees a monster, and it looks exactly like what her grandmother told her. Why is that the case? Why is that the case?
- 7.70** Well, it turns out that dream imagery and hippocampus and memory has a crucial part in our dreams. It plays a crucial part. It colors and taint what she sees. So she now sees exactly what her grandmother was saying. The monster comes, or the hallucination, monster attacks her, chokes her, strangles her, does all these things to her. She wakes up the next day. She's terrified. "What do I do? Do I go tell my grandmother? Do I go tell my mother? Do I go tell my family, my friends?" She doesn't know what to do. She keeps it secret.

- 7.71** Two days later, she has it again. Why? Well, she was anxious and stressed out. Does anxiety and stress predispose you to sleep paralysis? It does, right? To any of these conditions. Because of that, she has sleep paralysis again. Four days later, five days later, she has it again. Six days later, again. Now it's chronic. A month later, she develops potentially anxiety, trauma from this experience, PTSD.
- 7.72** And she might even go tell her friends about this at this point. She goes around in school, "Oh, oh, Joe. Oh, Maria. I saw this and that. That monster." And then they now have a cultural idea for that and can potentially go down the same route. Does that make sense? That's how we think that it can spread, sleep paralysis, and why we know now that it's more prevalent in certain cultures because of these cultural ideas running amok and getting a life of their own, and how the brain can escalate out of hand in this way.
- 7.73** I've heard that brain scientists can stimulate the brain. I was curious how many volts you use. That's my first question, to stimulate the brain. But then the second question is: I've heard that they can stimulate memories. I wonder if there's ever examples of when there was an area of the brain they stimulated that generated random or dream-like pseudo-memories that were not real. That would be an interesting thing. But maybe if they're serotonin and the—I can't remember the other inhibitors, maybe they couldn't generate something. But I wonder if scientists have generated with shock and at what voltage to maybe create a pseudo-memory.
- 7.74** That's great questions there. In terms of electrical voltage, I don't know the exact voltage. I've never done electrical stimulation. This is a little bit older using electrical stimulation. What we use these days is transcranial magnetic stimulation. It's a magnetic pulse. It's much more—I mean, it's much simpler, much easier, and you get high effects.
- 7.75** This has certain intensities. High intensity is, let's say, 15 to 20 hertz. That's called high intensity. That makes the brain more excited. And below five makes the brain less excited. The brain is pretty simple. It's either more excited or less excited. There's a random cutoff of five. And above five, if you zap the brain, you go [sound], you just become—the brain becomes more excited and you activate some of the excitement neurotransmitters. Below five, if the brain is hyper-excited, you make it slow down. You slow down the brain.
- 7.76** And so, yes, there is, in the sensory regions, it's right here, there's a strip of cortex right here, a strip of brain tissue. It's called the postcentral gyrus. Just adding that as a detail. If you stimulated—and Penfield, a Canadian neurosurgeon, I think in 1950s, did this for the first time. He stimulated that part of the brain and the patient would experience—they would have dream-like experiences. They would say, "I felt like this." And other cortical regions, too.

- 7.77** When you mainly do the sensory regions, it's mainly sensory experiences. When you do the motor regions, as you would guess, mainly motor. But then as you move about in the cortex, then you can experience all kinds of dream-like states. But in fact, the most interesting I think, is the amygdala, the emotional core. Because when you stimulate that—the amygdala is deep inside the brain, so you have to have an electrode. But if you stimulate that part of the brain, you will have dream-like experiences and fear and out-of-body experiences and all kinds of bizarre sensory feelings, dreams, everything mixed in one.
- 7.78** And crazier than that, if you have it on the right side of the brain—so you have two of these amygdalae, the amygdalae, if you got to do it on the right—I had a patient tell me, we saw a patient saying that they felt fear on the left side of their body. The fear was on the left side. What does that mean to have fear on the left side of your body?
- 7.79** Now, if you take the electrode and you're mischievous like me and you put it on the ACC, guess what happens? The patient starts laughing uncontrollably. Okay, so you activate the laughter circuits in the brain. There's actually laughter circuitry, and it's in the ACC area and a nearby region called the SMA. These are just fancy name for structures in the cortex. If you stimulate them, excite them, the patient would just start laughing. And he can't stop. He just keeps laughing.
- 7.80** I saw a video probably 10 years ago, and it was about sleep paralysis. And this guy in the video said that you're more likely to experience sleep paralysis if you're laying down flat on your back, staring up. And I was wondering if there was any truth to that statement, and if there is, why?
- 7.81** Sophisticated question. You're very smart. Brance, you have some questions. Absolutely. It's much more likely to strike when you're sleeping on your back. We have found that in our research, about 80% of cases is when you're sleeping on your back. Sleep paralysis can either hit at sleep onset, as you're about to fall asleep, or more commonly in the middle of the night when you wake up. It can actually occur as you're about to fall asleep.
- 7.82** That has to do with the fact that if you are sleep deprived and stressed, and especially deprivation of sleep, it can cause you to rush into REM sleep. You just want to go to REM as quickly as possible. That can also lead to decoupling and having sleep paralysis early.
- 7.83** Now, why does it occur mostly when we fall asleep on our back? It has to do with breathing patterns. And how we breathe and gravity and our lungs. We're more likely to have disrupted sleep as we sleep on our backs. That's why you have things like sleep apnea also occurring when you sleep on your back. You want to challenge me with these questions? Fair enough.

- 7.84** It has to do with breathing patterns. Does it have anything to do with the way that the back of your brain is hitting the back of your skull instead of— if you slept on your side, this side would be—you know?
- 7.85** In terms of hallucinations, you mean?
- 7.86** When you're laying on your back, your brain's free-floating in there, like when you're standing. When you lay on the back, the back of your brain, hits the back, and it rests on the skull. So does that have anything to do with it as well?
- 7.87** Good question. No, not directly. So your brain, luckily, is not affected by the pressure on it. We have the cranial, the skull is thick, right? You have this thick skull. So in terms of how you— that's going to have no effect on it at all.
- 7.88** But you could say that if you sleep on the left side or on the left side or like this, are you more likely to see your room and your things and the environment? Remember, the first step of my treatment is to close your eyes. I could say, if you have sleep paralysis while sleeping like this and your head is like this on the pillow or sleeping on your stomach, then you're less likely to see surroundings and so forth. But it doesn't have a direct impact on the brain.
- 7.89** I think about this a lot, which is, it might be a little philosophical, but thought it would be good to ask. I find that we tend to anthropomorphize our experience as true experience whenever our brain is what's— it's like a super computer of what's creating our reality around us. You said there's this part of the brain that gives you the sense of self. If that part of the brain wasn't there, then we wouldn't have a sense of self. You might blend with the rest of the world. Kind of like a oneness.
- 7.90** A oneness, yeah.
- 7.91** I know with people who do psychedelics experience that; they feel oneness with the universe and all that. And all that, yeah. I understand that has to do with serotonin and this empathy connection. Is there's anything you had to say about that?
- 7.92** In terms of the oneness?
- 7.93** Yeah. Well, it's hard to gauge which one's true. Is it true like our experience is true, or is there an experience outside of our experience that is true?
- 7.94** The thing is, yeah, I think you have to keep this in the context of prefrontal cortex. Yes, you have various regions of the brain that's active when we are dreaming and when our mind is wandering and so forth. These, let's say, the dream circuitry. When you are awake, these are tightly inhibited by the prefrontal cortex. The prefrontal cortex is the "shh" of the brain. It just calm things down. It makes you more likely to say, "Oh, this is you, this is me. This is the world, and I am separate from the world. I have this entity that I call me, Baland, and there's an external world around me with external agents."
- 7.95** Which maybe has to do with ego?

- 7.96** Yes, absolutely. When you have dissolution of ego, you take psychedelics and things like that, what happens is that the prefrontal cortex shuts down. When you take a psychedelic drug, not only do you have things like these serotonin ghost receptors become hyperactive, but you also have less hierarchical activation of the brain: A, B, C, D, so forth. It becomes like A, D, X. Far-flung regions of the brain start hyper-communicating. That's what psychedelics does. That's what dreams do. That's what the prefrontal prevents you from doing. The prefrontal is really what inhibits—but it's important to have a prefrontal. Otherwise, we'd all go around and blend.
- 7.97** Which is also to the point of the memory thing where you experience life in memory. In that state, you also don't experience life in memory. You start to see things as new again instead of—
- 7.98** 100%, right? Look, in your brain, you have certain algorithms, you have certain codes for life. I have a view of myself: Baland is this and that. I'm a good athlete. I am a good runner. I am a good this or that. You have a certain—or "I'm a bad person" if you have depression. "I am a bad this" if I have a psychological disorder or whatever. You have a lot of negative self-images, like talk. And that's all encoded.
- 7.99** And so these algorithms get a big blow when you take psychedelics because you think in a less rigid algorithmic way and start thinking in new ways. Then what happens as well is when you take these drugs, you have what's called growth factors in the brain. You got BDNF, GDNF, this fancy name for fertilizer that makes the neurons and synapses connect in new ways.
- 7.100** Normally, when we are adults, when we go about, we don't have a lot of this BDNF in our system, these growth factors. That's why our brain doesn't keep changing its form all the time, its structure. It stays the same most of the time. When we are kids, though, we have a lot of these growth factors. We have a lot of them. We need a lot of growth and plasticity.
- 7.101** When you have depression, your brain has plastically wired into a depressed brain. Now you have a depressed, anxious, traumatized brain. Now, when you do, when you take a psychedelic—serotonin boosters and so forth, antidepressants— you shake up the system a little bit.
- 7.102** You, first of all, change these fixed patterns of thinking and codes, you go, "The hell with this. The hell with that. I'm not going to think this way anymore." You can get away with it because the system is now thinking in divergent way. Then you add fertilizer on top of that—using a metaphor here with growth factors—and then new synapses can be stitched together, and then you have a new self coming out six months later, or with psychedelics, even faster. That's what these are doing. Insightful question. Thank you, brother.

Lecture 8: Spirit, Science, Sleep

At a glance — A final reflection weaving together spirit, science, and sleep.

- 8.1** So for this final lecture, I didn't know how to start, what to start with and what to talk about. Then I came out and Lizzie mentioned something, and it's a good point. I mean, she said, "Look, so many people, when they are in sleep paralysis or even when you have certain nightmares, you might feel it's a demon or it's something that is actually spiritually real. And as a scientist, how do you deal with that?" Because having a faith, having beliefs, believing in God—I believe in God. I think a lot of Americans do, and a lot of people around the world, frankly, believe in these things.
- 8.2** In academia, we tend to hide it, like brush it under the carpet, everything that has to do with God and spirituality and all. It makes sense from the perspective that it's hard to do science with a certain judgment, like prejudgment of what you're doing. But as a human being, you can't help but wonder, right? What's going on? Is it really a demon? Could there be demons? Could there be things from other realms? Now, my personal view—if I were to look at it completely, objectively—I do believe that spiritual entities from other worlds can communicate with us.
- 8.3** I do, frankly, believe it. It's going to cost me my job at Harvard. There you go. I don't care. I do believe that's possible. Why do I believe it? Well, it's a belief. It's faith. It's my own personal thing. When I go to the laboratory, I'm not going to start communicating with demonic beings. I'm going to keep it to the science. But I can have my own belief about the world of other realms and things like that.
- 8.4** I'll give you a good example. I think this is a good example. Look, 100 years ago, if I was to say on Brance's skin right now, there's trillions of animals crawling right now all over, and you can't see them, but they're there. What might they do to you if I said that? They'd burned me, right? But then they invented the microscope, and then they could look closer and they saw that, well, there's actually a lot of stuff going on there under the hood we can't see right now.
- 8.5** For me, the analogy of that is similar with a spiritual world or another world that we cannot see. Science and the instruments of science have not reached that point where we can communicate with other beings. But my own experience is that, well, these things could exist, and science doesn't necessarily negate that.
- 8.6** We just explain the brain structures. The brain structures are there. I think as people who do believe in God or have faith, I think it's important to say, "Look, we have to be honest to the science and say, well, these are the brain structures involved and these are activated when these things happen and we feel this and that. But you could still have an actual demonic entity—an energy that's evil out there and still have the brain patterns." Why not? I mean, if it's another world, it's not a physical world, obviously. So

that's what I'm saying. We don't have to be that rigid in our thinking. And I think that's my own personal perspective. So sure.

- 8.7** In a sense—I wanted to share this with you. Now we're going down the personal route and personal stuff. I just want to share something. A few years ago, something happened to me, personally. Things were going very well. Things were fine. Life was great. And then I took some medicine. See, medicine is good, but not always. I took some medicines—a cocktail of antibiotics and other stuff—and it afflicted my brain in a very severe way. Okay, so I got brain damage from this.
- 8.8** It's not brain damage. Obviously, I'm doing well, but I had like, there was some toxicity. And overnight, I could not see well. My vision—I had visual problems. I couldn't look at screens. Any type of light would affect me in severe ways. Severe ways. Imagine a year—for a year and a half, I couldn't look at any TV, any YouTube. For six months, I couldn't read. Now, you know about my life, reading was one of the most important things. I couldn't read now.
- 8.9** For six months, I couldn't read. I had sunglasses everywhere. I couldn't listen to sounds, from electronic sounds and radio and things like that was extremely overwhelming. It was really bad. At the same time, other things happened. I had my eye on a girl that I wanted to marry. That whole thing fell apart, and my life was just down the drain. It goes to show that you're a scientist— it doesn't mean your life is perfect. My life was at the lowest of low. I couldn't read, I couldn't look at screens, I was wearing sunglasses. People thought I was Terminator T1 or whatever you call him, walking around. It was a very hard time.
- 8.10** Around that time, I was called up to lecture. I was called up to lecture. Peterson Academy said, "Come, Baland, we need you to come and give a course." At that point, I thought, "I'm not going to ever lecture again." I thought, "It's impossible. I can't see—my ears, light, nothing. I could never go back to science. My life is over." But I thought, "Let me give it a try." So I came back. I gave lecturing a try. I had said no to other lecturers, but I thought, "Let me do this. This is important. Let me try to do this—educate people—let me give it a try."
- 8.11** I came back. And the reason why I could do it, even though I had lost a lot of weight, I was sickly—I was really hiding it. I was hiding my illness and all that. I was trying to keep calm and just do the content. But it was very hard. Like any light, have you seen "Bloodsport"? When guy in the final fight—have you seen that movie? It's an older movie, probably a different age. Anyway, being blinded at the very end of a fight, you can't see—the other person you're fighting with throws something poison in your eye, and you can't see or something like that.
- 8.12** I felt like that. It was very hard. But the reason why I came out of it was because of one thing. And Larry, you asked me when we came out—you said, "Baland, something about the lecture style is interesting," or something. I don't want to make it too big

because I don't think it's that good. But you know what I mean? You said it was okay. It was okay.

- 8.13** I think you know what works? It's all passion. If you really love what you do and you're passionate about what you're doing, that's the key. That's what I wanted to tell about that little anecdote to let you know about importance of passion. Even if you are through the biggest of obstacles—if you have that passion, you can do it and you can overcome it.
- 8.14** I was wondering—because you said something about how our waking states are similar to our dreaming states, and that maybe they're not so different—I don't know, I thought about this one theory, which is that consciousness is the fundamental reality, not the material reality. Say when you see a demon, it's not a materially real, but it's maybe more consciously real, or I don't know how to—
- 8.15** I understand. There's this idea that consciousness is everywhere and the human brain is a recipient of consciousness, for example, and that's all that's going on, right? The thing is with these theories is that—okay, where is consciousness? Are you here for the full? Are you here for the next course? Okay, but consciousness, and where is consciousness? What is consciousness? During dreams, consciousness is being processed, right? We know that. It's obvious from the dreaming state.
- 8.16** Now, consciousness seems not to be located in your liver, for example, or your kidneys or your knees. Why do we know that? Well, if you have damage there or your kidneys or your liver, there's no changes to consciousness. Consciousness stays the same.
- 8.17** So it seems like the brain in—consciousness is localized inside the human brain. Is it inside the entire brain? Probably not, right? Because if you damage some parts of the brain, consciousness stays is the same. So there are certain structures within consciousness—within the brain—that seems to where consciousness comes about. And so this is for a different course.
- 8.18** My point is, is consciousness everywhere? And is the brain just the recipient of that consciousness, like a radio getting signals from the outside and hitting the radio? We don't know. We have no way of knowing. We can't test it. We can't say it's true. We can't negate it. Could it be true?
- 8.19** Well, technically, it could be true. Could there be a ghost right here? And I'm not being frivolous. Could there be an angel or ghost right here? There could be. I just can't see it. So it gets to a point where you can't really be sure. But look, as I said with the insects on your body, who knows how a super scientist, a thousand years from now, will think about the universe? Certainly much different than you and I are doing right now.
- 8.20** I imagine that the cells in our body are all individual. They have some potential consciousness that they are all united by to serve us as this—we are like, I guess you could say, their God, for instance. They all serve each other in order to serve us. It feels

the same way on the macro level with us, all the same way. When we serve God, you serve the people around you in order to do so. It's like a same as above, so below instance. I imagine that's possible that the conscience is just integrated into something. It's hard to put into words.

- 8.21** It's hard to put into words. The answer is: we don't really know. It's very hard. But the question is: the chicken or the egg? Body or brain first? What do you think? What came first? Body or brain first?
- 8.22** Well, not brain, but consciousness.
- 8.23** It actually turns out the brains came to support our bodies. That's how scientists look at it. Isn't that crazy? That goes back to what Amanda was talking about—gut bacteria—because we have trillions of bacteria in our gut. And so from a scientific lens, our brains are supposed to have come to support the body. And so some would might argue that the real masters of us, Baland and all that, is the bacteria inside my stomach. There's so much signaling going back between bacteria in my stomach and the brain all the time. I know this is off topic, but it really is not.
- 8.24** If you have any bad foods or anything that changes the flora of the bacteria inside your stomach, well, that will completely change. It can change, instantaneously, the emotional core. It does bi-signal—the pathway is going from the brain to the gut, and within seconds, you can affect how your emotional state is.
- 8.25** Have you tried this? Sitting in a restaurant, you get very bad food, and then you instantly feel annoyed, and irritated, and things like that. There's something about—that's the gut flora and the bacteria being affected, and that can profoundly affect sleep, too. There is that. But in terms of consciousness and where it is and all that, it's very difficult to pin down. It's philosophy, but it's science, too. We have to be open-minded and ask these questions.
- 8.26** We talked a little bit about dream monsters when we got into that, and had an idea about sleep paralysis, and dreams and what's going on there. Now, I just want to add—the dream monsters that we see in dreams are very different from the ones in sleep paralysis. We said they had a much more vivid structure in terms of how they look, their visibility, and so forth. So they are very different.
- 8.27** They tend to be located in a spatially bizarre scenario and space. So you see them, for example, in a mountainous place with castles, and rainbows, and planets floating, and things are bizarre. And that amplifies the whole experience so you remember it much better. So that's one key point. One key point of it is the environment that's mixed up and messy—and we mentioned why that is the case—the structures in the brain that has to do with logic and concept—creating concepts.
- 8.28** I think that's a key point to make that concepts itself, like having concept of me, Baland, chair, wall, a house, a studio. These are concepts, right? And these concepts are

created in the brain and woven together in a way that makes conceptual sense. Now, if I'm a schizophrenic, I'm going to mix that up and mix it all up and say, "Well, I am me and the wall is me and things are just blending." That's psychosis. That happens during psychosis. But again, that outer layer of the brain, the DLPFC, the logic part of the brain, when it shuts down during sleep, that's why concepts get mixed up. And the monster you see is embedded in that bizarre context.

- 8.29** So in the context that's bizarre, in a hyper-cosmic concept because of the ghost receptors and the serotonin 2A receptors being hyperactivated. So the scenario as well is strange and bizarre. And then we have the monster itself that is memorable, a memorable monster. And I think— so you have—the monster looks a certain way.
- 8.30** Now, it's important that the monster has to be bizarre and strange in order to remember the monster and go tell your fellow people. I think that's a key thing. Now, going around and telling your fellow people you saw a monster serves as a potential warning for threats in the environment. Let's put it that way. That's why monsters appear. I think monsters appear in dreams as a warning and making ourselves ready to face potential monsters in real life.
- 8.31** I'll give you a good example. Kids, for example, children, their dreams are highly populated by monsters. They have—I think it's 70 % of their dreams or something that's populated by monsters, animal-like creatures, monsters. Whereas when you grow older, it tends to be more human-like figures. And also you have the scenario of the little child that is telling you, "Mom, there's a monster under my bed." Have you heard that before? Seen that, right? "Mom, there's a monster." What is that about? Why do children have this innate ability to say, "Mom, there's monsters?"
- 8.32** Now, in the cave—in the caveman days, having that inclination to spot monsters in my closet, and under my bed, and under my chairs in the dark, would that be a powerful survival mechanism? Like you had children like that. I love children. If I was a caveman and there was potential, like tigers and lions running around and I was hiding in my cave, it might be good to have a few of those running around, the kids seeing monsters everywhere. Even though 90% of the time, the monsters weren't really there, but the one time the child was right, the whole bloody tribe survived. That is because of this innate ability to be overly protective and threat-detecting in that way.
- 8.33** So that's a part of it. But generally, the monster that you see in your dream has to be memorable so you can pass on the information. But at the same time, it cannot be too bizarre. It cannot be too strange, too complex. It has to be enough for us to remember it and have a salience effect on us. So it's, "Yeah, it's strange. Oh, it looks bizarre. Yes, it looks this." But not that much that we go, "Oh, this is so bizarre. I cannot even put words on it. Like, I saw a monster with 100 heads. I can't even put words on it." That's the point. You don't want that. You want to be able to describe what you saw. So that's a key point of that.

- 8.34** Now, dreams serve as healing. That's why you need it. We mentioned before that as you cycle through the different stages, deep sleep provides the physical healing so you can run from the tigers and all that, and you can go hunt. So that's the deep sleep. But then you have REM for emotional healing and cognitive clarity and cognitive rigidity—cognitive flexibility. You have that in REM sleep. And we also talked about why that might be the case. Because of the neurochemical environment and the changes in the prefrontal cortex. For example, that the prefrontal cortex shuts down.
- 8.35** We talked about before—with the question about psychedelics. We said psychedelics and drugs like this can make our brain less hierarchical in processing. So the prefrontal is less in charge and different parts of the brain become activated in a more flexible and more linear manner. Now, in the dream state, that occurs naturally. Things are blended and the prefrontal is shut down, and so you can process things in a more flexible and linear way.
- 8.36** Now, we also said the stress chemical noradrenaline, the stress hormone, is not available. For that reason, because it's not available, you face threatening scenarios without physical arousal and stress. And imagine this—imagine if a serial killer was to walk in right now and walk towards you with a spear, or a sword, or something, and he comes towards you and you have no stress hormones. You're just completely calm. That'd be pretty cool. You could be more flexible, couldn't you? You could be like, "Okay, let me think now. I'm not going to panic. I'm just going to slowly go get some a tool to protect myself" and so on.
- 8.37** That's the point in dreams. In dreams—because you don't have stress and chemicals like noradrenalin—you can tackle threatening situations in a calm, just relaxed, completely cool manner. I said that so you learn how to process it in a different way. And then the chemical that's responsible for plasticity—acetylcholine—it's highly active in the REM state. And so for that reason, your brain can then wire itself. It wires itself in a new way. It says, "Oh, okay, I'm dreaming about this person trying to kill me. I'm not stressed. Maybe it's not that dangerous. Maybe it's just what it is." And then your brain learns to take away the threat and the sting and the terror of it all.
- 8.38** Rosalind Cartwright did some studies, I think in 1950s, that were very interesting. So she had depressed people, she had people that were going to get a divorce. Actually, they had just had a divorce, recently divorcees. They had been in a marriage, they didn't work out, and then they had a divorce. She looked at their dreams. She had a dream diary, and she looked at their dreams, wrote them down. "Okay, here you have this person is dreaming of this dream, this person is dreaming of that, this is this dream." And then one year later, she came back and assessed their dreams, looked at their dreams.
- 8.39** Interestingly, she found that people who have dreamt of their divorce and themes around the divorce, like "I saw my ex-husband," or "I saw my ex-wife doing this," or "I saw this and that," they were much less likely to get depressed one year later. Much

less likely to get depression. They were buffered and protected from depression by simply dreaming about their divorce and the divorce situation. So this shows us it's not merely about sleeping and dreaming, but ideally, if you can dream about the negative event and have it come to mind, well, that can be even more powerful. That seems to be the key ingredient. Even though you probably will forget that dream in most situations, having you dream about the event is what seems to be very powerful and effective.

- 8.40** AJ wanted to know about depression. Well, depression—look, depression is an interesting disorder. It's highly rampant in society. There's 80 million people or something worldwide. It's spreading like wildfire. Debilitating disorder. You have a lot of fatigue. Chemicals like noradrenaline—you don't have a lot of that, for example, so you tend to be very sluggish. You have a mask-like expression often. So you go like this. You humbly smile. You're very—it's like that. You don't have a lot of energy. You don't have adrenaline and noradrenaline, and that's a problem. You don't have energy.
- 8.41** And so it turns out that depressed people have a lot of problems with sleep. They tend to wake up in the middle of the night, for example. So that's one thing. They tend to wake up in the middle of the night and just be—they can't fall back to sleep. They just wake up and can't fall back to sleep. So that's one symptom of depression.
- 8.42** Now, in terms of REM sleep, and this is critical, in terms of REM sleep and rapid eye movement sleep— if you have depression, and if the world is horrible, if the world is terrifying in some ways, and you are scared of the world, and your life is not worth living, and things are just bad—might it be useful to have a lot of REM sleep? Would REM be important for these people? What do you think? What would your guess be? Pretty important, right? You want to have REM because who would want more REM than somebody with depression, right? It might be the most important—REM sleep for them is key. It's critical.
- 8.43** So in fact, what happens for people that are depressed, you would often find that—I mentioned before that sleep, the deep sleep comes before REM, and you have more deep sleep in the first half of the night, right? Remember, you want to have physiological benefits first, right? You have more deep sleep, and then by the late part of the night, you have more REM. Okay?
- 8.44** It turns out if you have depression, REM will interrupt your sleep cycle and jump to the initial stages where you have a lot of deep sleep and just eat away at all the deep sleep and just interject itself in there and just—you have a lot of REM sleep in the beginning of the night and throughout the night. A lot of REM. This is REM, REM, REM, REM because you want to process your emotions, which makes sense, right? When you think about it. So you have that.
- 8.45** Now, here's the problem, though. What might be the problem if you have a lot of REM sleep instead of deep sleep? We mentioned that depressed people are very tired.

They don't have energy. Sometimes their skin looks very tired. They look very— low energy. What is deep sleep doing? Physiological benefits—skin, hair, energy, metabolic, glucose, all the benefits you want as a human being, having energy and being invigorated and walk around in the world—you don't have that. So it comes at a cost.

- 8.46** Even though you are getting REM, you're getting more REM, now you cannot really—you're still very tired and fatigued. And that shows you how depression is really a biological disorder in a profound and crucial way. And how that deep sleep is affected in depression. Does that make sense? Am I being clear or am I being— is it good?
- 8.47** In the dreaming state, prefrontal shuts down, you think more holistically. You have that. And then we said stress chemical, we don't have that, but there's another interesting pattern of activity in the brain. When you're wakeful, we said you have these erratic beta waves, quick and fast. Deep sleep, you have what? You have big, huge ocean waves. This is deep sleep. Why do you have big ocean waves? What is the reason for that? Yes, the brain is less—the speed of the brain is lower, and the big ocean waves represent the fact that far-flung regions of the brain are not active. That's deep sleep. Different parts of the brain are not active.
- 8.48** And then you had the spindles on the end, we said—spindles, which tells you that the memory part of the brain is storing information to the safe vault of the outer cortex where it's stored for long-term storage. Then enters REM sleep, and you have the interesting, intriguing fact that then, waves speed up again—looking like the person is wakeful, looking like the person is now awake, and aware of the surroundings, and all that. The person is dreaming, and you have dreaming consciousness mimicking the waking state.
- 8.49** But then occasionally, what can happen is you have a strange type of wave called theta wave, a very slow type of wave emanating from the amygdala, the emotional, fear department the brain, and the hippocampus, the memory part of the brain, coloring the brain, affecting it. You have these theta waves coming out from these two structures. And guess what they might be doing? Well, they are weaving our emotional memories that are stored in the hippocampus and the amygdala, and aiding in the process of massaging our emotions, and our emotionalities, and taking this thing out of negative memories.
- 8.50** They're like the whole—this is where we say wisdom emerges from the brain in REM sleep. That's the point that you have wisdom, but also emotional wisdom coming from the amygdala and hippocampus—that's now completely going into a more slow mode and really weaving memories into the fabric of your brain in a slow, wise manner, so to speak, in a holistic and meaningful way. So you have that.
- 8.51** Theta waves. All you need to know is you enter REM sleep—fast waves— and then occasionally you have these theta slow waves emerging from the hippocampus and the amygdala, slowing things down and weaving the whole net—the whole narrative. So

you have that in REM sleep. And it's clear—the role of emotion in REM—it's clear. It's quite profound.

- 8.52** I mean, you wake somebody up from REM sleep, their amygdala might be 60% more active— if you actually deprive them, excuse me, of REM. If you deprive a person of REM, amygdala is highly active. If you have somebody—you show them negative pictures, let's say a snake or let's say a cobra snake or let's say a burning house and show them these images—one group, you show morning, and then you show the images again in the evening.
- 8.53** Obviously, at the first exposure, you feel terrified, I mean, the amygdala becomes highly active. Fear centers become highly active from this visual exposure, from these pictures. Somebody in REM sleep, somebody who has gone through REM, you show them these pictures in the evening, and in the morning, you show them again. What happens? Clear-cut decrease in the amygdala just from sleeping. Just from sleeping.
- 8.54** Now, practically—this is interesting to know what happens in the laboratory—but practically, what does it mean for you and I, Larry? It means, for example, sometimes if you have an argument, you should sleep on it. You should just have a little nap, sleep through the night, perhaps. I've had this many times. When I was a kid, I remember always, not always, but I would fight with my sister. I have a sister, sometimes we fight, and I get angry with her. And then I would sleep. And then when I would wake the next morning, I found that I was not angry at her anymore.
- 8.55** Have you had that before? You feel like you're supposed to be angry. I was angry at myself for not being angry at her. I was like, "I should be angry at her. She really annoyed me yesterday," but I wasn't angry anymore. And that is precisely what REM sleep is doing. It really soothes the emotions, really calms you down, and you look at things much more holistically and flexibly. So that would just be an example whenever — if you have something, if you're angry at somebody or you want to react emotionally, sleep on the thing. That can really help.
- 8.56** Sleep. I want to touch a little bit on sleep and the importance of sleep and what we can do to improve our sleep. Because in order to have REM and dream, you need to have sleep. But before I do, I do want to touch upon, a little bit upon, the conditions in where sleep is compromised. I think that would make sense. So depression—it's compromised in depression, in people that have anxiety disorders, and it seems to be a common threat now, we've seen similar conditions. Anxiety, people with PTSD—you can imagine now, we talked about this at length—have trouble sleeping.
- 8.57** People with neurological disorders—if you have brain damage in the areas of the brain that control sleep, obviously, you can have problems sleeping and dreaming in some cases. As we said, if the dopaminergic centers in the brain are affected, then you can't dream, remember? You can't have REM dreams. In fact, in the part of the brain that has to do with juggling up images in the brain—it's up here in the parietal lobe— again,

a damage there, and again, you can't dream. So it shows you that dreams in many ways are—even though we see them as very complex, and epic, and greater than life, in terms of the brain, they are pretty simple. You can turn them on and off in simple ways by just turning down a part of the brain, or having damage to a part of the brain, and so forth.

- 8.58** To improve sleep—the reason that people with anxiety disorders and depression and all these conditions have poor sleep is because of the emotional arousal of the brain, the nocturnal arousal element that we've been talking about. Now, if you have those conditions, if somebody has this condition, they should go to treatment. But once the emotionality and the hyperarousal becomes better, then sleep will become better. And dreams—you will have more dreams and more healthy sleep.
- 8.59** It turns out if you go and have CBT and therapy, then you can have 12 weeks of that and the emotional core of the brain—the amygdala we talked about before that can become bigger in these conditions, can actually shrink again. You take antidepressants—antidepressants can be a bit hard with the side effects and all that, but antidepressants or other medications can also cause the emotional part of the brain, the fear centers, to shrink as well. The effect of that is, of course, better sleep and more healing dreams.
- 8.60** But it tells us something, for all of us, for all of us, if we have stress anxiety overall, not in the pathological sense necessarily, but if you have a lot of stress, that can affect sleep, too. So one thing that you want to do is, one of the first key steps, is to have more healthy habits.
- 8.61** Exercising, for example, can reduce stress. That's one key thing, right? And just taking things more calmly—all the things we mentioned, really. I don't want to repeat myself too much, but walking in nature, social interactions, all that stuff. All the things that we know to reduce stress will make the emotional part of the brain less active, and we have better sleep as a result. So that's a key thing.
- 8.62** But more than that, what we also need to do is—and this is important, I think—is that we all have a chronotype. Meaning, one-third of the population tend to be people that sleep early. So they sleep around 9:00 p.m., for example. 9:00, 10:00—once the sun sets. One-third is in the middle of the night, so they are in the middle. Naomi, are you in the middle? Early sleeper? Okay, good. Like me. Actually, it's funny. I was a late—my whole life, I was always a 2:00 a.m. sleeper. Then later in life, I suddenly became a 9:00 p.m., 10:00 p.m. sleeper.
- 8.63** I don't know how it happened. There's supposed to be genetic reason for this. Did you know that? It's supposed to be genetically wired in us. They would say—scientists say that, "Look, the reason why one-third, one-third, one-third—one-third late sleepers is that when you were living in the caveman situation, you want to have some people

awake at certain times to protect the tribe from the lions and all that. So it makes sense in terms of protecting the gene pool.

- 8.64** But personally, it doesn't make sense because I always slept—my whole life as a kid, I was always up at 2:00 a.m. My mom would get angry and—"Why are you not sleeping? Get some sleep." And I was like, back then we had TV. We had TV on and that was my life. And then suddenly, later on, I found, "Okay, if I sleep early, I just feel more energized. It feels better, and it's practically better." I felt like life wasn't designed for the late sleepers. That's what I really found. Yeah, it's an interesting thing. It turns out that teenagers, for example, have a natural inclination to want to sleep later, too.
- 8.65** That could have been a part of it. That tells you that the whole school system is messed up. What do you think? You're in the school system, university business, making young people rise so early. I think it's not good. You're tired, right? And there's a natural, genetically predisposed reason for this. As a teenager, you just want to sleep until 10, 11. Trust me, I have an extensive experience. That's a part of it. Finding your chronotype is important.
- 8.66** Alcohol, for example, is a REM suppressant—it will suppress REM sleep. Did you know that, Amanda? Did you know? If you drink alcohol, your sleep becomes fragmented, and you have less of REM, and the healing effects of REM. So alcohol—reduce, ideally. Ideally, even remove if you can.
- 8.67** Other medications, too, can—antidepressants can actually suppress REM, too, but that's a different story. There's so much benefit—in fact, if you can tolerate the drug—I mean, they are very hard to tolerate. I have seen patients that get hyper-affected by these. I mean, they just can't tolerate these antidepressants. They become even more depressed, sometimes even suicidal, especially in the first few weeks before the drugs really get their effects.
- 8.68** So they are very hard to deal with. But if you can tolerate them, then there's ways that you're—they are so powerful that they even, in some cases, can outweigh the fact that you're not getting REM in some cases. But generally, I think there are better medications now with psychedelics and all that. But that's still in the workings. We're still working on that. Those are some points.
- 8.69** What else we have on my list? I have a whole list here of sleep improving. We talked about the fact that—find out what your chronotype is. You don't have to do a genetic test, but just see what seems natural to you and stick to that. Stick to that. And one reason why it's important to stick to that, Heavenly, you know why it's important?
- 8.70** Because, remember, in the first lecture or maybe the second, I mentioned that we have in the hypothalamus, the sleep part of the brain, we have a clock there. And that clock looks at light intensity, but also what we call zeitgebers, meaning time-givers. It sets your internal biological rhythm based on other information, too. So this could be when you had a last meal, when you had the last social interaction that was arousing. When

did I go to a—I don't want to say party, but I went out and went about and things like that. What time was that? When did I—even certain sounds can affect your, like loud sounds, can affect your inner clock there.

- 8.71** So what you ideally want to do is that you want to time your—you want to plan social interactions, plan your meals, plan your exercise, plan when you go to sleep, depending on your biological clock, and keep it consistent over time. When you keep it consistent like that, it gets the suprachiasmatic nucleus—the hypothalamus, gets the message and stays within that plan. It keeps the schedule, so to speak. And then occasionally, of course, you can mess it up and you want to go to, I don't know, with your friends and out and have fun. Yeah, sure. You can mess that up. You don't have to be a rigid robot over here. That's not what we're saying. But try to stick to a schedule. Try to keep it as clean, as we would say, for most of the time. That's one key thing.
- 8.72** Now, that clock—light is very important in terms of programming it. So programming—light hits the eye—eyeball, optic nerve— and then it goes there to the hypothalamus, even bypassing the other visual centers. It goes there and tells us, "Oh, now we are seeing a lot of light. That means it's day." Set the clock based on that. That means, if you want to set the clock in a way where it's scheduled correctly, when you wake up, go outside. If you exercise, even better— an extra signal—get a lot of sunlight at that point. That could be very, very important, too.
- 8.73** Now, do you use a lot of iPads late in the evening? You don't. I don't think you do. Good. You don't use iPads and phones and all that. If you do—if you have homework or something you have to do important, then sun-blocking— like the blue blockers can do. Even though that technically there's other light waves as well you want to block, but you want to keep the light dim. Keep the light dim. Does that make sense? Those are important cues.
- 8.74** By the way—an interesting experiment—I have to tell you about this. Guess what happens if you have somebody and you put him in a cave for a month—no sun whatsoever. How would he sleep? Would he sleep like three hours, four hours, two hours, one hour? How would he sleep? What do you think?
- 8.75** I think if you put somebody in a cave, they would experience multiple nocturnal experiences in the 24 hours, and they would become disoriented. But they might get their own pattern of four hours sleeping, five hours up.
- 8.76** It's a good insightful point. But it turns out, Larry, you are wrong. Dead wrong, okay? People sleep for nine hours in what would actually mimic how we sleep now. Like nine hours, nine hours, nine hours. It's pretty profound. No light whatsoever, no cues telling you when to sleep. But here you are in this cave—scientists are very passionate, so they will do the experiment and they sleep these nine hours. So there's this innate natural inclination to sleep in these types of—

- 8.77** You know what? Yesterday, I was having my walnuts, right? And they actually helped me sleep. They did help me sleep. I think it was that and then a mix of that and a little bit of—"God, help me sleep. I can't sleep. Can you help me sleep?" Just trying to calm myself down, soothe my system, meditate a bit, close my eyes. But I think the walnuts help. And I think the reason the walnuts help is because inside— so there's a lot of interaction between our gut and Amanda wanted to know this. And it's a very interesting point. Now, I think I slept because of the walnuts, because they gave me— there was some glucose there, and I went out of the semi-fasted state, and I could just —my stress hormones went down.
- 8.78** In fact, we know that if you—especially not only glucose, but especially sweets like strawberries and other things, they can actually calm down your system and your stress system, what we call the sympathetic system, nervous system. So it calms you down. But I think that happened yesterday. But definitely eating healthy overall and eating things like having fish oil, omega-3—for example, high omega-3, can help with your gut. I think potentially that can help you sleep in a much better way.
- 8.79** Staying away from processed food, staying away from foods that make you bloated and that make you feel tired. In the Stone Ages, people would rely a lot on just meats—the meats they can get when they fasted for a very long time. They got their body adapted to that in many ways so they could sleep. They could sleep through the night. So that's a point. Although there are actually studies showing that the people in the tribal days only slept about seven hours.
- 8.80** So, I think when you are fasted, your brain also operates at another level, too. So fasting actually has a very good effect on the gut. The good bacteria can flourish. So when you are in a fasted state and don't eat, it's not like the good bacteria necessarily will die, but you will have more of the—they will flourish more. You have less of the bad bacteria, for example. So that's one point to add.
- 8.81** And another point also when I couldn't sleep was—I think this helped, too. I think the temperature of the room was a problem. The room was hot. The heat here is unbearable. What do you think, Larry? Arizona heat. Yesterday, I went out and I walked about and suddenly this heat wave hit me and I was like, "My God, what do I do?" I seriously panicked. I was almost about to hide behind a building to get some shadows. It's too much.
- 8.82** I was obviously wearing a suit and all that. I wasn't dressed for the occasion. But the heat here can be unbearable, it can be intense. And so in order for you to sleep, you have to have your core temperature to fall to a certain degree. Having a cooler temperature in your body can be very helpful for sleep. So always keep your room as cold as possible, as tolerable. So I put the thing—I put it down on the Fahrenheit. I put a little bit down. So that helped me also, I think. That helped. Add it to the equation.

- 8.83** If you want to take a hot shower, that's important. You can take a hot shower before sleep. It sounds counterintuitive, but when you take a hot shower, you actually cool down the body. You can do that. Sometimes you can even do a trick. This is what I do. Sometimes I want to trick the system. Just by washing my hands and my face before I sleep—you can get some of the same benefits of actually taking a full shower because of the way heat is emanated from the body and on the surface—and how it works with your skin, and your face, and your hands. So I do that. That's important to do. To conclude, if you follow these tricks, I think you will have better sleep and more dreams.
- 8.84** Can you speak a little bit to the endocrine system's effect on sleep, things like cortisol or oxytocin and testosterone?
- 8.85** This guy always have the difficult questions—challenging. No, it's good. We need—look, it's very important. That's why I'm saying, AJ's questions are very important. Cortisol rises when we wake up. So there's a rise in cortisol as we wake up in the morning. That's why sometimes drinking a coffee as you wake up—that makes cortisol go up even more—can be a bad idea because you get a cocktail of cortisol already being high in the system and then drinking coffee. That's cortisol.
- 8.86** Testosterone is released during deep sleep. There's a lot of testosterone. For a long time, people thought that might be the place because obviously you have all the physiological benefits of sleep coming from deep sleep. But it turns out that REM sleep also is a place where a lot of—a state of sleep where a lot of testosterone is released. So we need that as well. Those are the two faces. And the other chemical you said?
- 8.87** I just said oxytocin as an example, but it was—
- 8.88** Interesting. Pass. I don't know. No. Oxytocin in sleep, I think the whole endocrine system and oxytocin, vasopressin, all these endorphins and these neuropeptides—I think that they are replenished in sleep. They are replenished in sleep. The whole idea of the way you can think about sleep is a little bit like charging your iPhone. So you have an iPhone—you use it throughout the day. We use these neurochemicals throughout the day. And then when we go at night, we want to replenish that system. So that's a part of the explanation for why these systems, these chemicals, stop being produced because we want to give it a rest, want to have them rest. I think in terms of these neuropeptides, though, I am not sure that—I have not heard of them being shut off in the sleep. There's no literature on it, to my knowledge. But it could be. It could be that there are some effects on sleep.
- 8.89** Thank you.
- 8.90** Sure.
- 8.91** You were talking about going through hardships and tough times earlier in the discussion lecture. I was just wondering if you had finished your points. I don't know if you were finished connecting the dots, or maybe I just didn't understand where you

were going with that—talking about wearing sunglasses, and going through that time for yourself, and how you were connecting that with sleep and dreams. Just a little bit—if you could touch on that, please.

- 8.92** How I connected with sleep and dreams: I think that for me was a harsh time. It was a difficult time. And I touched upon how my life was full of obstacles early on. A lot of trauma. My parents from Iraq—a lot of war, refugee camp—my life was literally going to nowhere. Prison, criminal—all my friends became criminals. But I kept going. I kept being resilient. I kept going at it.
- 8.93** And I also had what I would call moments where that was not my doing. Literally, at one point, we went to—there was this party, and I was supposed to go there. And my parents called me at that moment and said, "Oh, look, we want you to come home." All my friends went, and there was a fight, and there was just—all of my friends got stabbed, and things were really bad. There was moments like this that made me think, "Okay, I've been given a chance," and I think my parents, my loving parents, had a big role. My point with all of this was to mention that it's not necessarily directly related to sleep and dreams—although dreams in the more metaphorical sense.
- 8.94** So I'm currently writing a book about dreams in the metaphorical sense—when you have dreams as a kid, dreaming about Peter Pan and flying and meeting and facing obstacles and challenges, and then thinking it's going well and things are fine now. Finally, life is looking so good. And then you're hit by a blow from nowhere. And we all experience this in life. You think things are going so well, and suddenly the trial and tribulation hits you, and you can't see, you can't read, you can't hear, you're shattered emotionally because what do you do if you can't read a book, right? You're physically—
- 8.95** And so, overcoming that, I think, is something that all of us—having passion and having hope at the very end, I think it's relevant for all of that. And ultimately, what I'm doing here—I think this project here is about passion, about science and passion, not following a linear way of looking at science. The best scientists that I know in my life have always been the people that were out of the box thinkers and saw things in a different way.
- 8.96** You said in dreams, we react to threats calmly because of the lack of noradrenaline. Is there a way to practice managing noradrenaline in waking life in order to react to threats in a more calm, maybe faithful manner?
- 8.97** You're practicing noradrenaline in the waking state? Oh, absolutely. Absolutely. So this is a part of classical conditioning and things of that nature. So what a therapist might tell you is if you get that email from your boss or something, or that email or something that's threatening—if you can calm down your system at that point and calm down your nerves and meditate and stay calm, then the neurons—the whole circuitry gets rewired. So every time you experience and see that cue, you don't get aroused. And so that's a way of utilizing that same thing. And it works very well.

- 8.98** So do dreams give you that opportunity to face those things? And then does that help with waking life, lowering of noradrenaline?
- 8.99** Absolutely it does. Yeah, it's exactly the same. So in the dreaming state, you get that as a gift. You get that as a gift. Your brain picks the most threatening scenarios and exposes you to them, and then you practice them without noradrenaline. In the waking state, though, you choose yourself. You say, "Look, I have these scenarios. Whenever I have that thing happen to me, I'm going to relax and stay calm, and that will be helpful." Now, you can even imagine some meditation as well. You can close your eyes and imagine a certain scenario and meditate on that and keep your adrenaline levels low. All right. Thank you very much.